

Shatter Clustering: Surface Density Threshold (alpha_MAD) Analysis

Sweep over $\alpha_{\text{MAD}} \in \{0.25, 0.50, 0.75, 1.00\}$ with Default Defaults

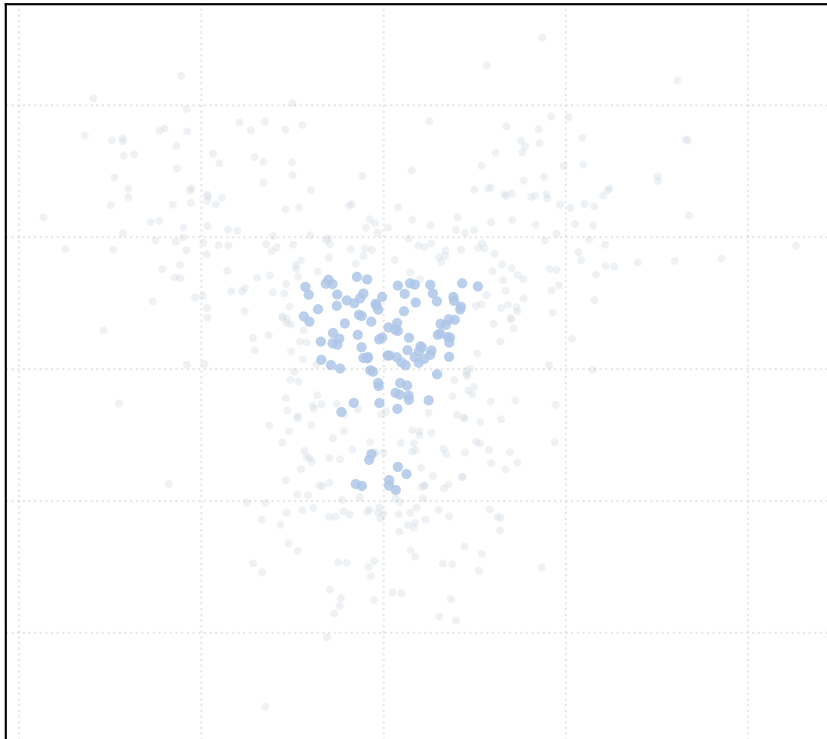
Experimental Configuration

- Total Evaluated Datasets: 30
- Tested alpha_MAD Values: [0.25, 0.5, 0.75, 1.0]
- Total Executions: 120 (4 runs per dataset)
- Fixed Default Parameters: alpha_accept = 0.5, t_percentile = 0.15, quant = 0.15
- Target Output PDF: benchmark_report_alpha_mad.pdf

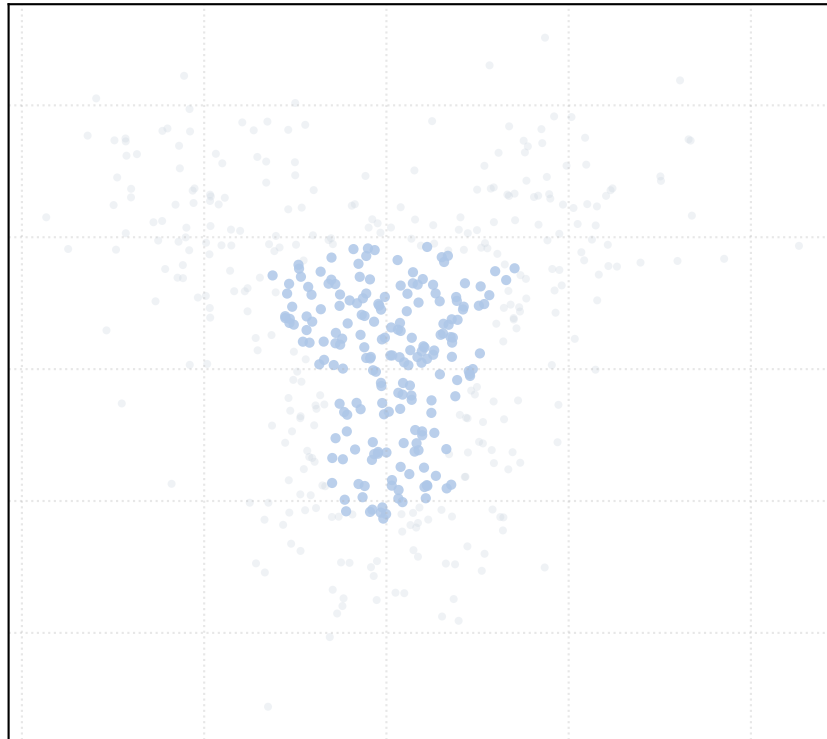
Parameter	Value / Range	Algorithmic Role
alpha_MAD	{0.25, 0.50, 0.75, 1.00}	MAD multiplier for surface growth compatibility in grow_ense
alpha_accept	0.50 (Default)	Log2 fold-change acceptance cutoff
t_percentile	0.15 (Default)	Top 15% densest seeds
mean_dist_quant	0.15 (Default)	Quantile for adaptive search radius r
min_intersection	1 (Default)	Overlapping vectors to trigger merge

Dataset: bifurcation_2way_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

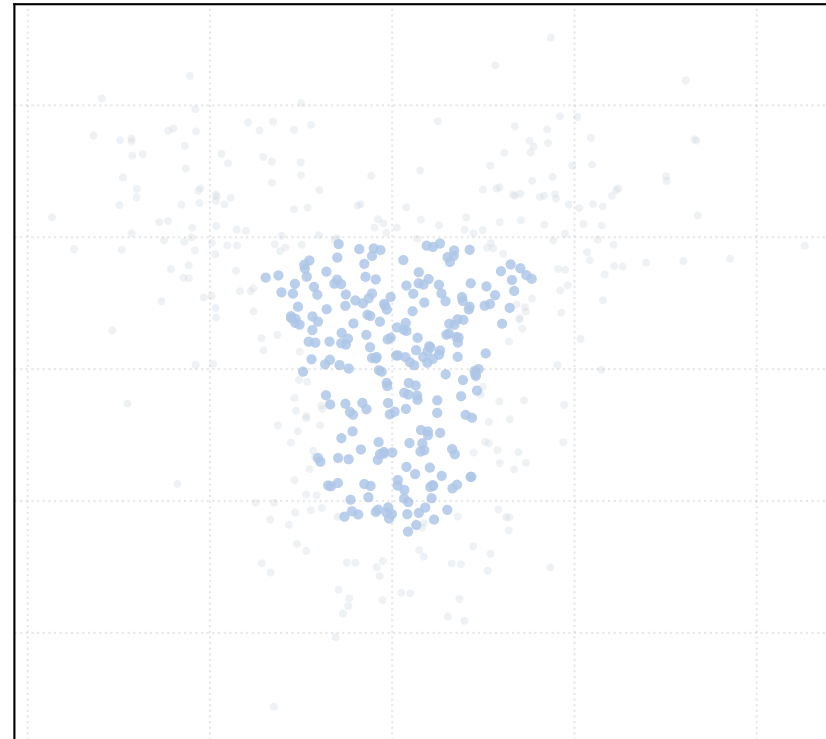
alpha_MAD = 0.25 [1 Clusters]



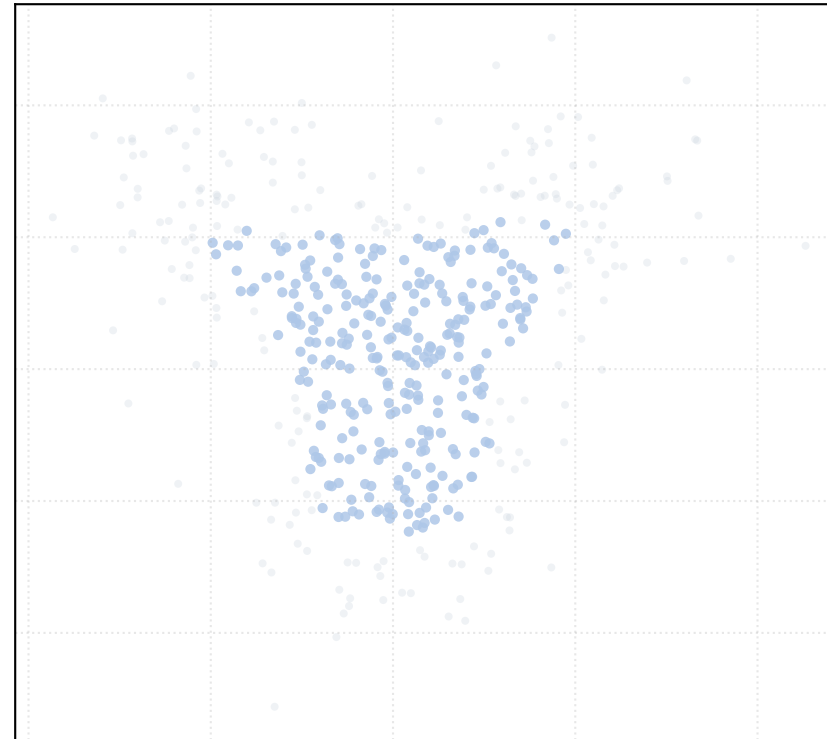
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

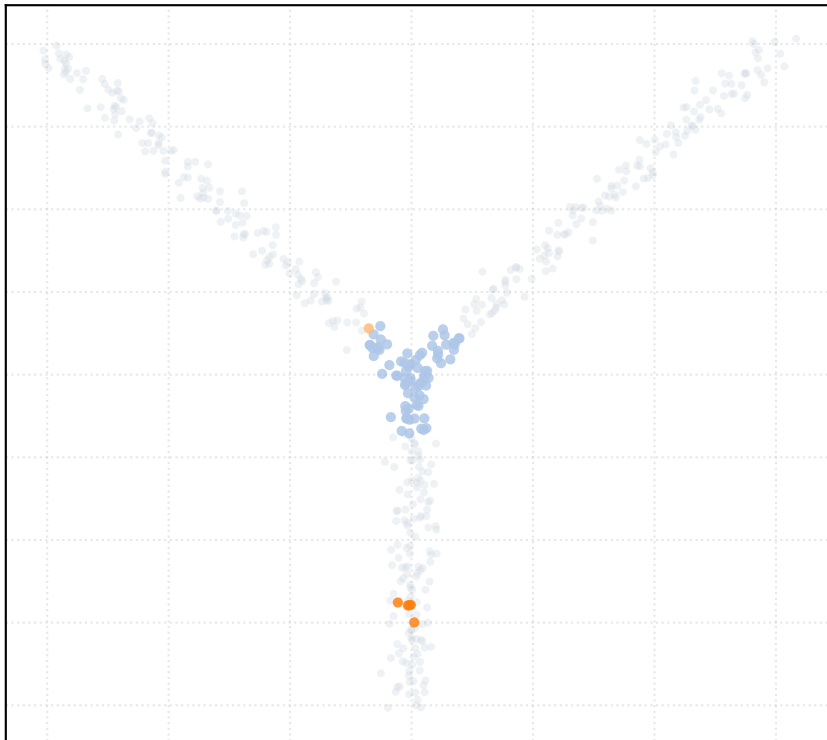


alpha_MAD = 1.0 [1 Clusters]

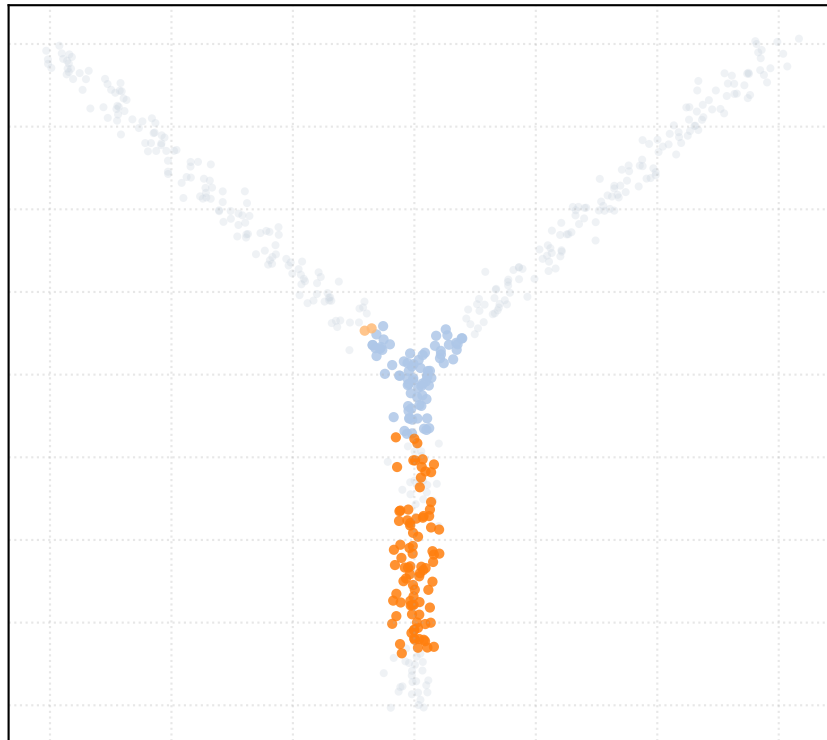


Dataset: bifurcation_2way_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

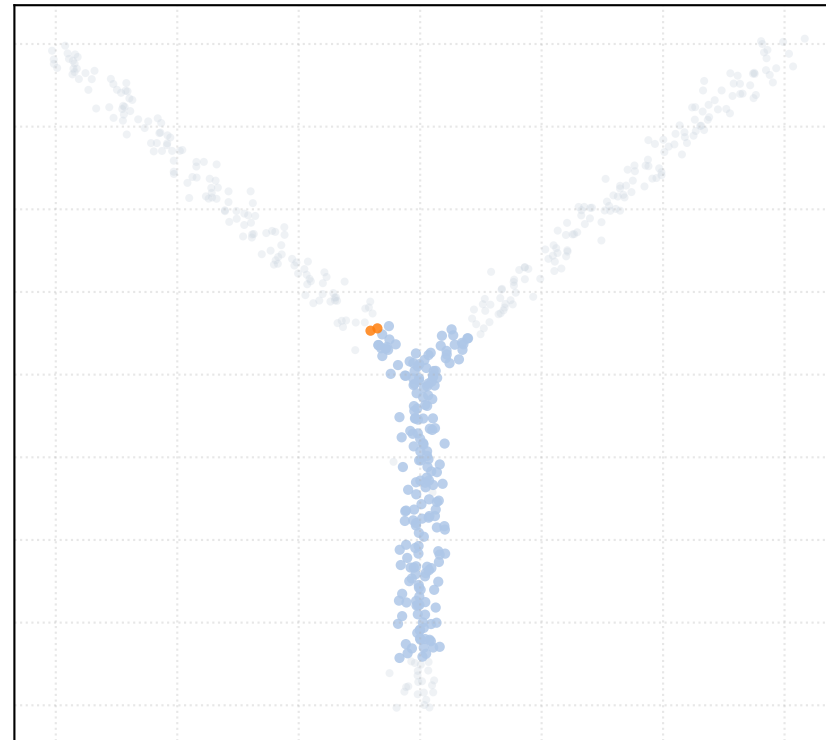
alpha_MAD = 0.25 [3 Clusters]



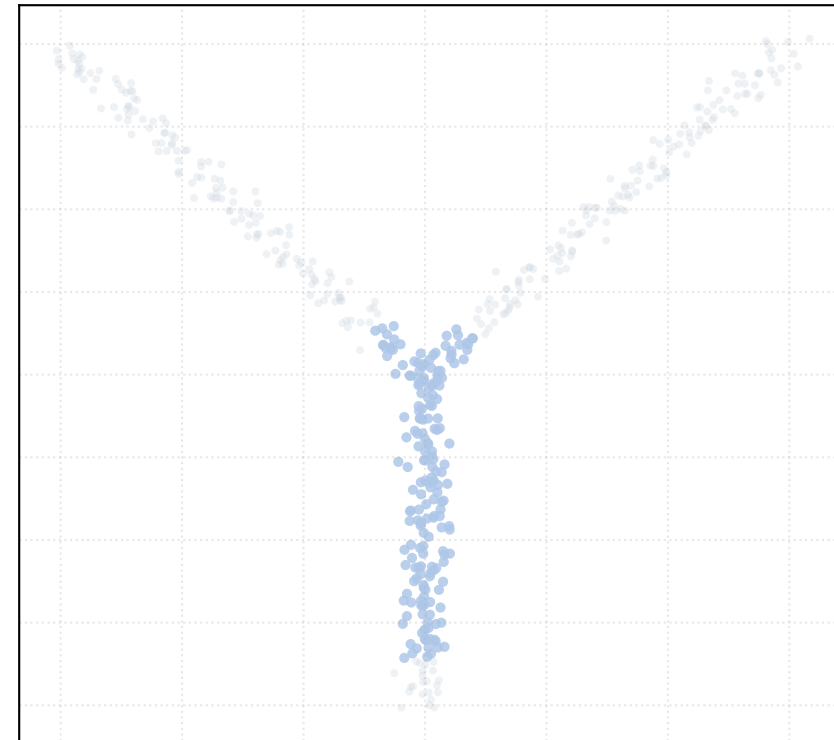
alpha_MAD = 0.5 (Default) [3 Clusters]



alpha_MAD = 0.75 [2 Clusters]

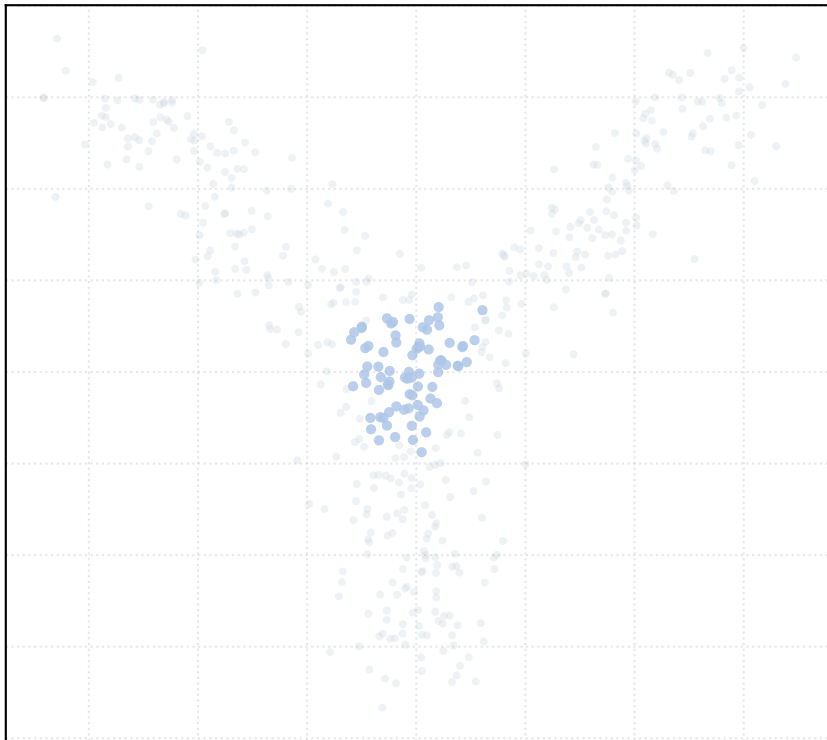


alpha_MAD = 1.0 [1 Clusters]

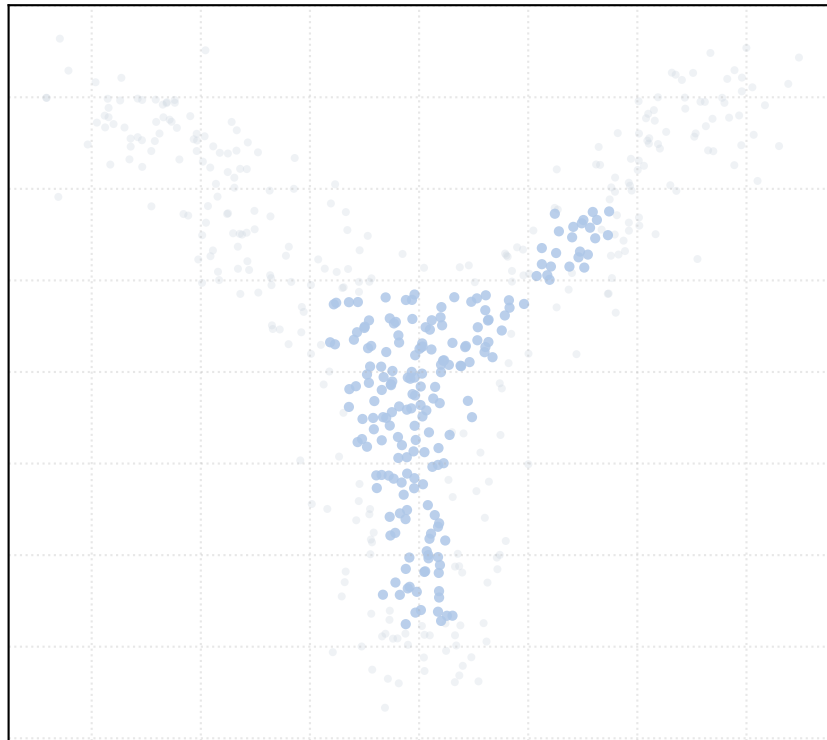


Dataset: bifurcation_2way_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

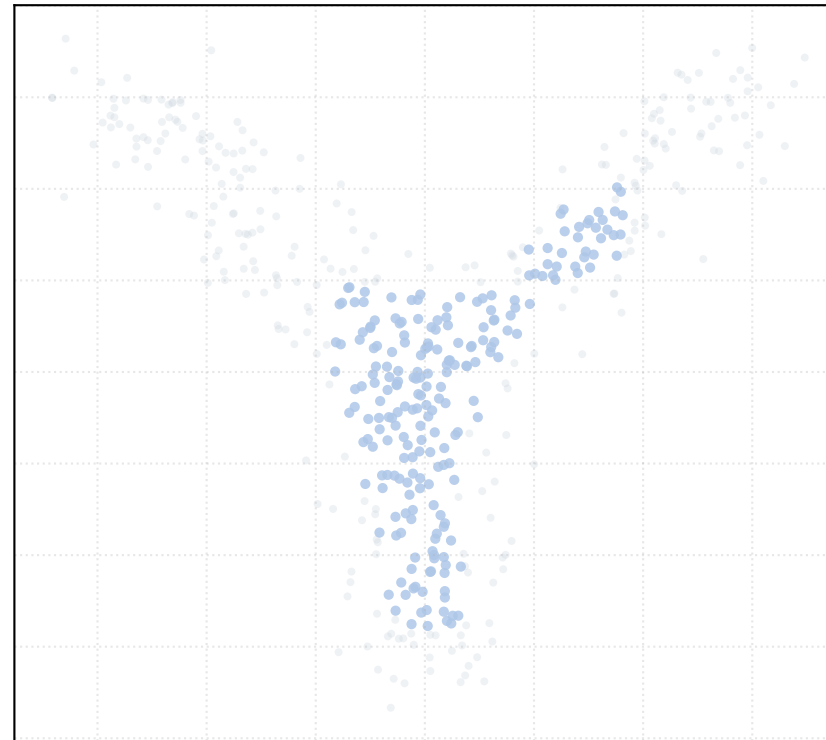
alpha_MAD = 0.25 [1 Clusters]



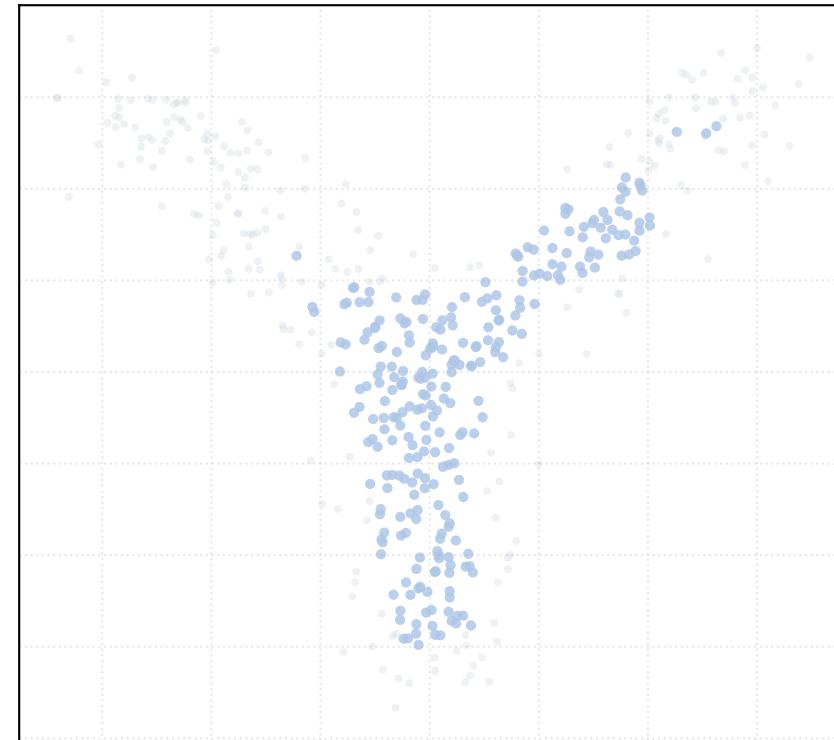
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

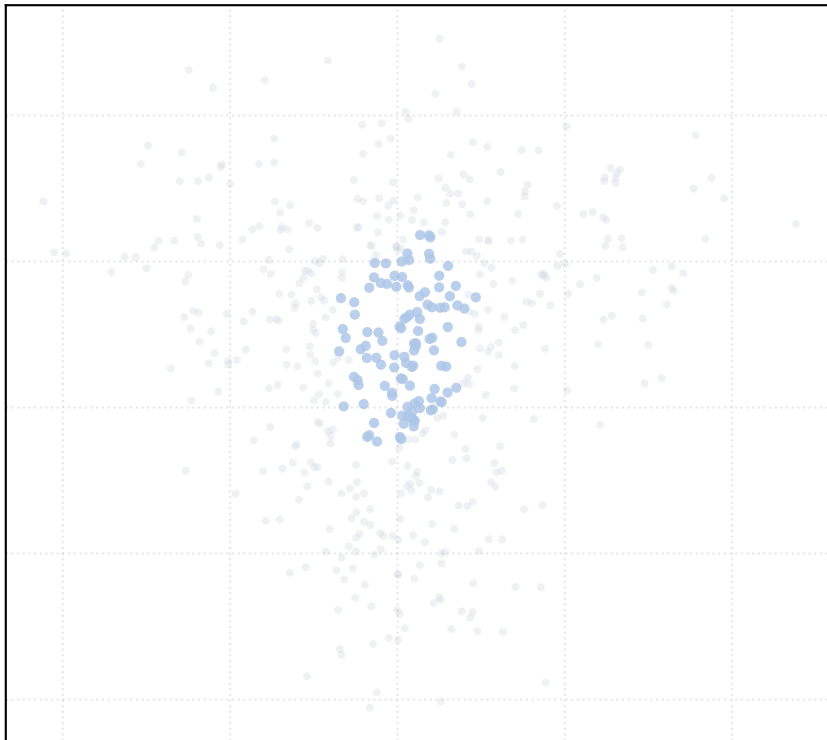


alpha_MAD = 1.0 [1 Clusters]

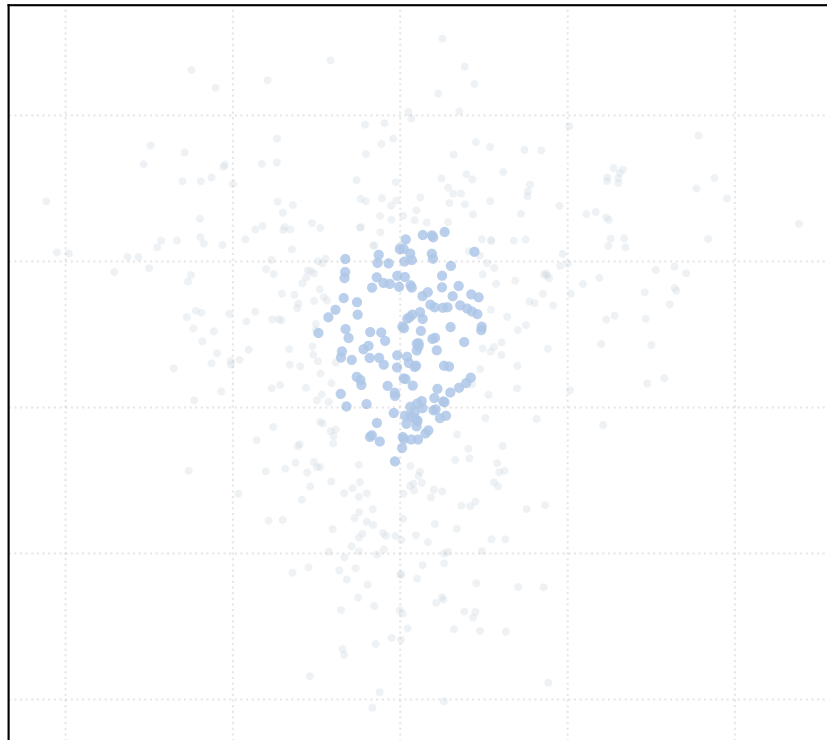


Dataset: bifurcation_3way_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

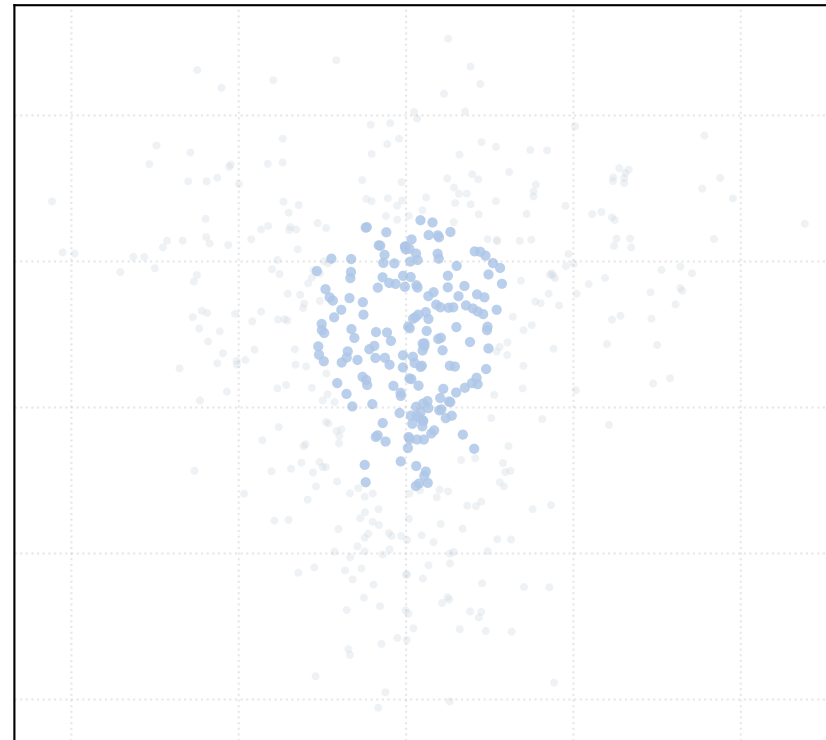
alpha_MAD = 0.25 [1 Clusters]



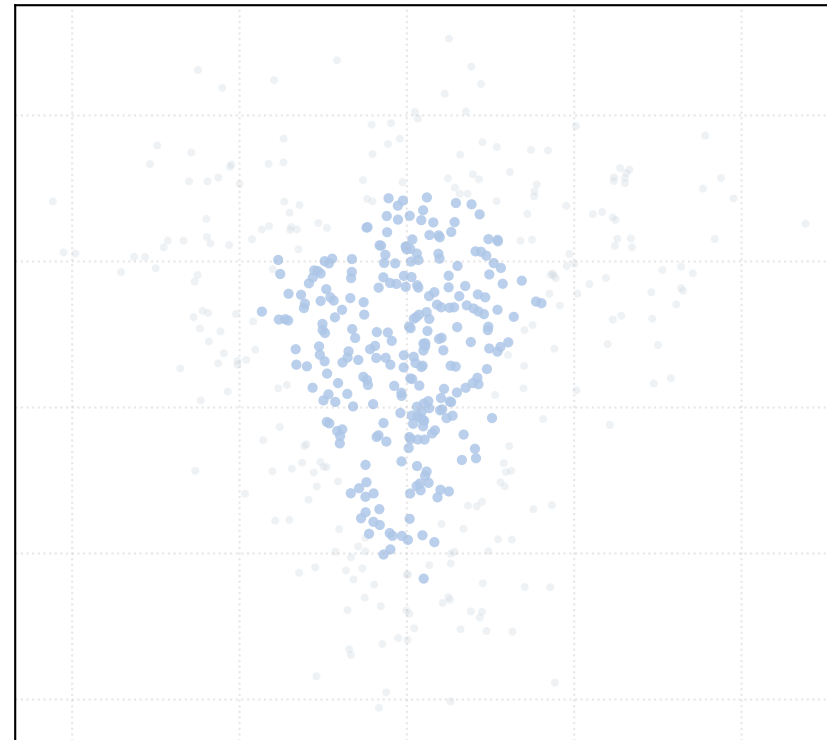
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

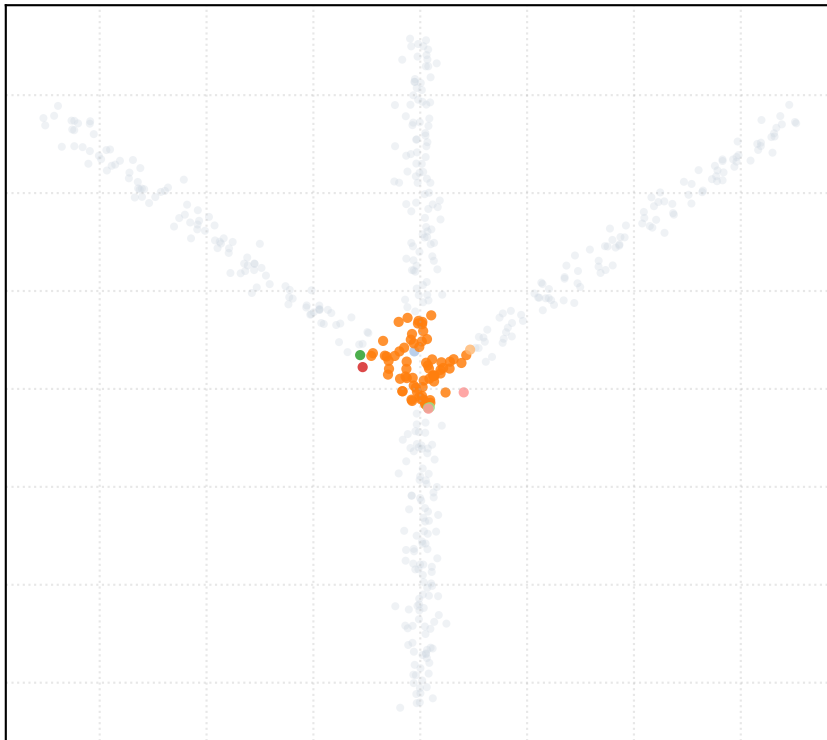


alpha_MAD = 1.0 [1 Clusters]

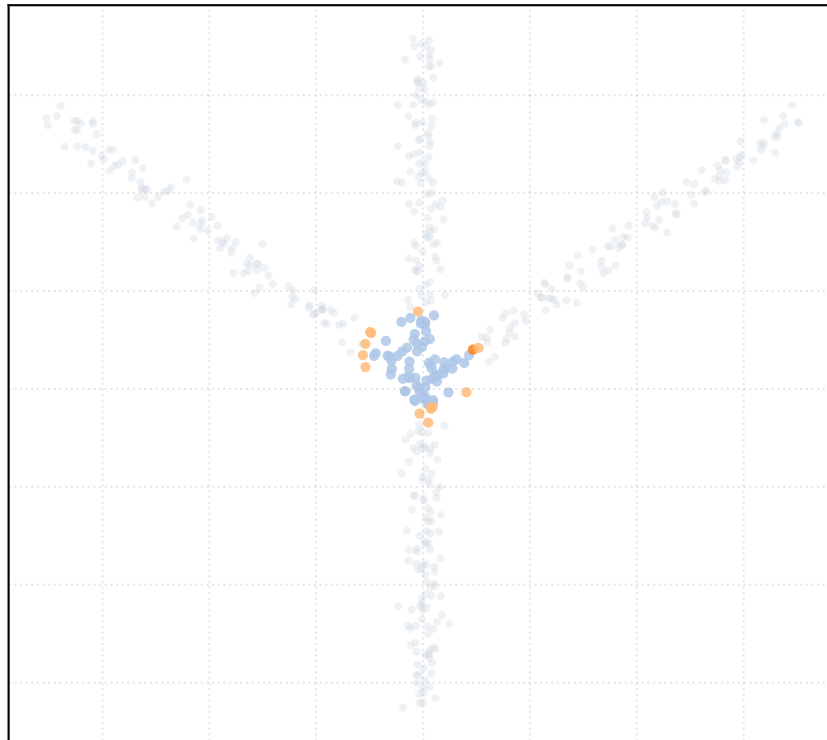


Dataset: bifurcation_3way_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

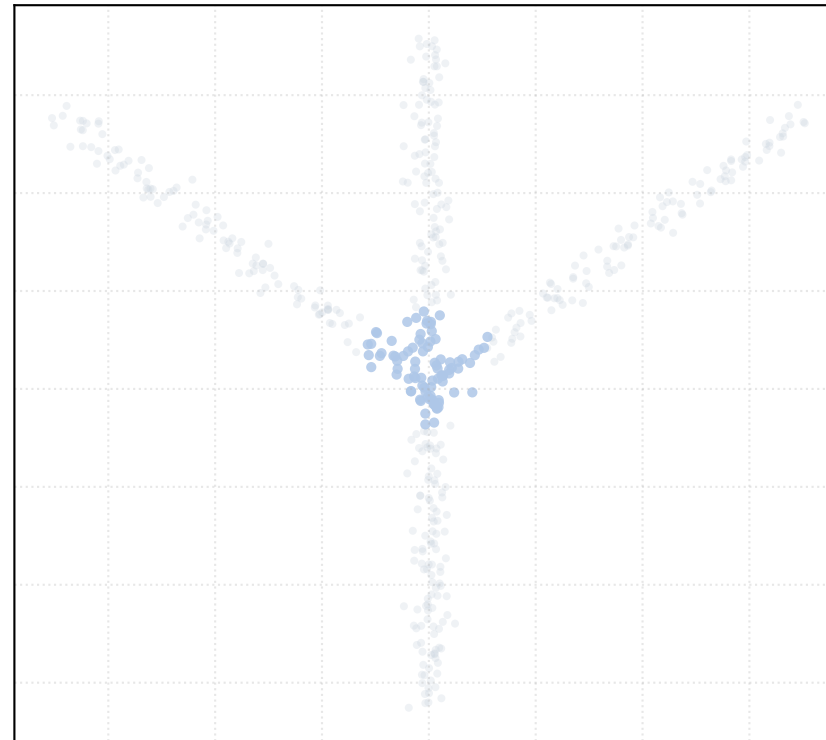
alpha_MAD = 0.25 [7 Clusters]



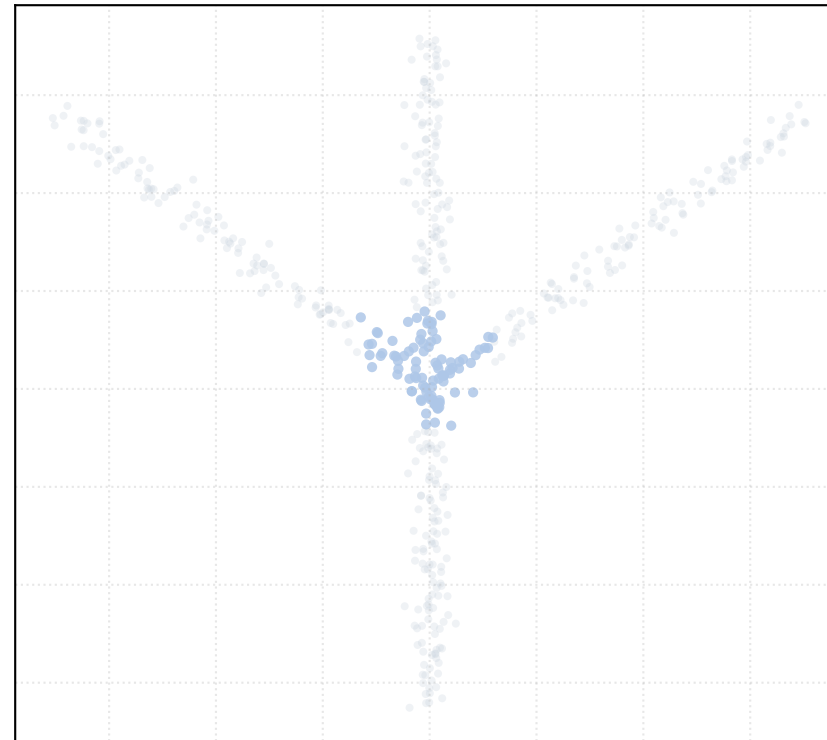
alpha_MAD = 0.5 (Default) [3 Clusters]



alpha_MAD = 0.75 [1 Clusters]

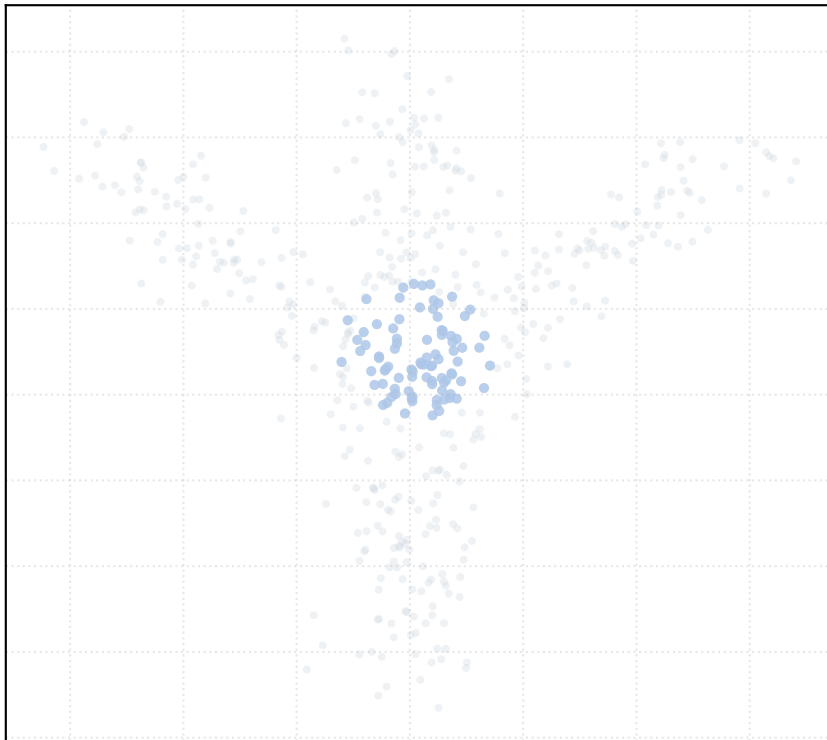


alpha_MAD = 1.0 [1 Clusters]

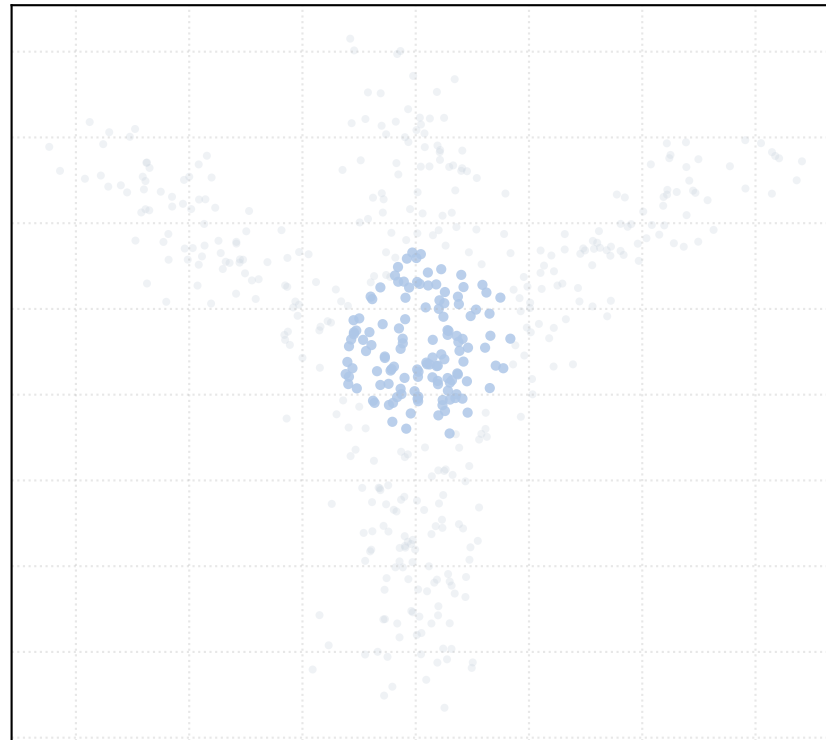


Dataset: bifurcation_3way_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

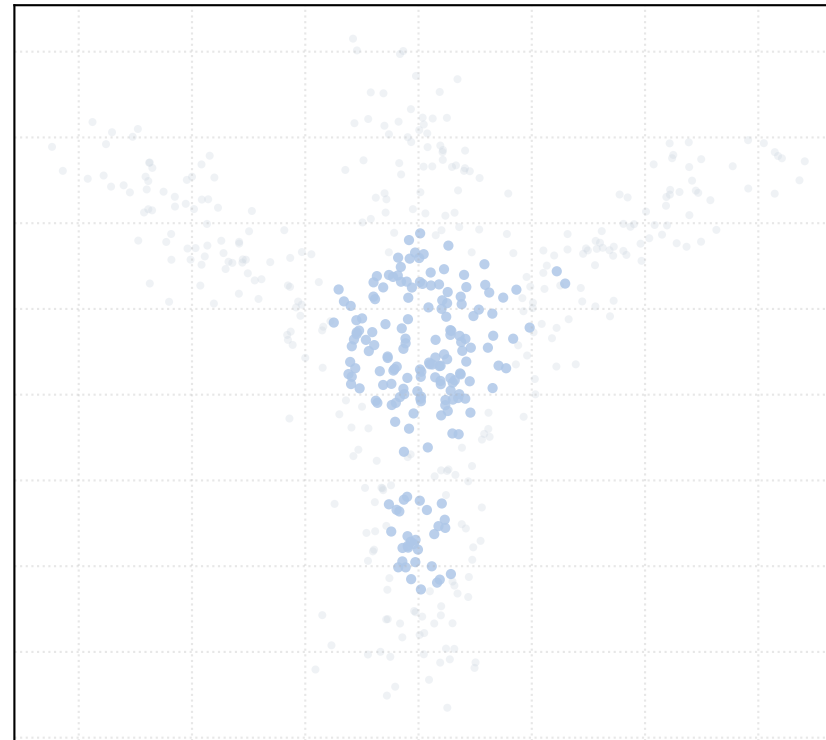
alpha_MAD = 0.25 [1 Clusters]



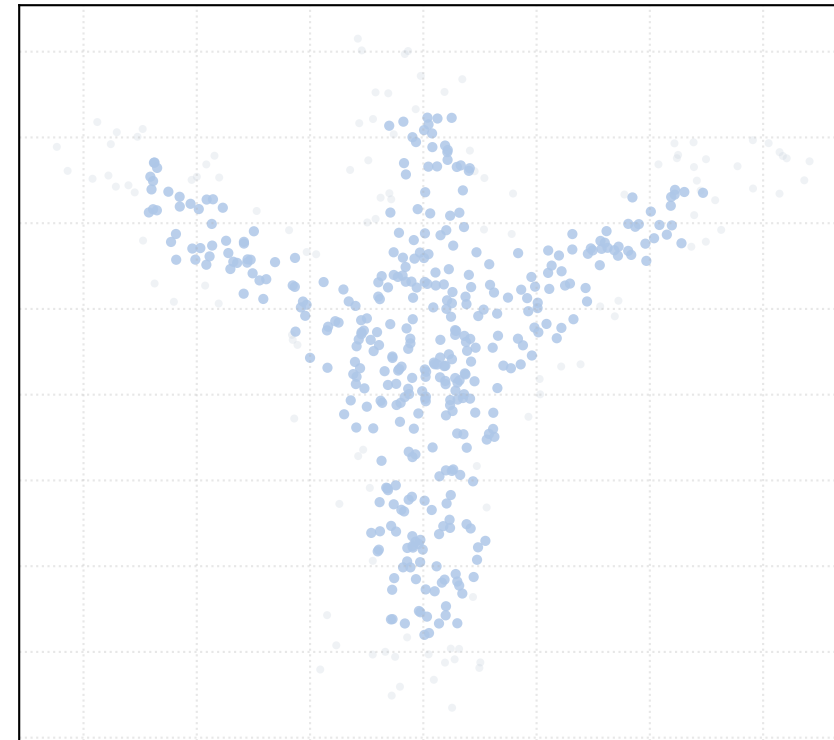
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

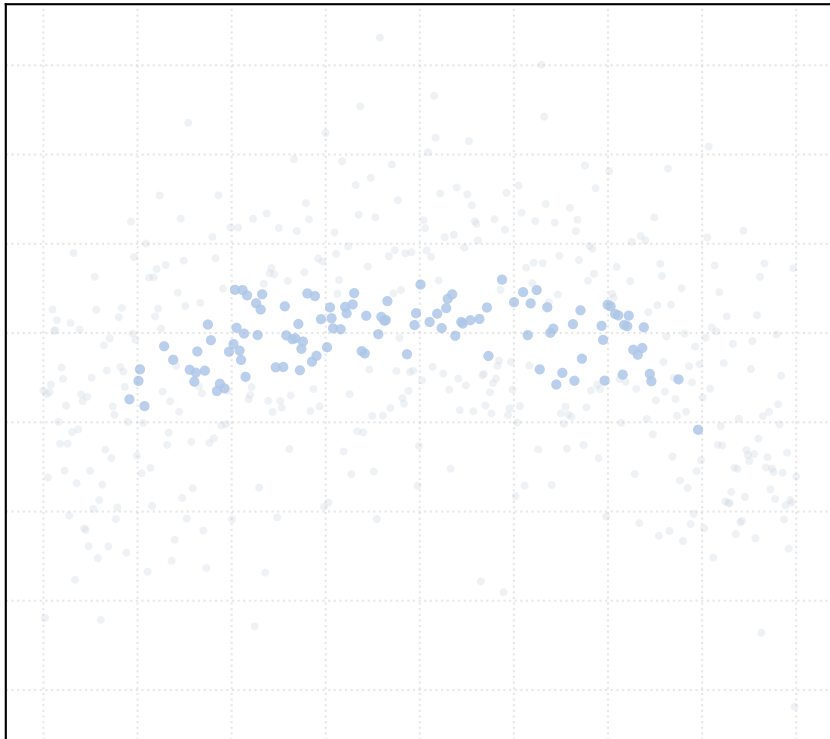


alpha_MAD = 1.0 [1 Clusters]

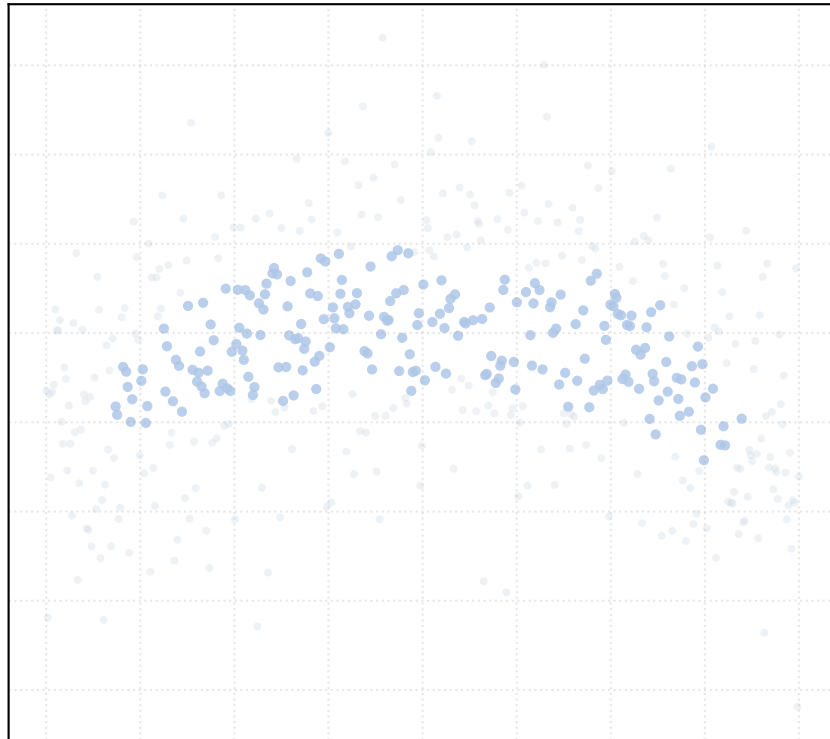


Dataset: circular_arc_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

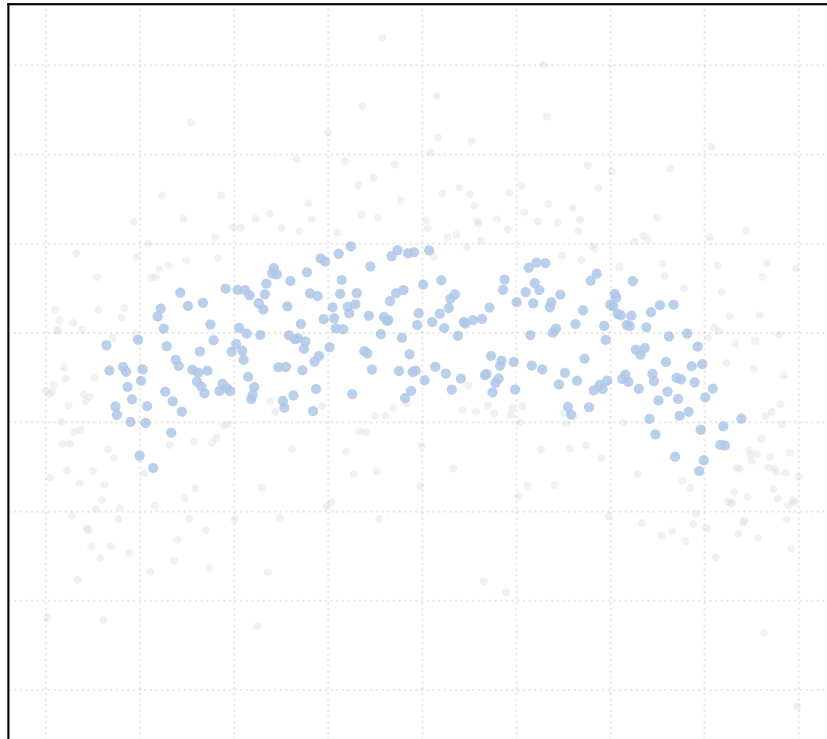
alpha_MAD = 0.25 [1 Clusters]



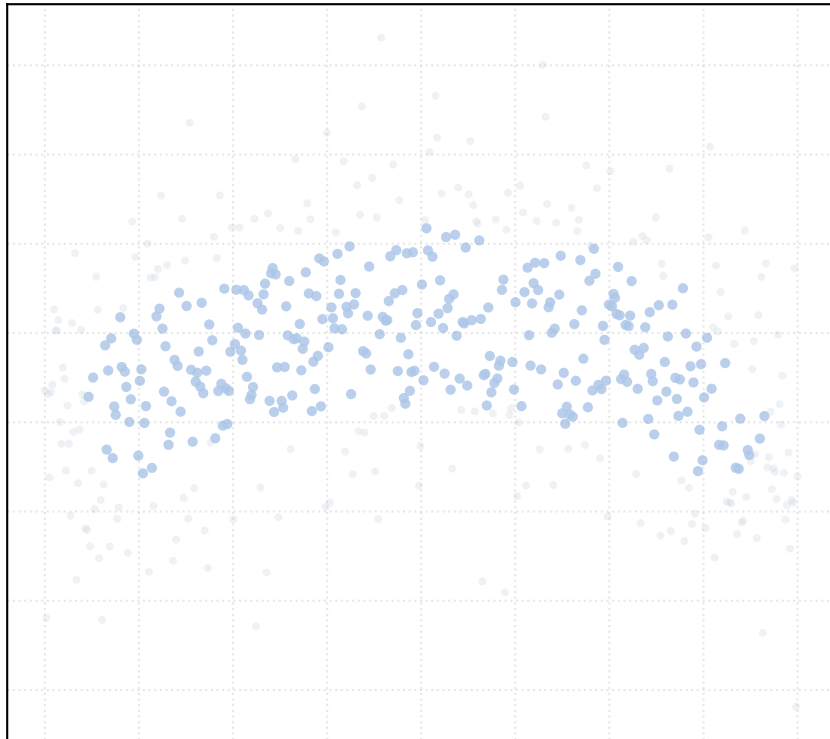
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

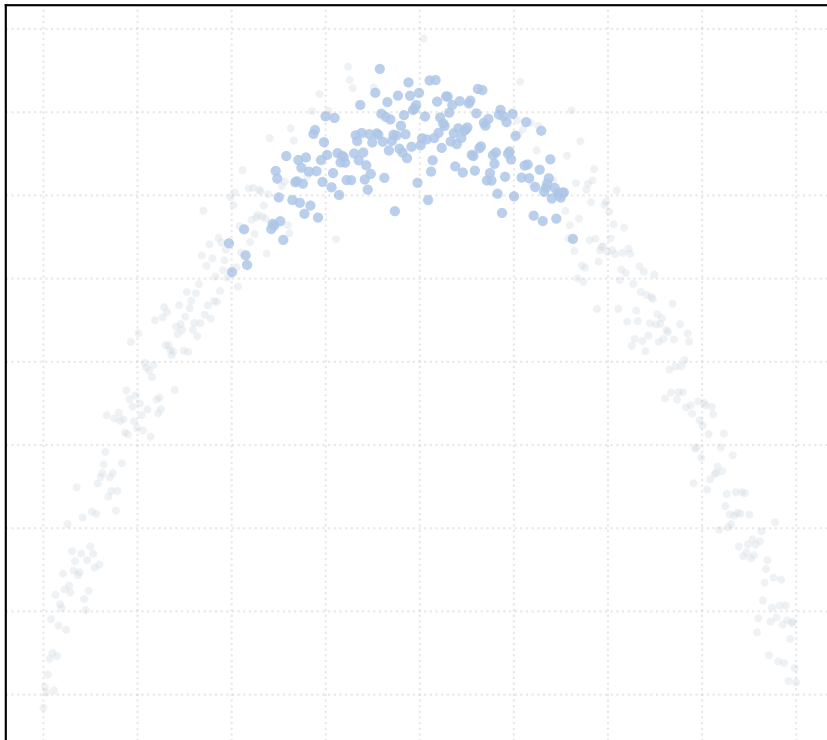


alpha_MAD = 1.0 [1 Clusters]

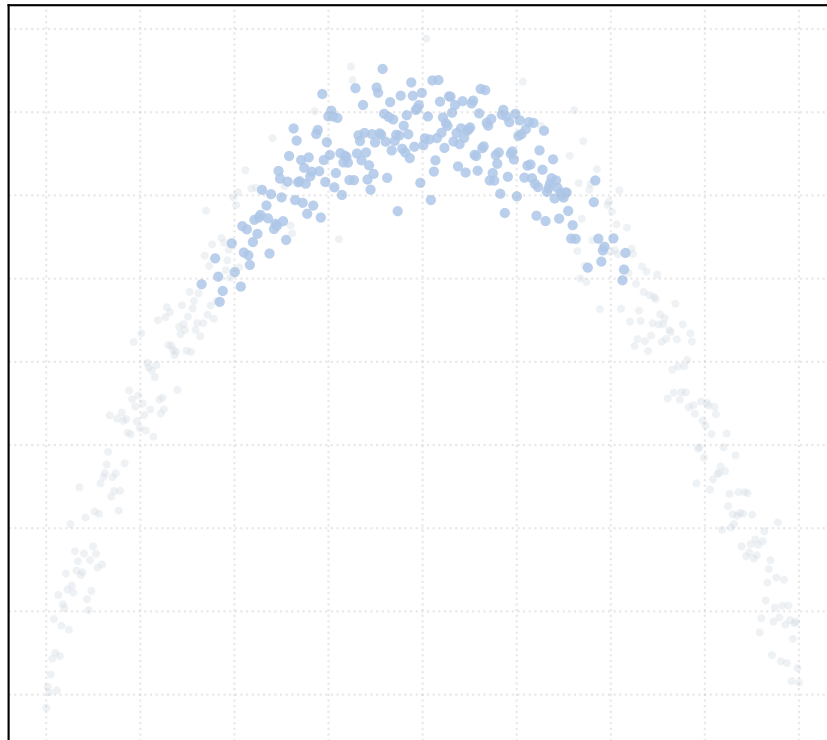


Dataset: circular_arc_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

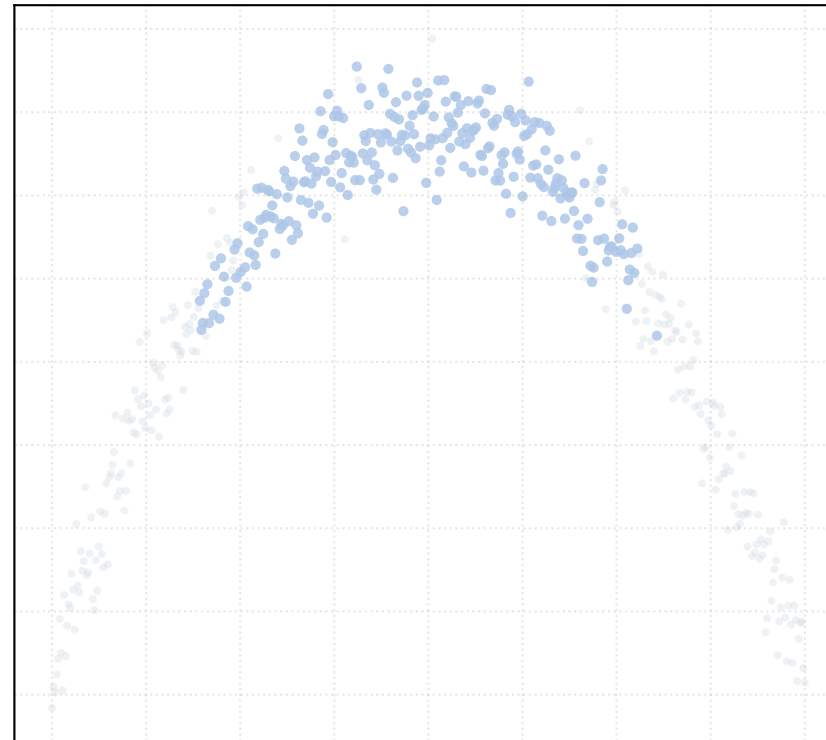
alpha_MAD = 0.25 [1 Clusters]



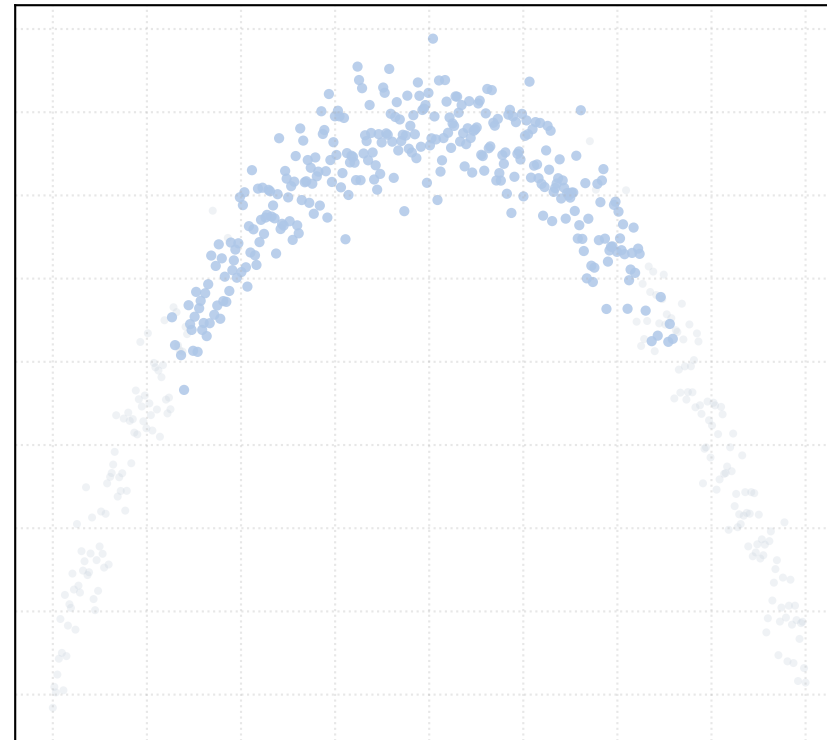
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

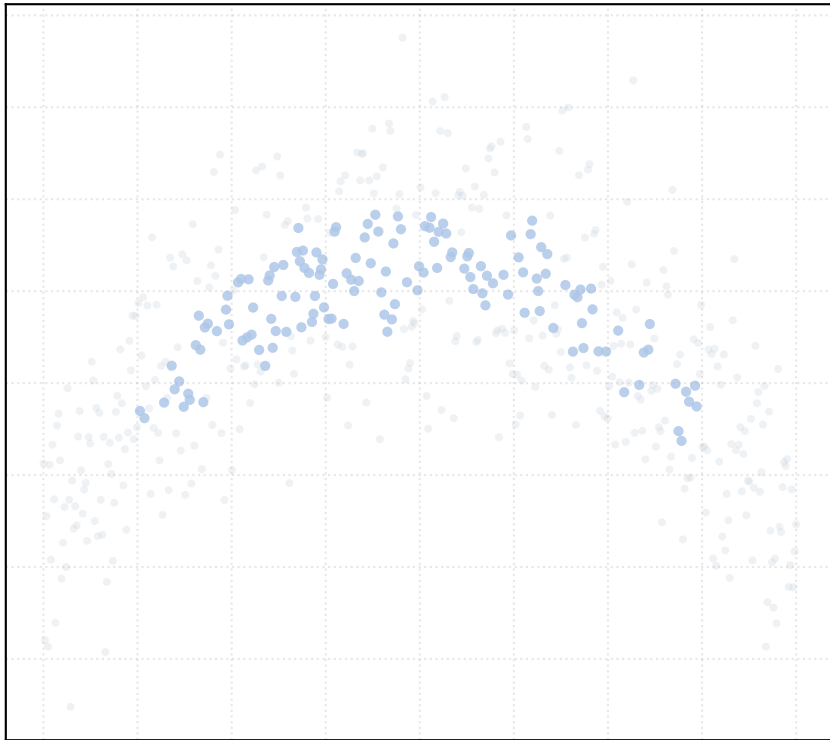


alpha_MAD = 1.0 [1 Clusters]

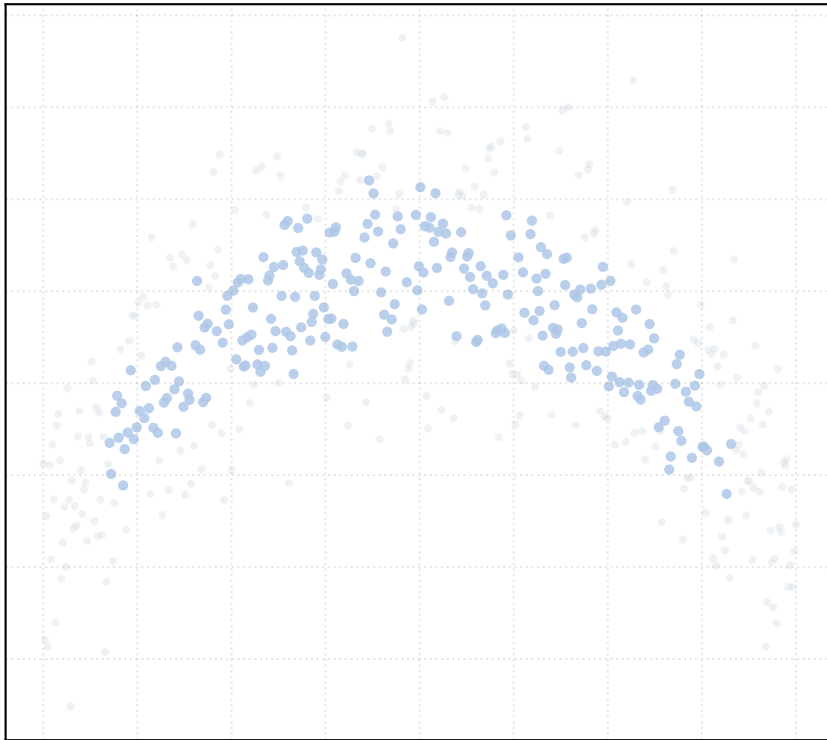


Dataset: circular_arc_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

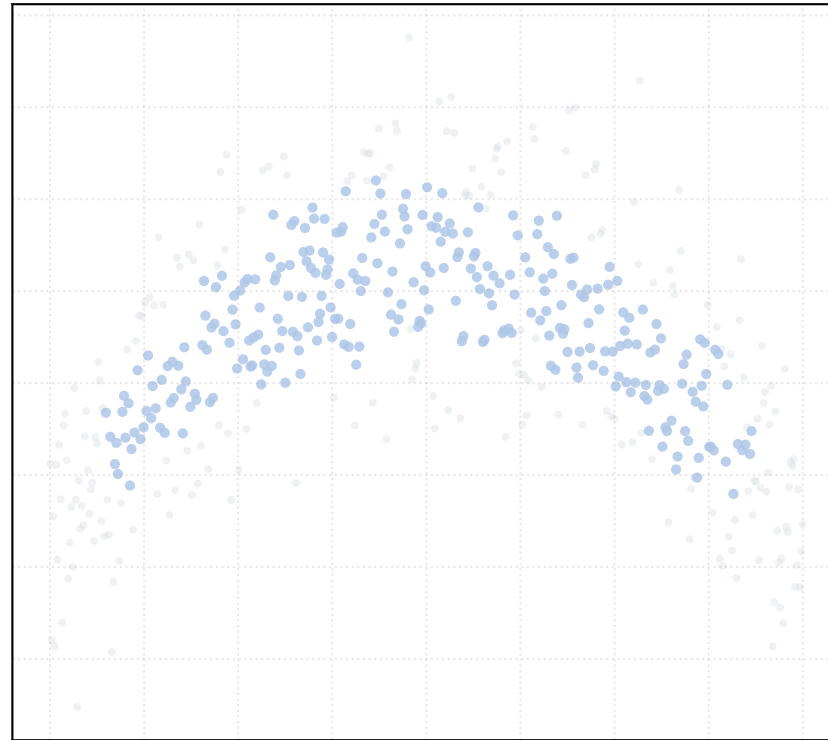
alpha_MAD = 0.25 [1 Clusters]



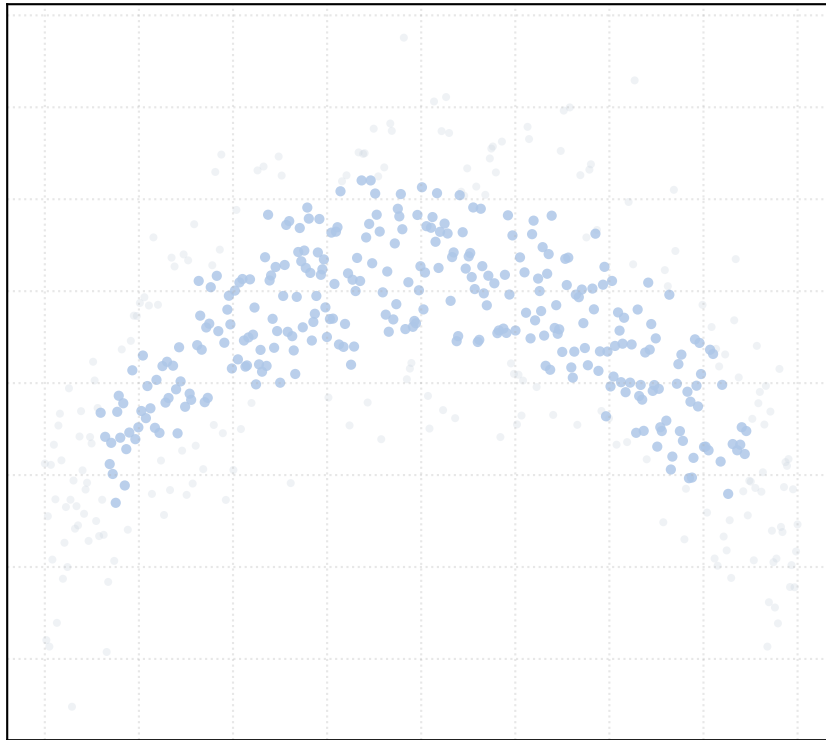
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

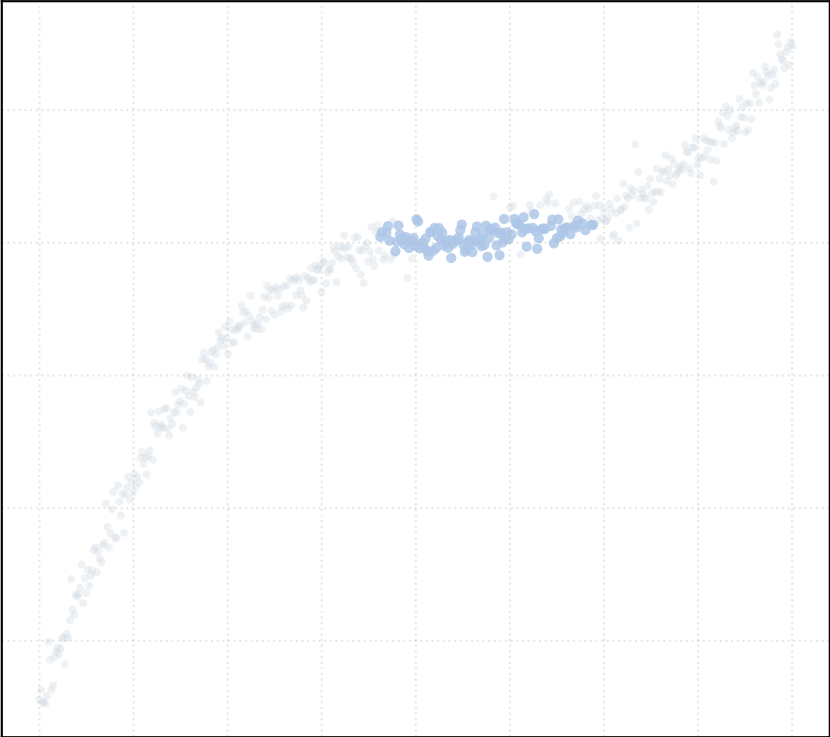


alpha_MAD = 1.0 [1 Clusters]

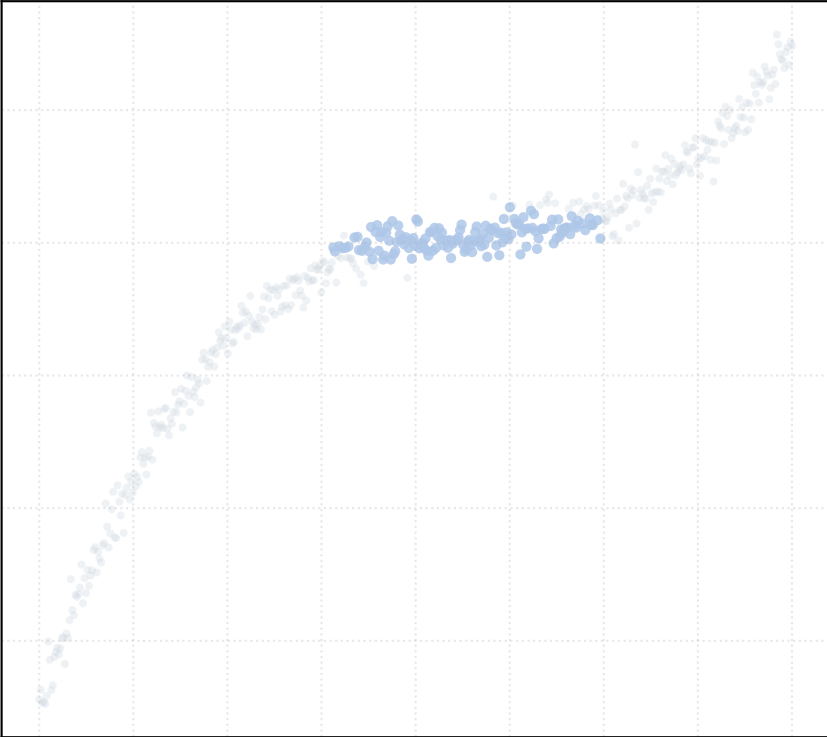


Dataset: cubic_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

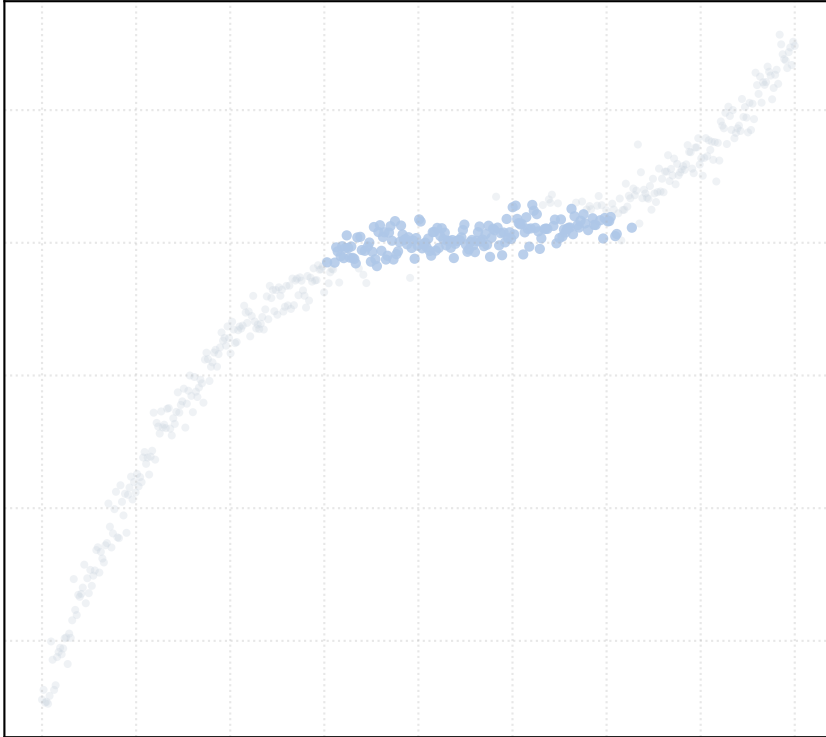
alpha_MAD = 0.25 [1 Clusters]



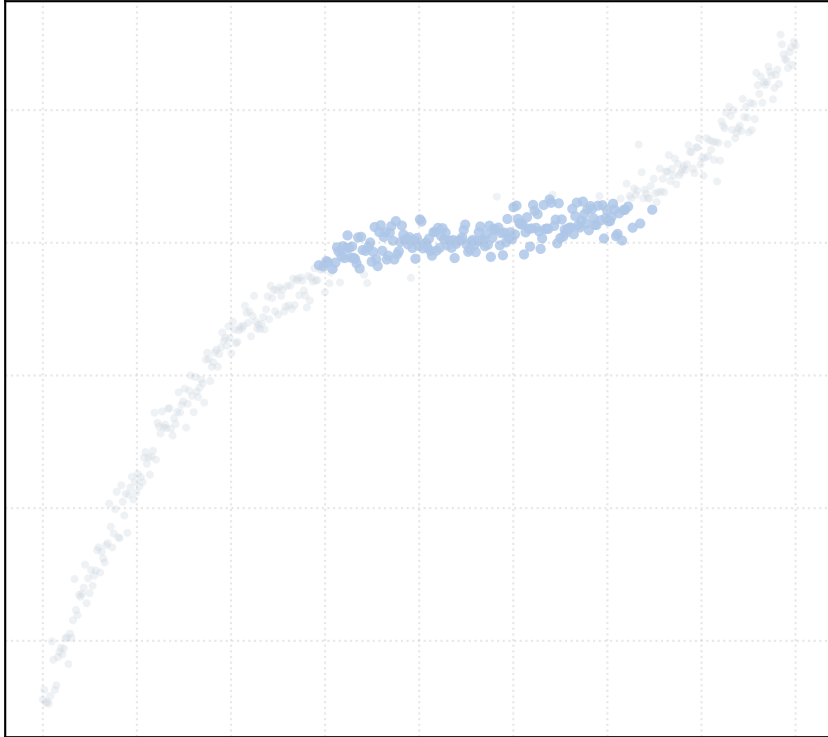
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

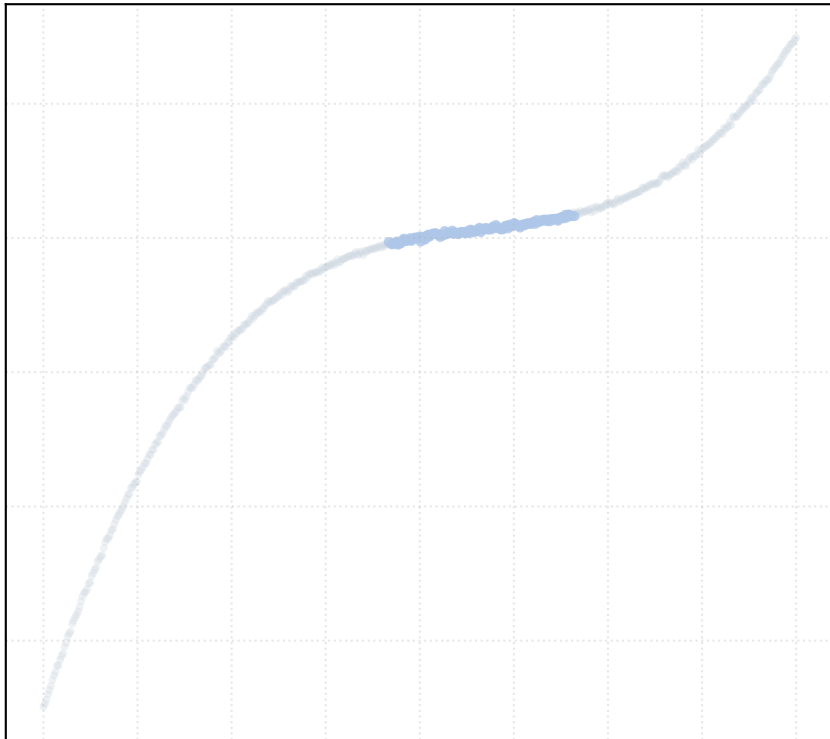


alpha_MAD = 1.0 [1 Clusters]

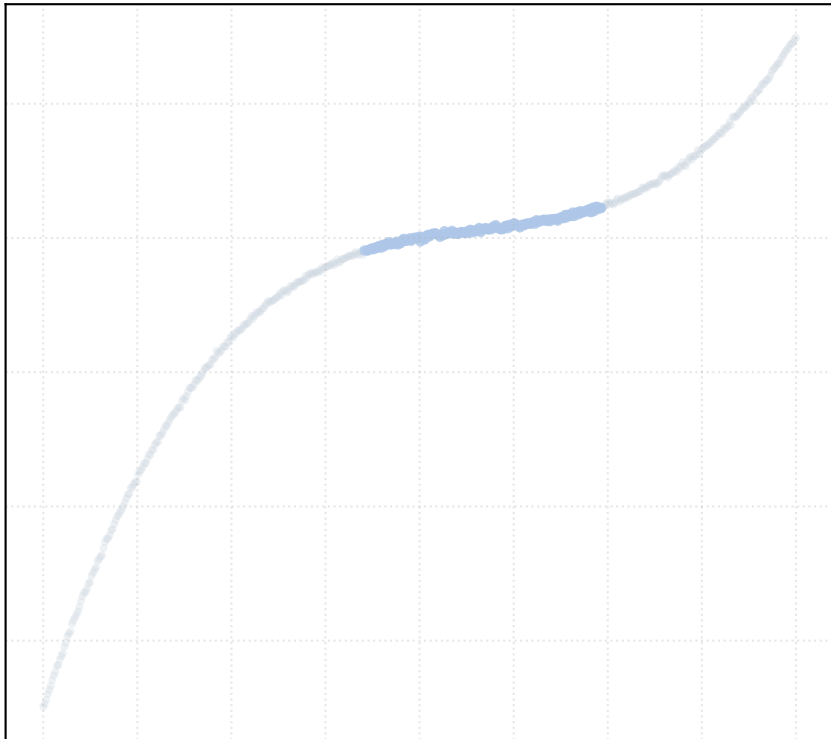


Dataset: cubic_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

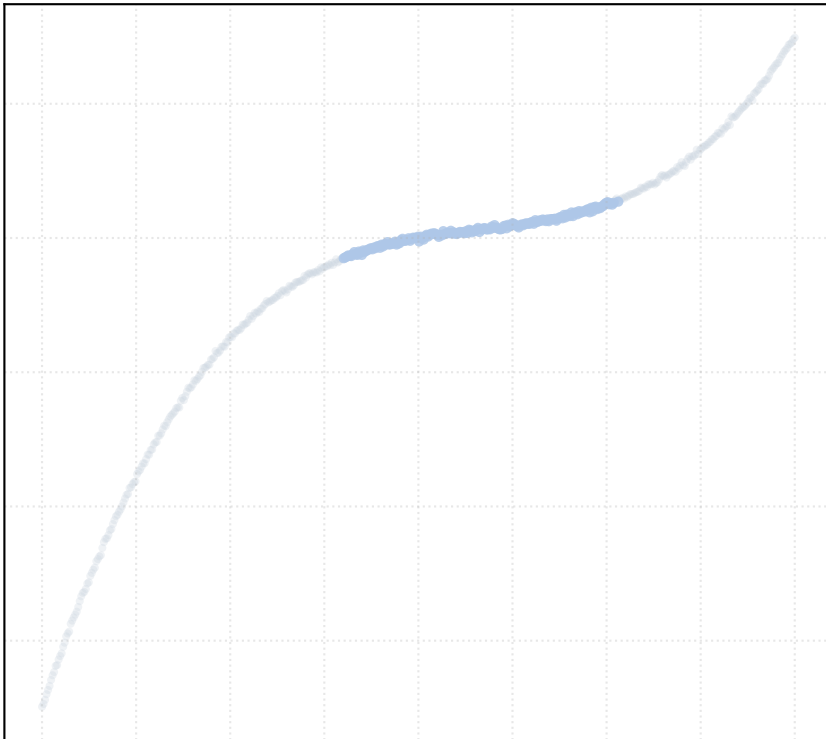
alpha_MAD = 0.25 [1 Clusters]



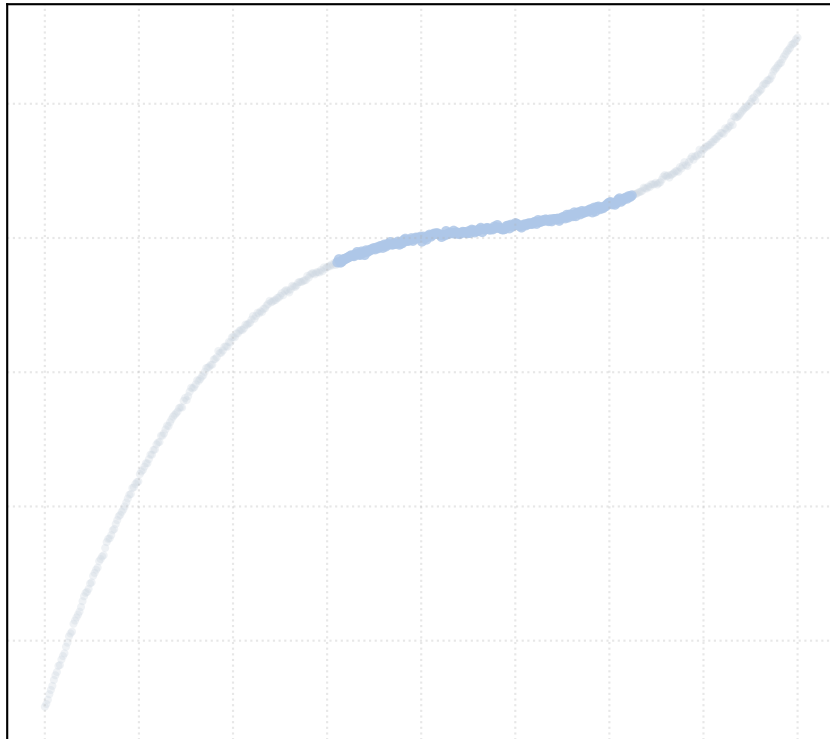
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

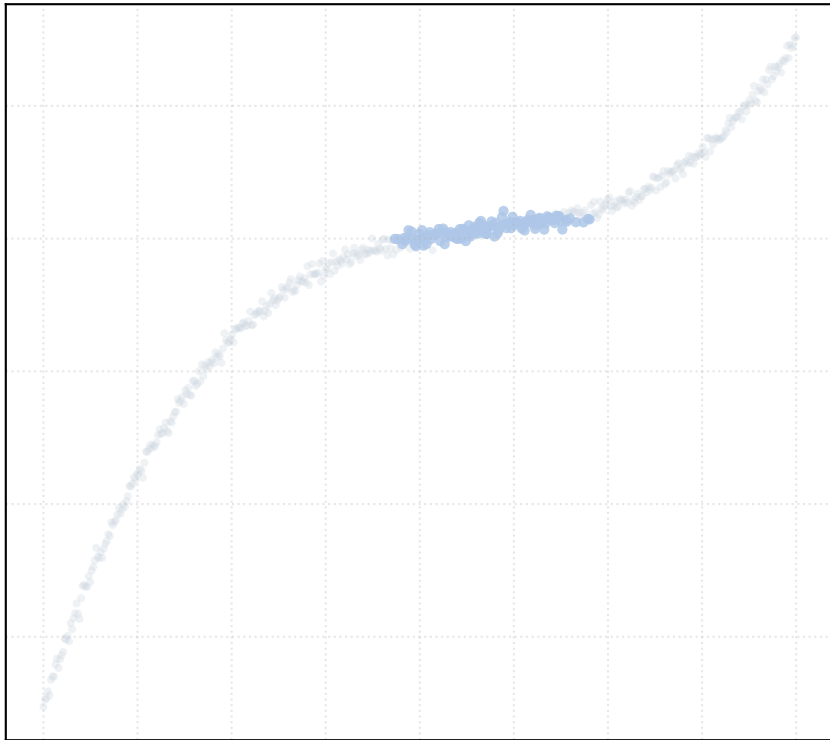


alpha_MAD = 1.0 [1 Clusters]

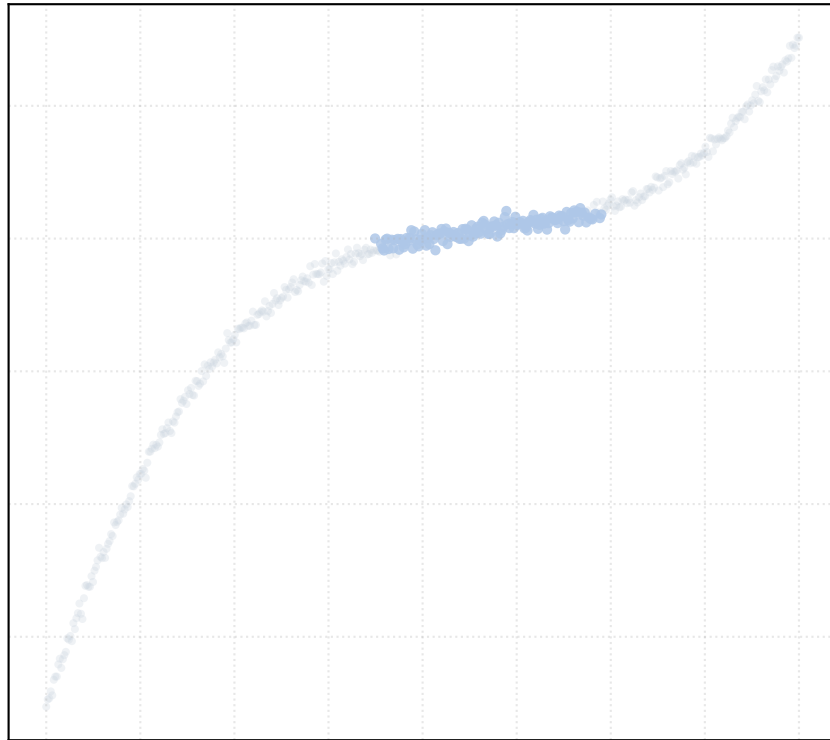


Dataset: cubic_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

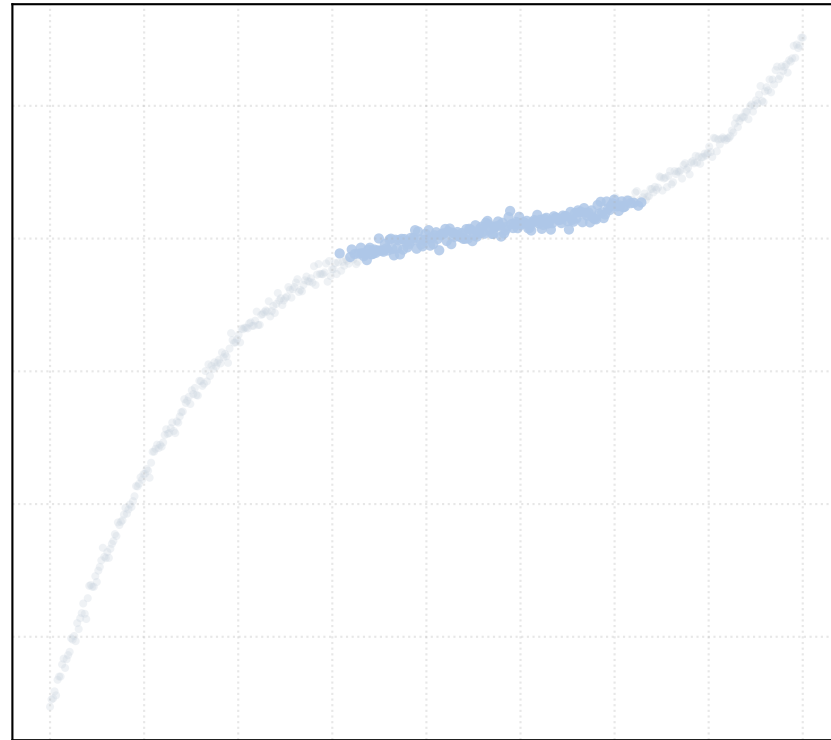
alpha_MAD = 0.25 [1 Clusters]



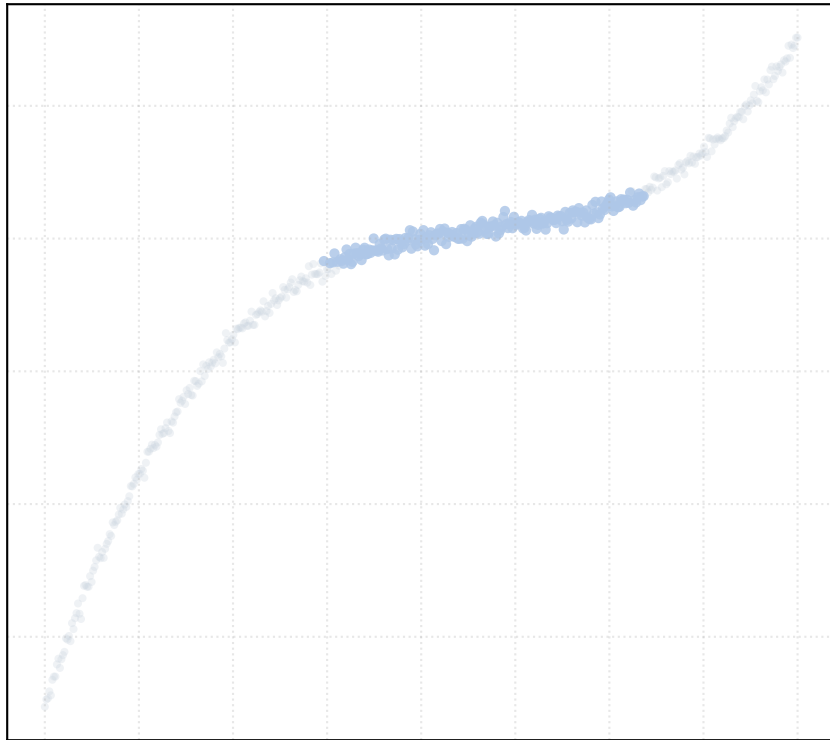
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

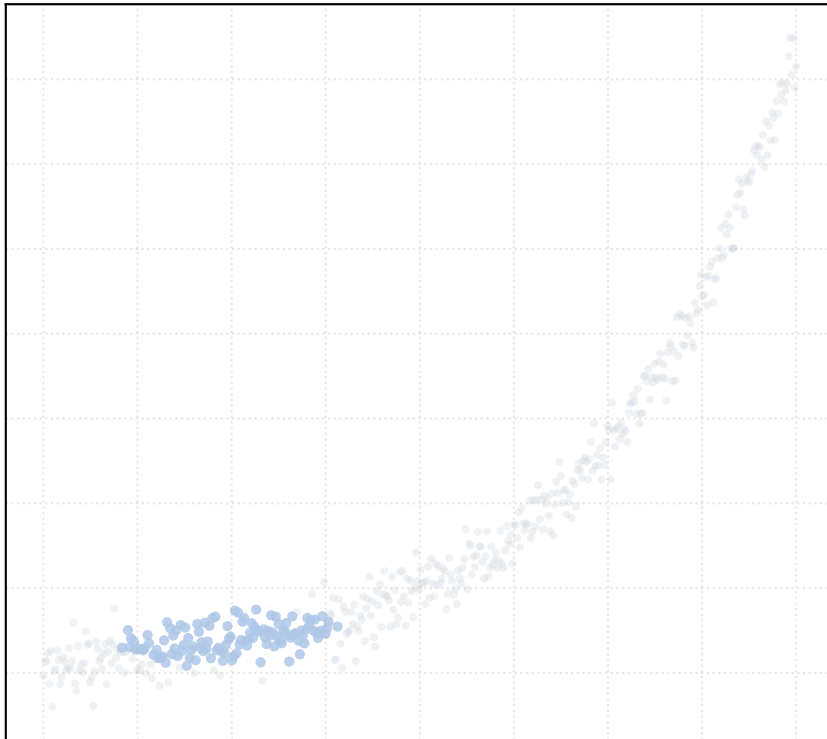


alpha_MAD = 1.0 [1 Clusters]

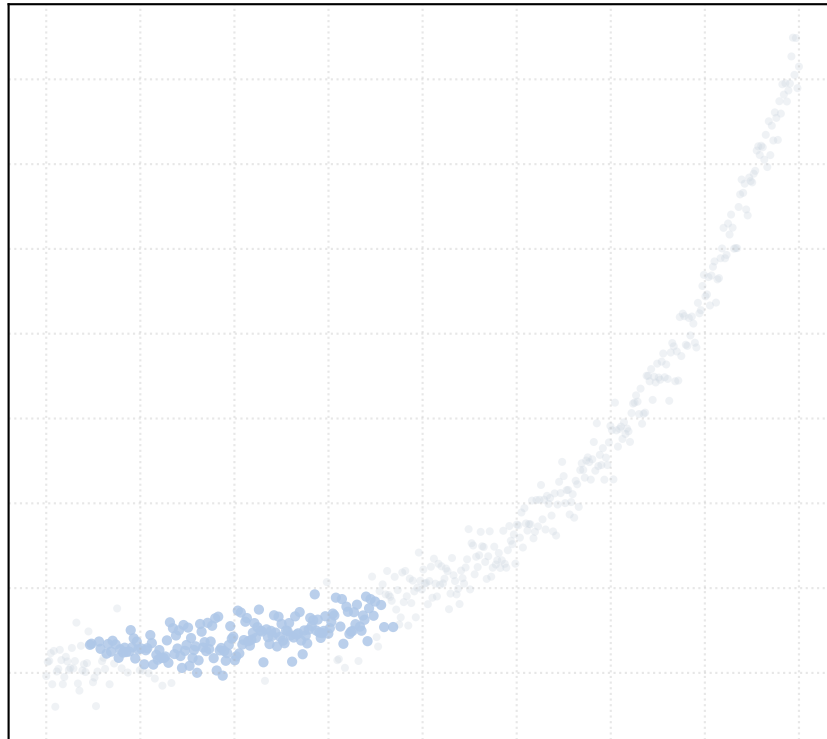


Dataset: exponential_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

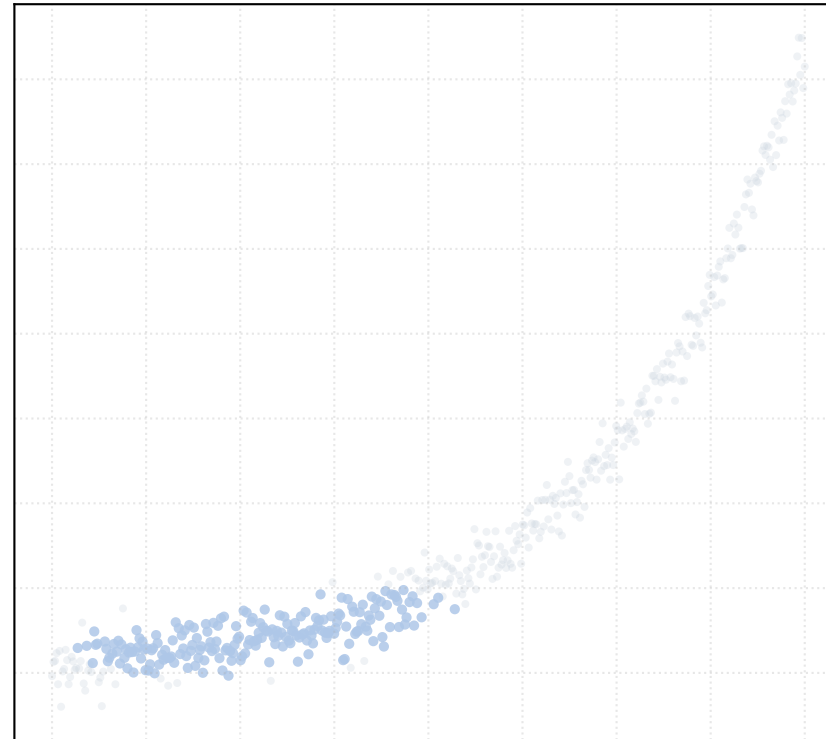
alpha_MAD = 0.25 [1 Clusters]



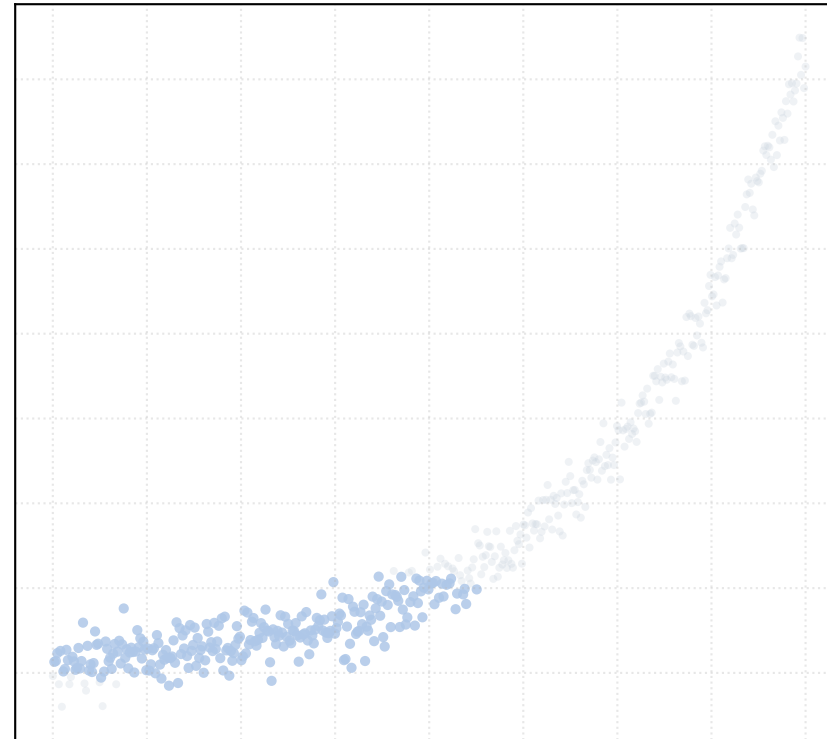
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

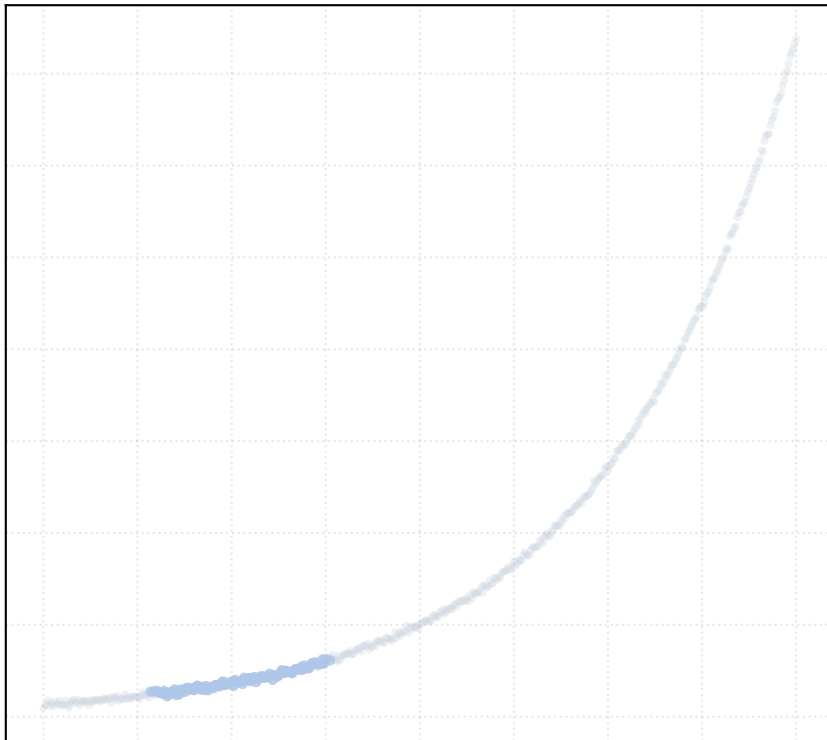


alpha_MAD = 1.0 [1 Clusters]

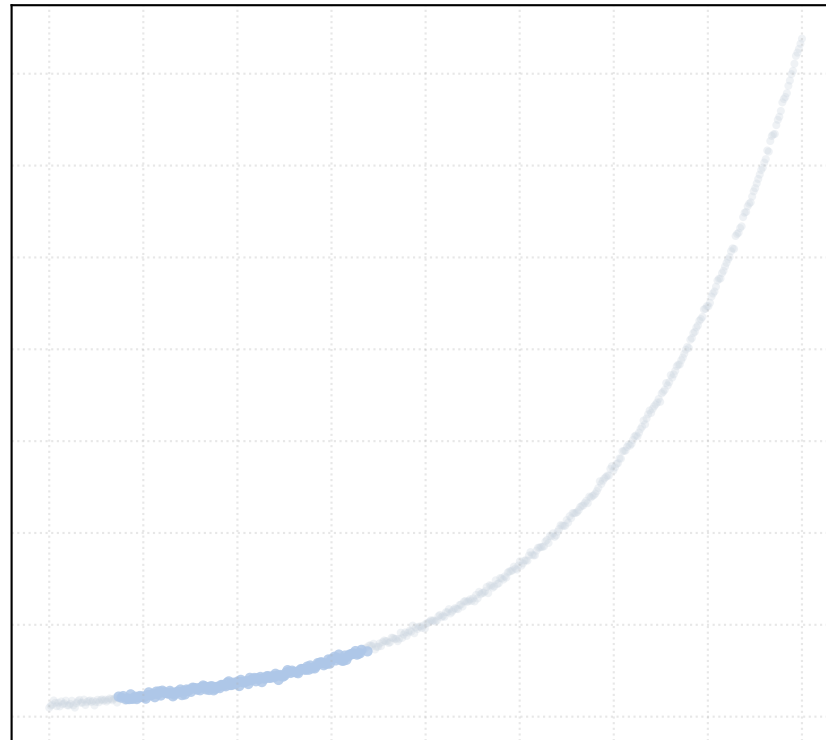


Dataset: exponential_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

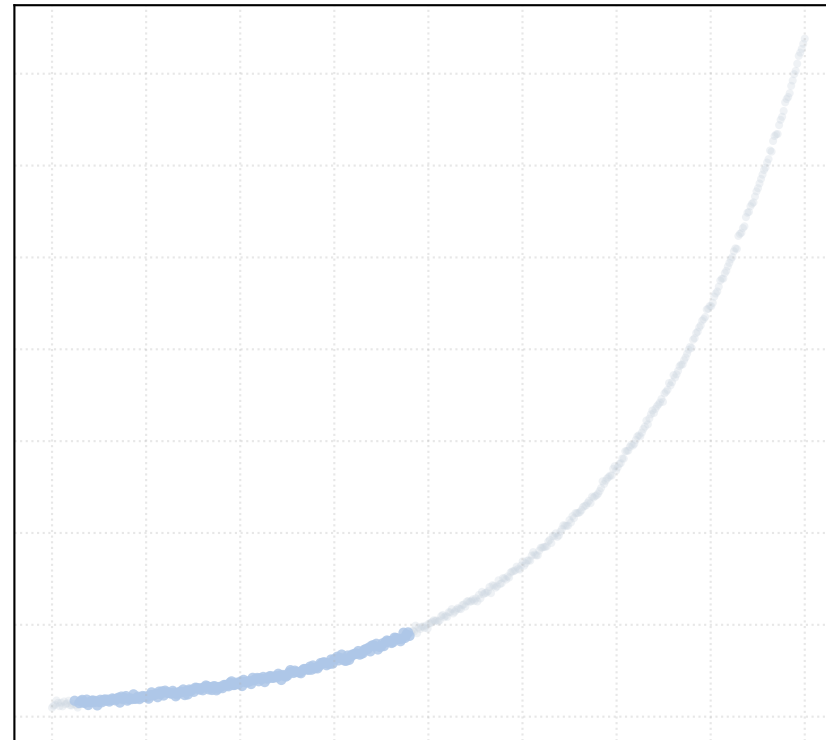
alpha_MAD = 0.25 [1 Clusters]



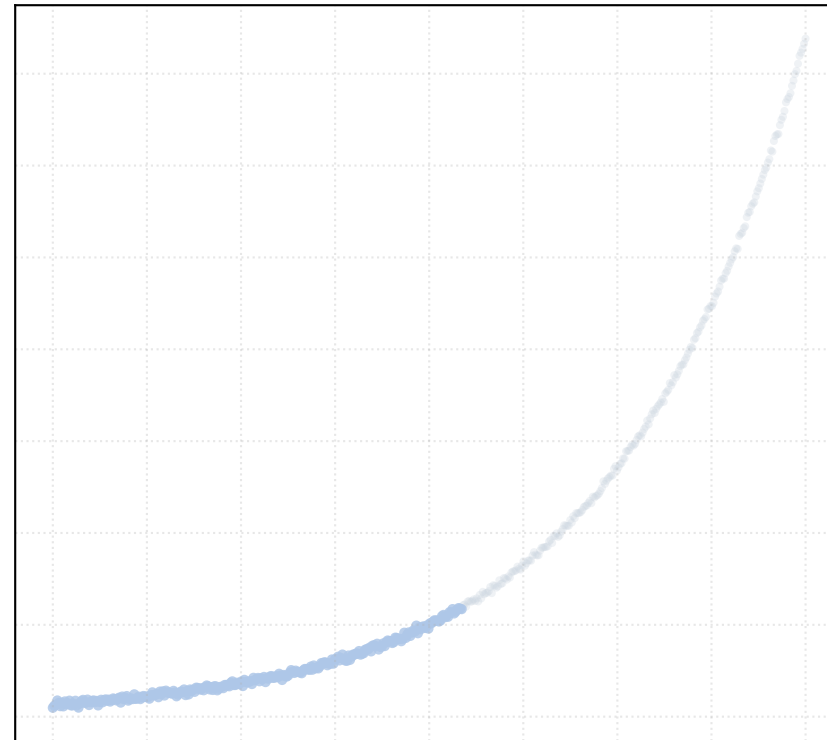
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

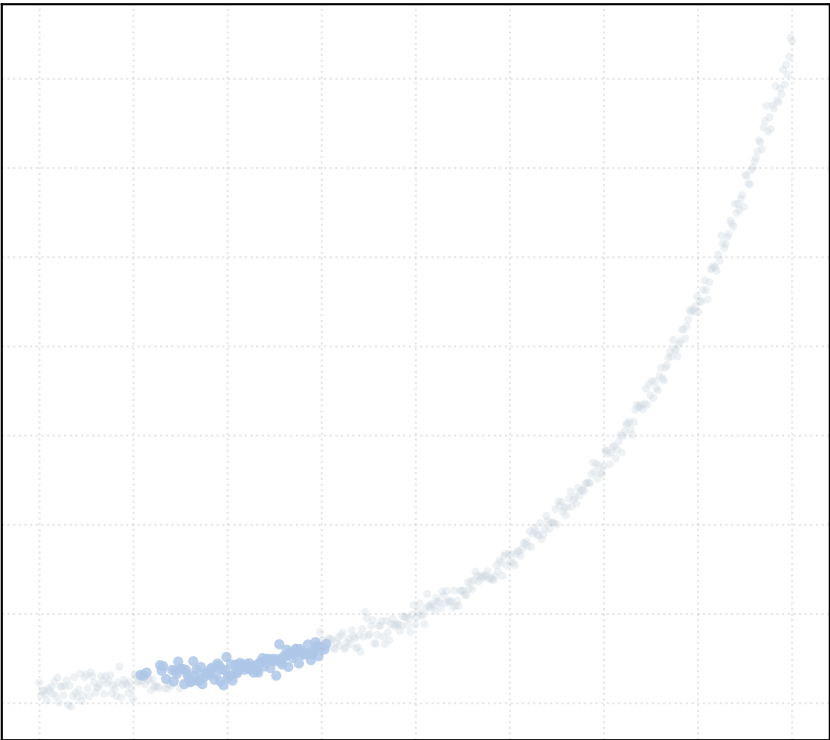


alpha_MAD = 1.0 [1 Clusters]

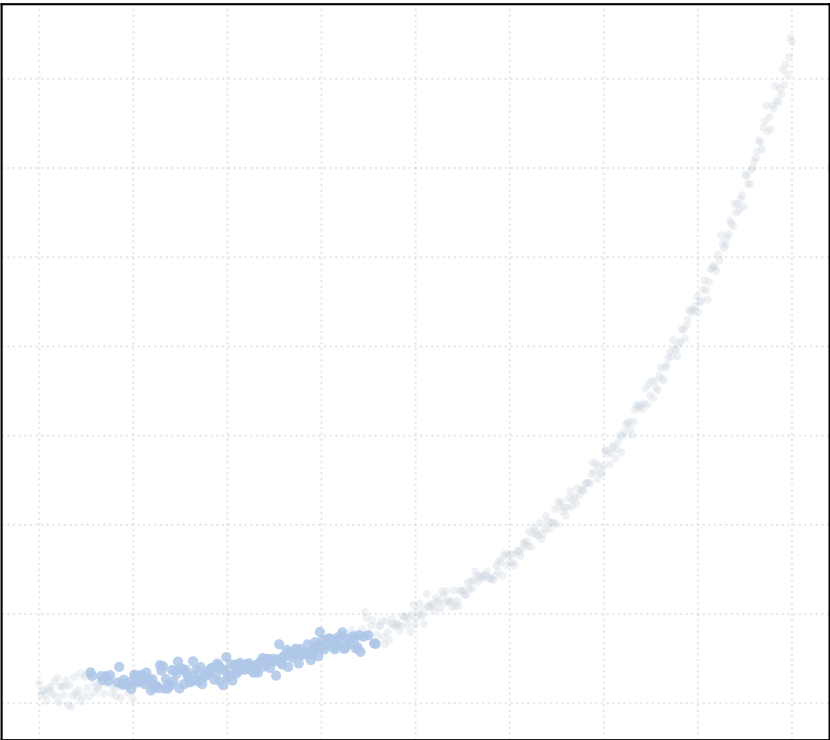


Dataset: exponential_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

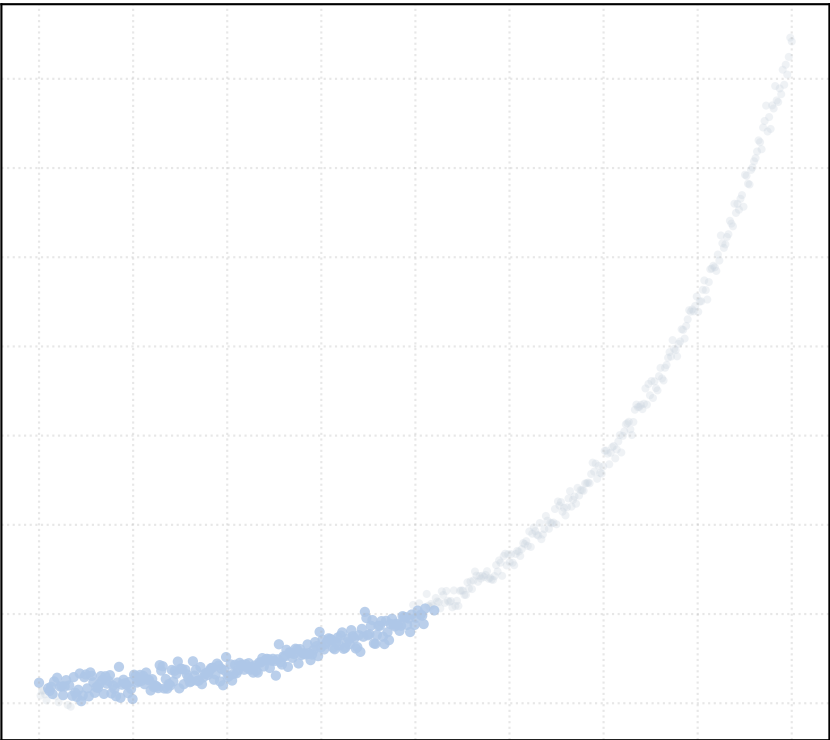
alpha_MAD = 0.25 [1 Clusters]



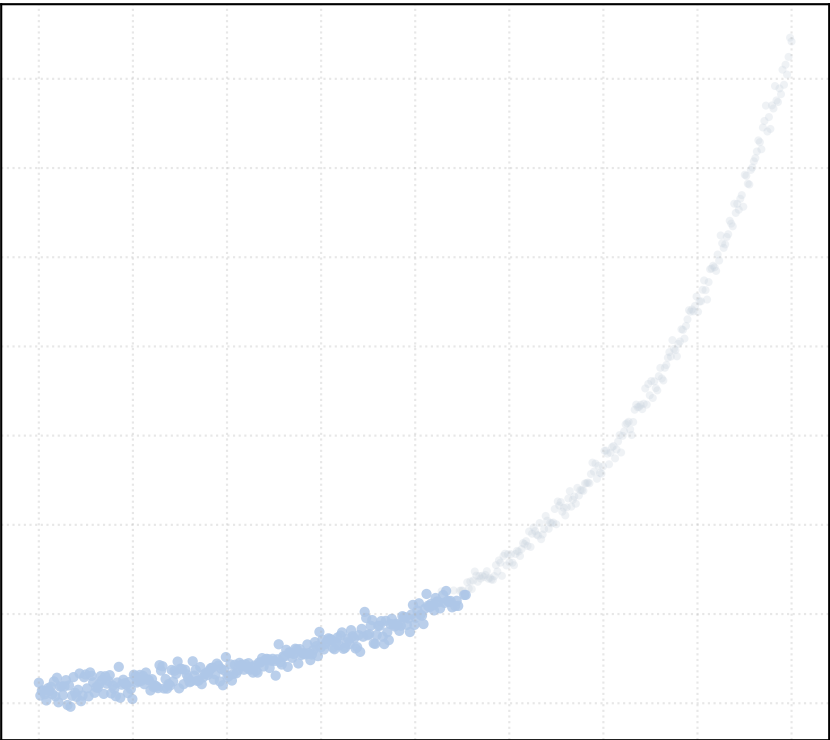
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

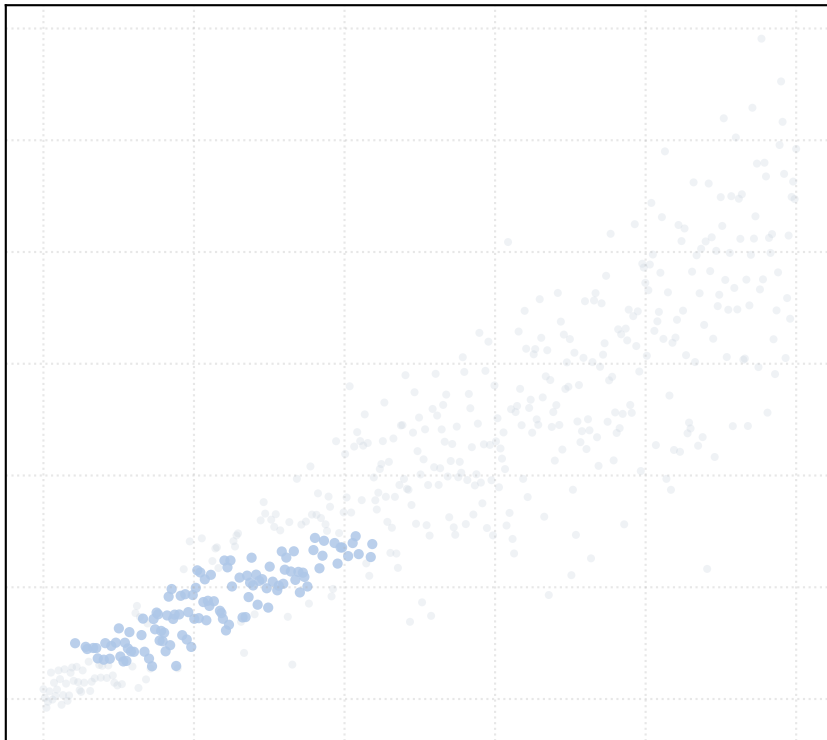


alpha_MAD = 1.0 [1 Clusters]

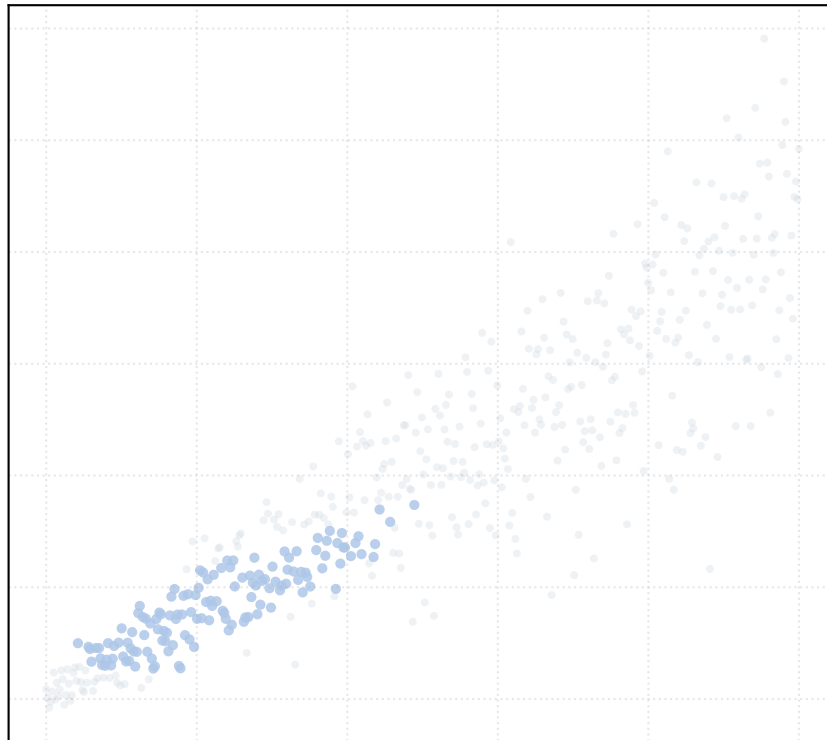


Dataset: heteroscedastic_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

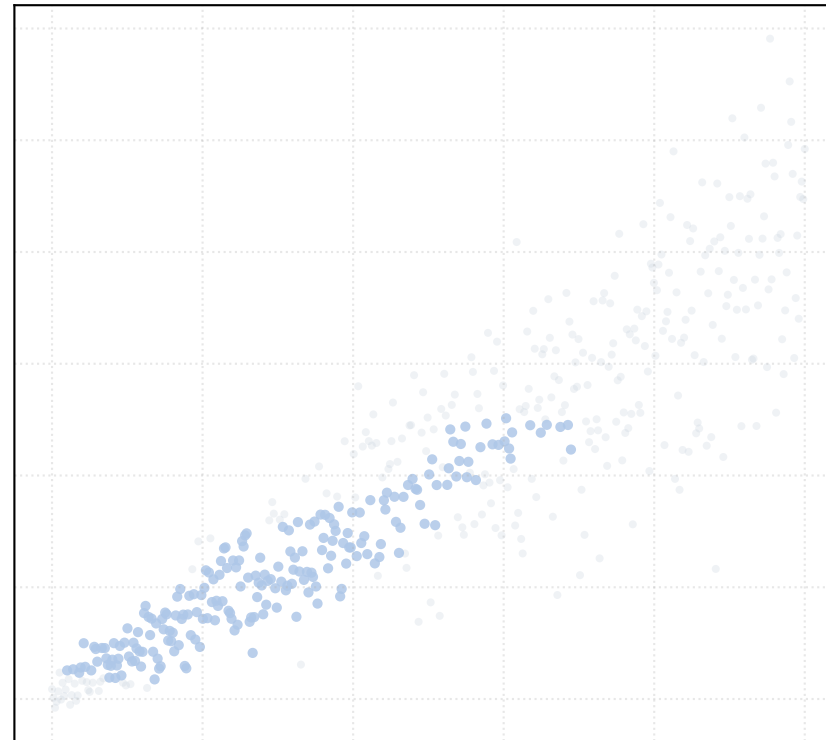
alpha_MAD = 0.25 [1 Clusters]



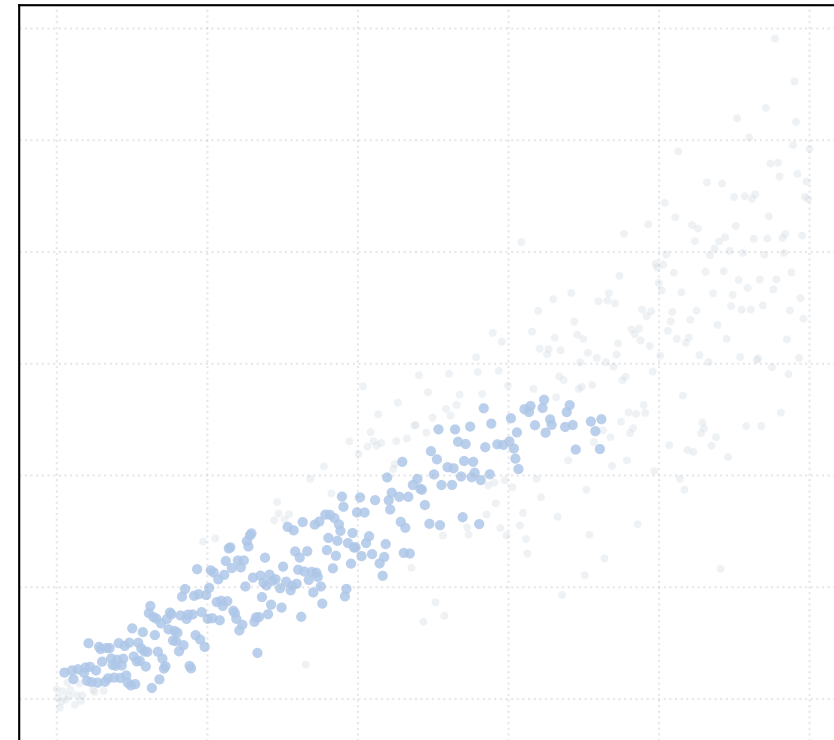
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

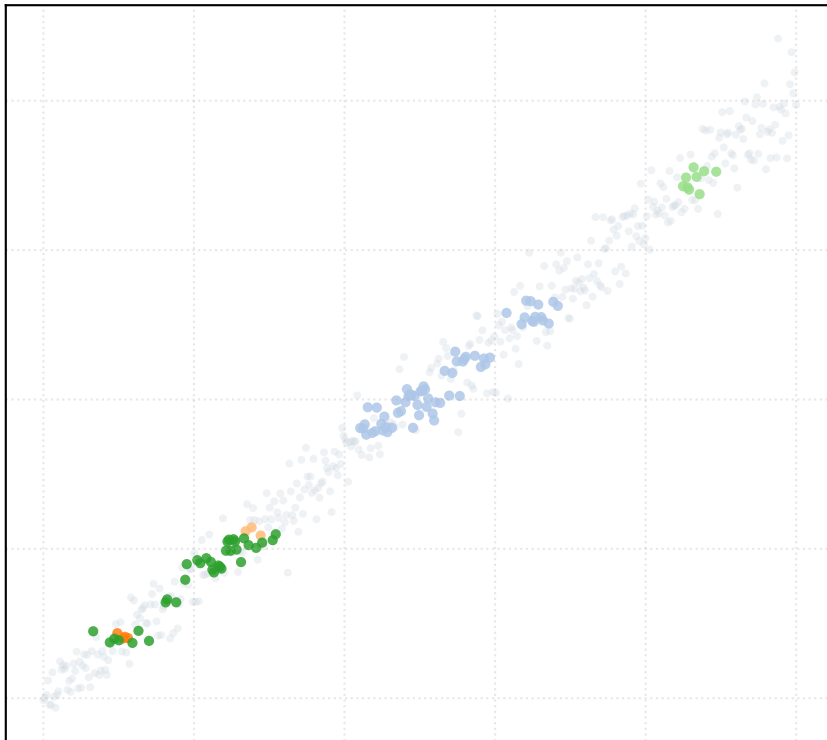


alpha_MAD = 1.0 [1 Clusters]

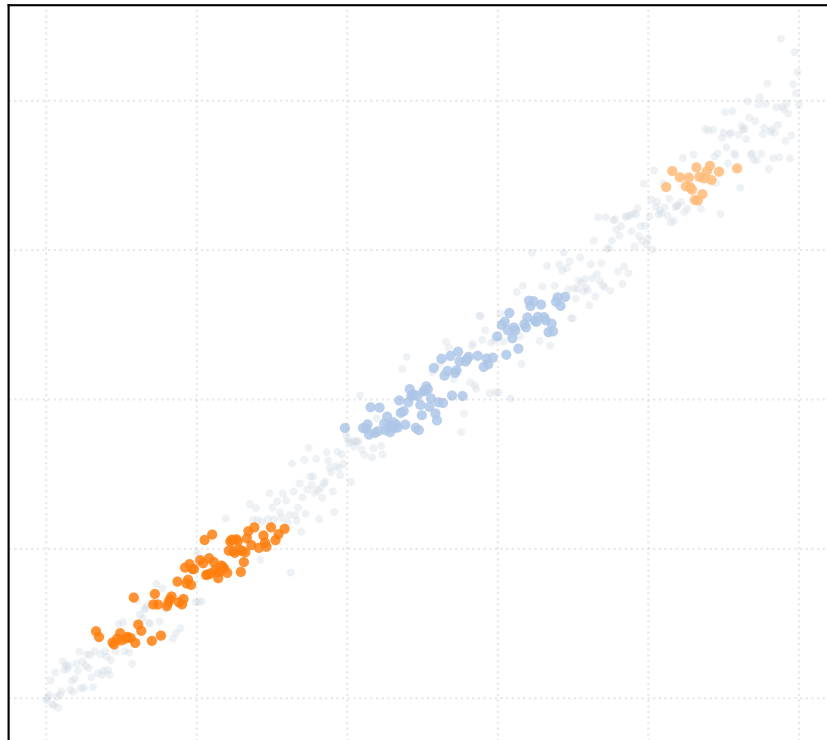


Dataset: heteroscedastic_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

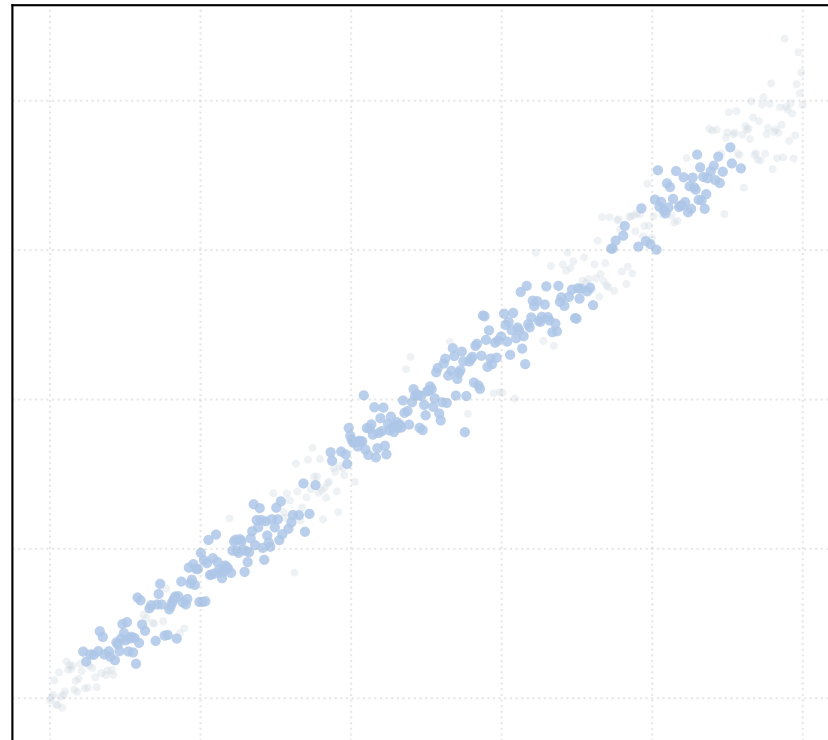
alpha_MAD = 0.25 [5 Clusters]



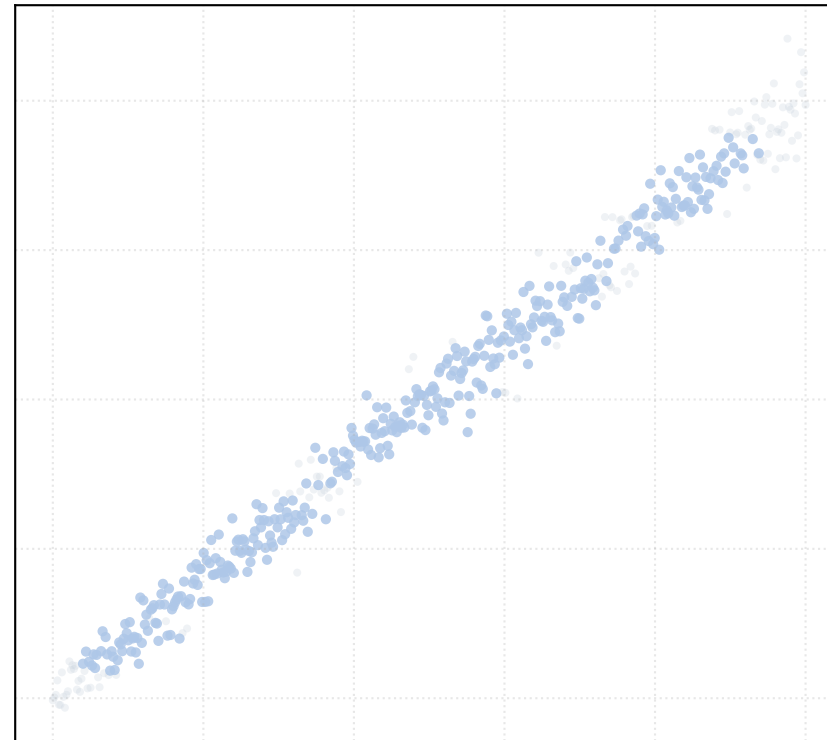
alpha_MAD = 0.5 (Default) [3 Clusters]



alpha_MAD = 0.75 [1 Clusters]

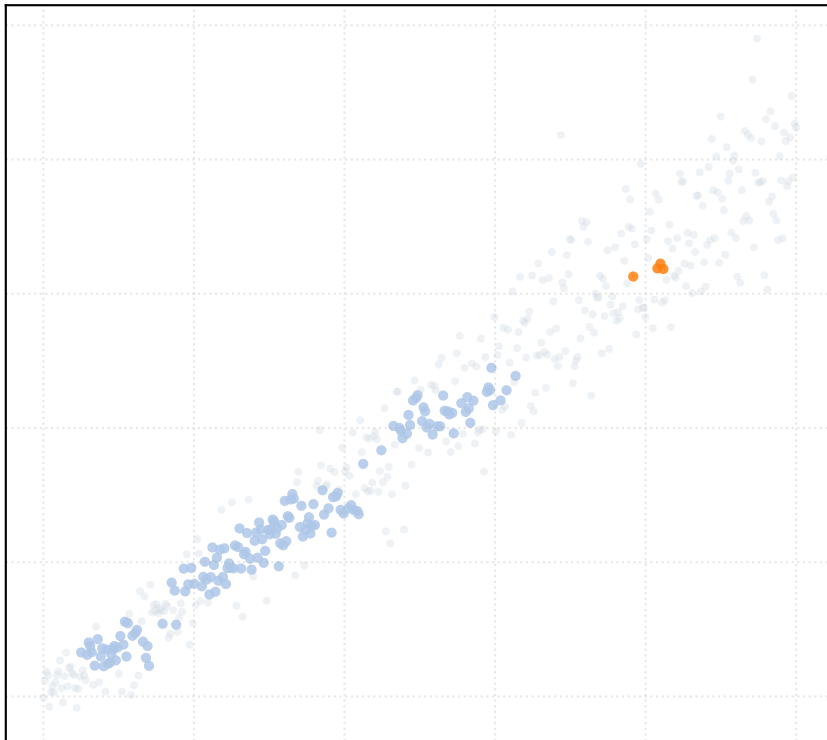


alpha_MAD = 1.0 [1 Clusters]

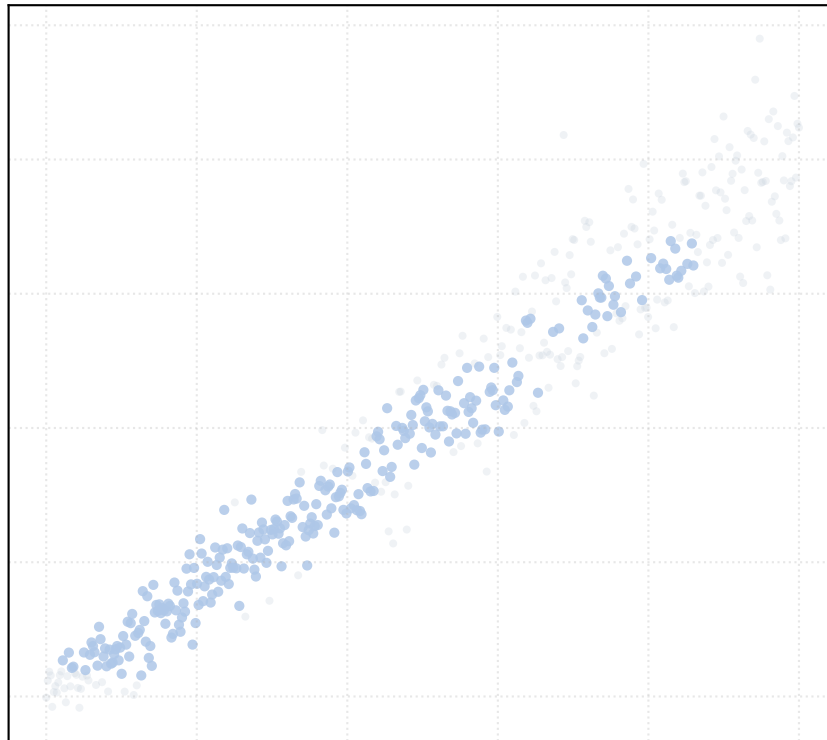


Dataset: heteroscedastic_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

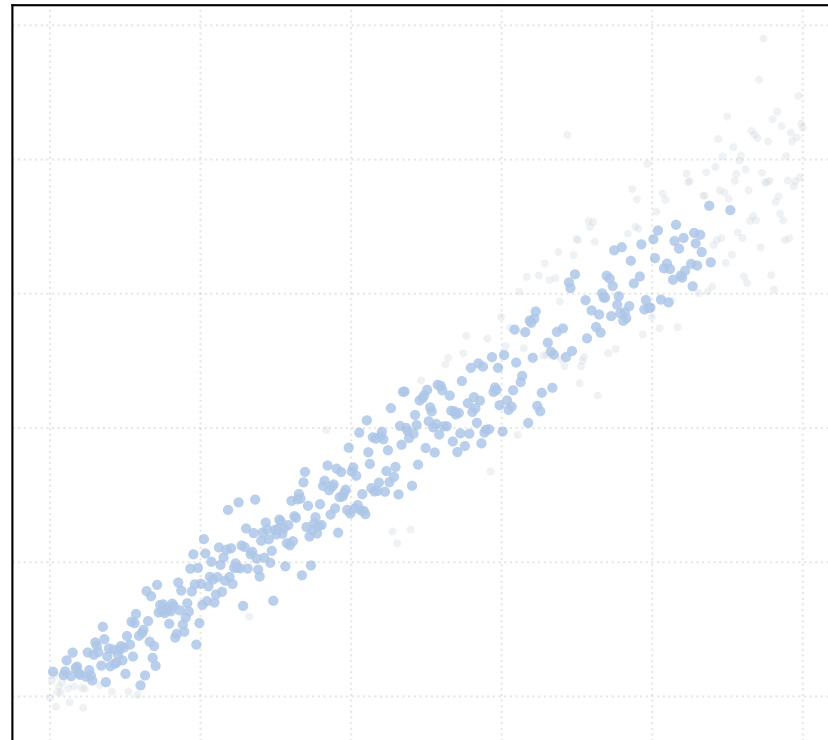
alpha_MAD = 0.25 [2 Clusters]



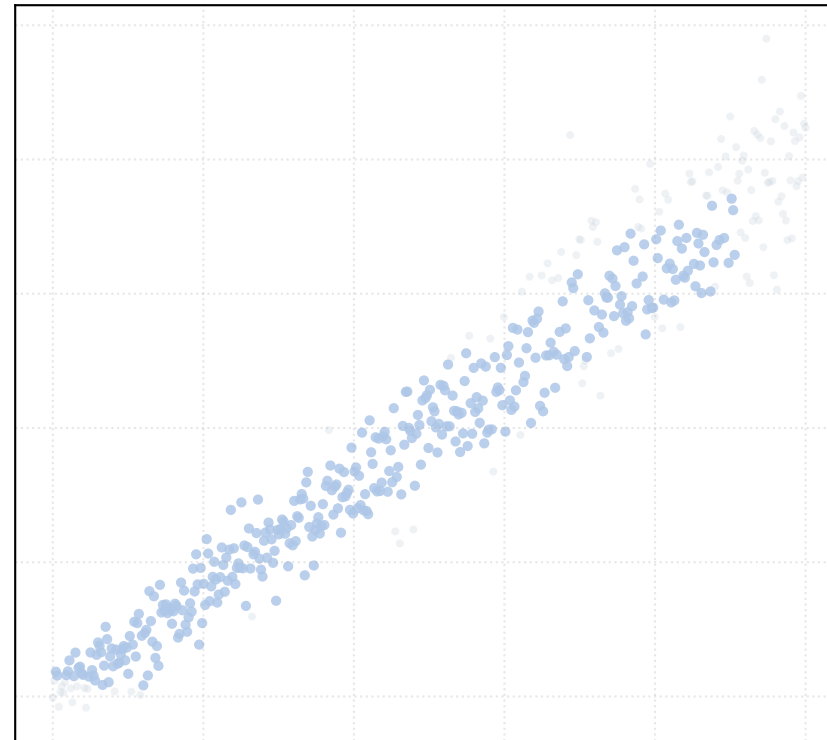
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

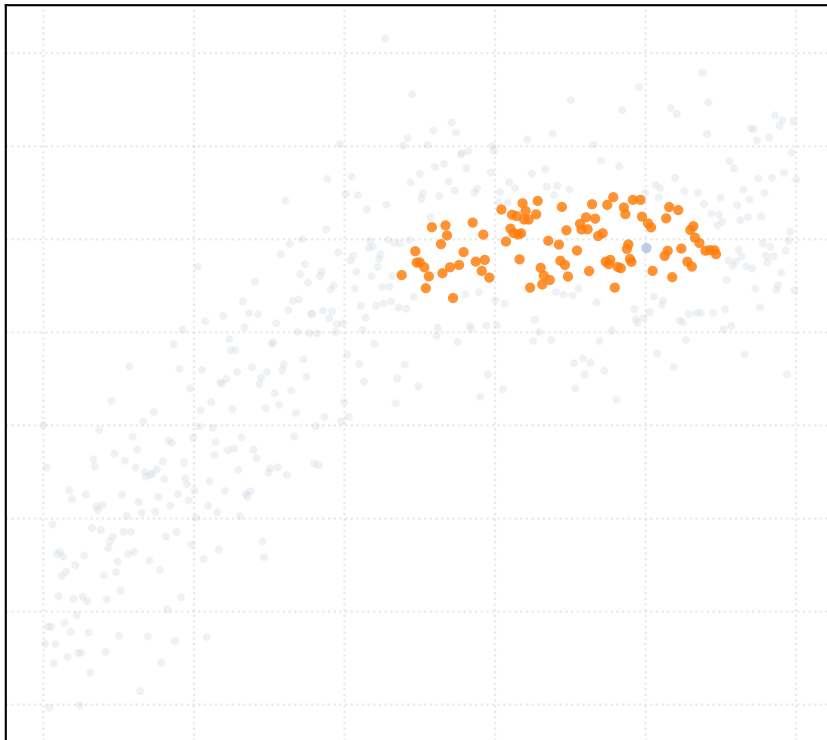


alpha_MAD = 1.0 [1 Clusters]

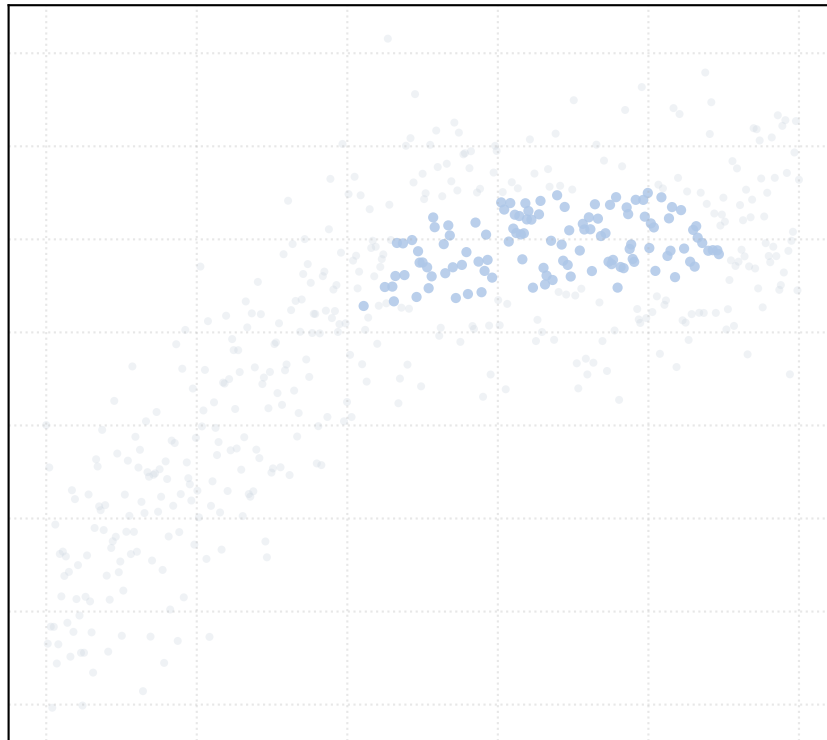


Dataset: kinked_curve_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

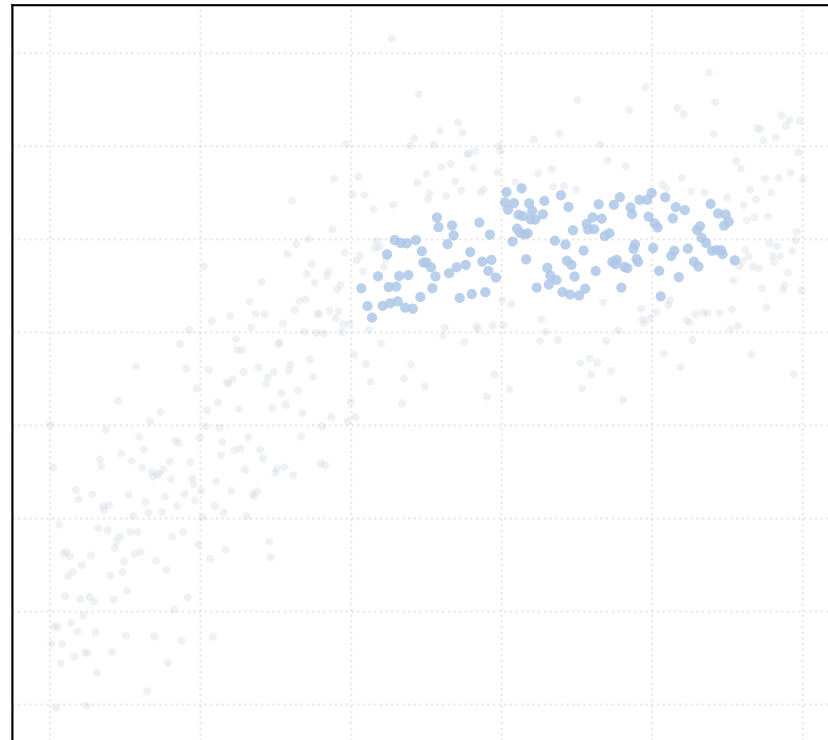
alpha_MAD = 0.25 [2 Clusters]



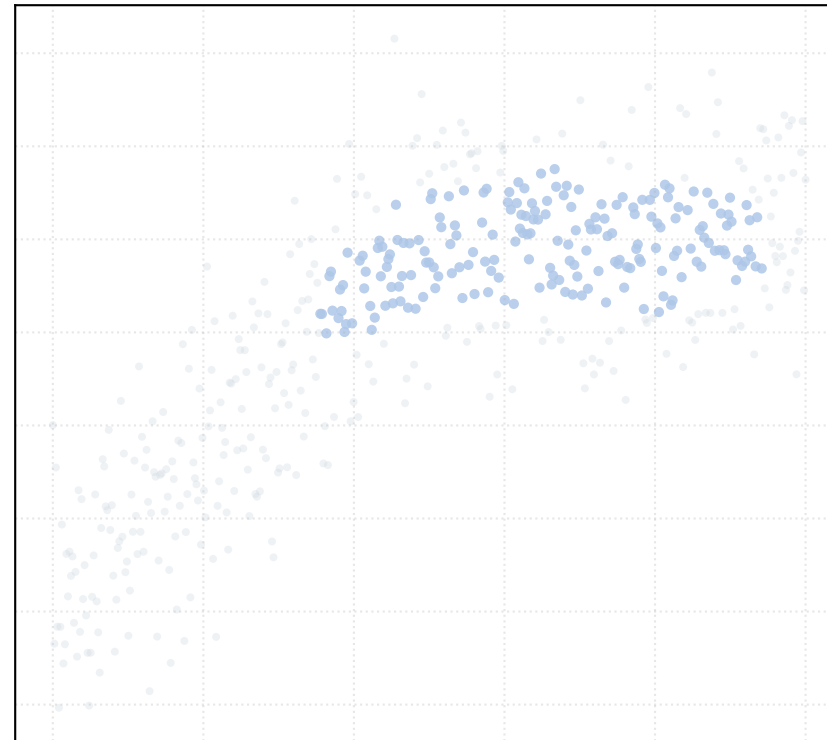
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

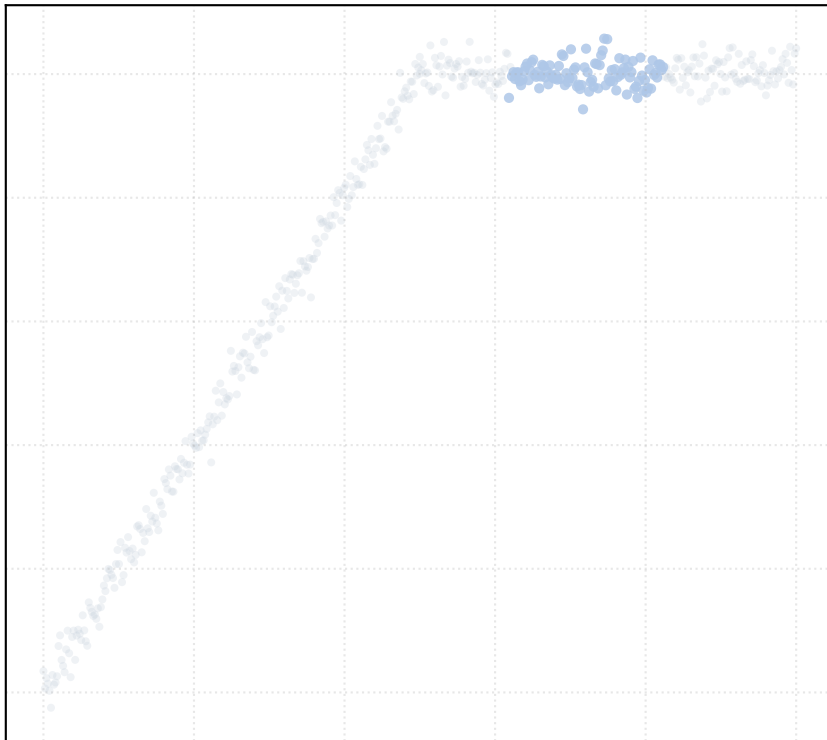


alpha_MAD = 1.0 [1 Clusters]

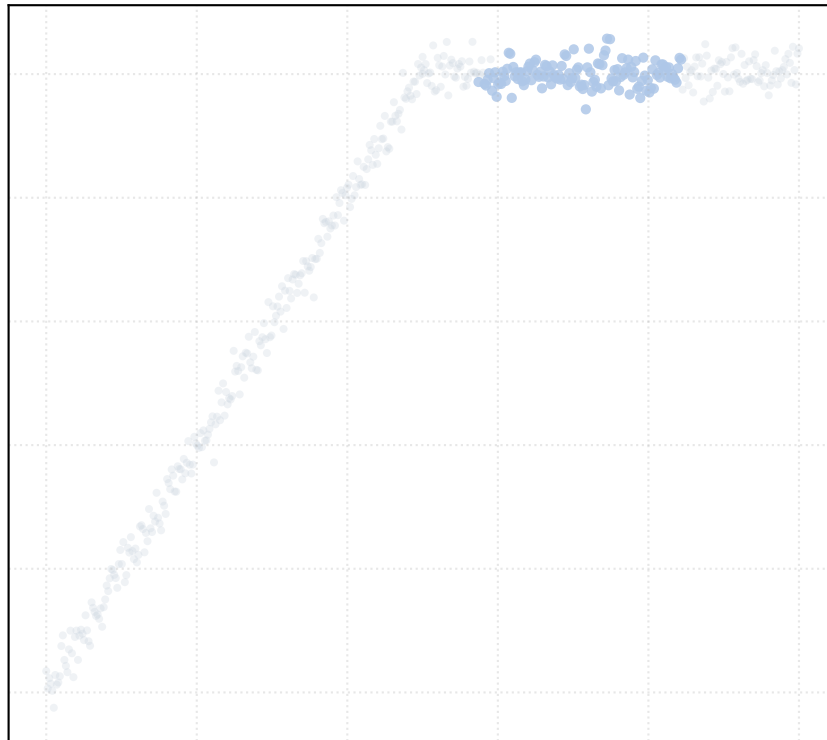


Dataset: kinked_curve_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

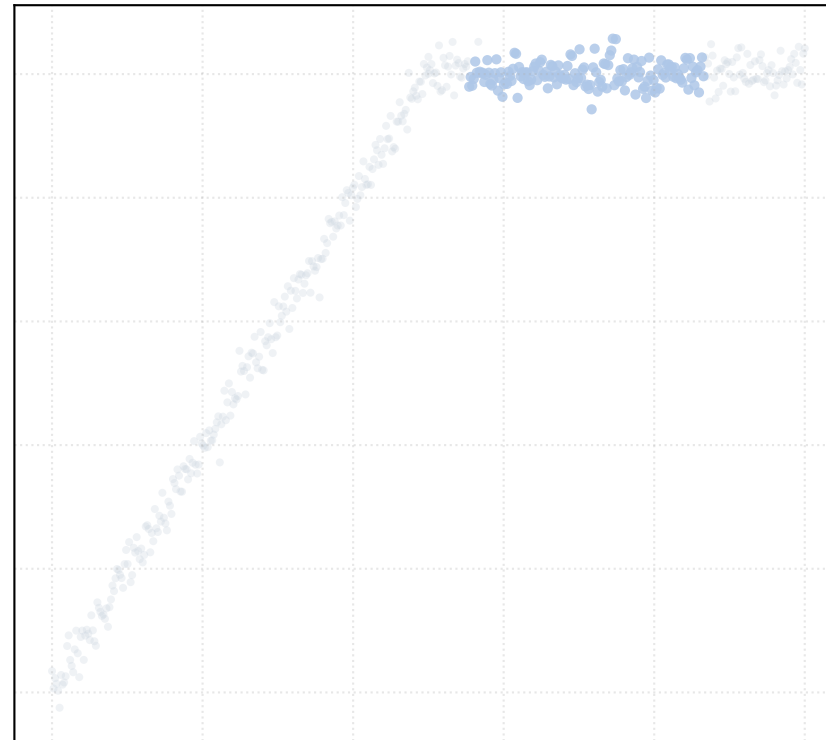
alpha_MAD = 0.25 [1 Clusters]



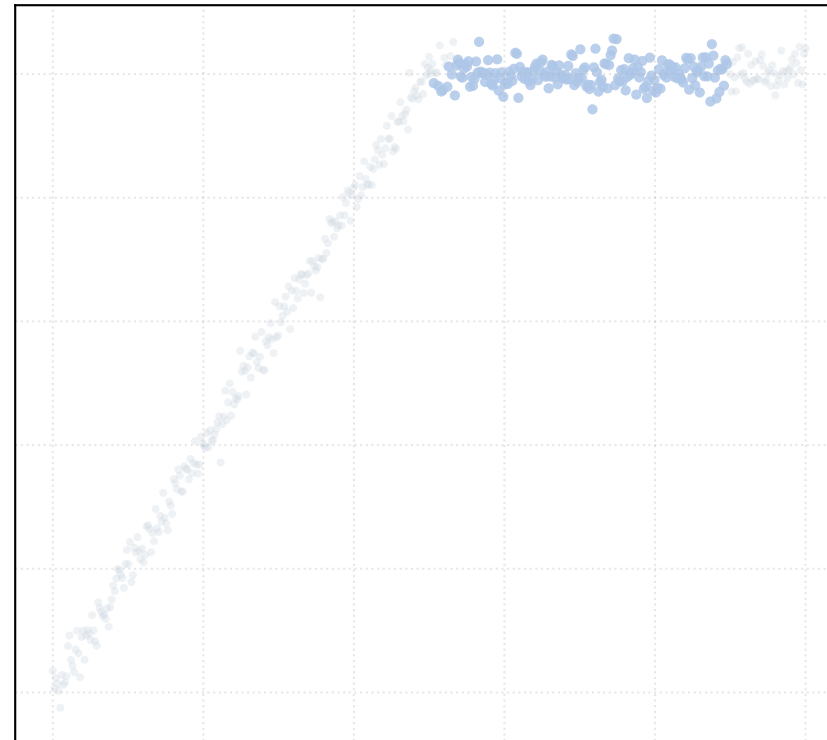
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

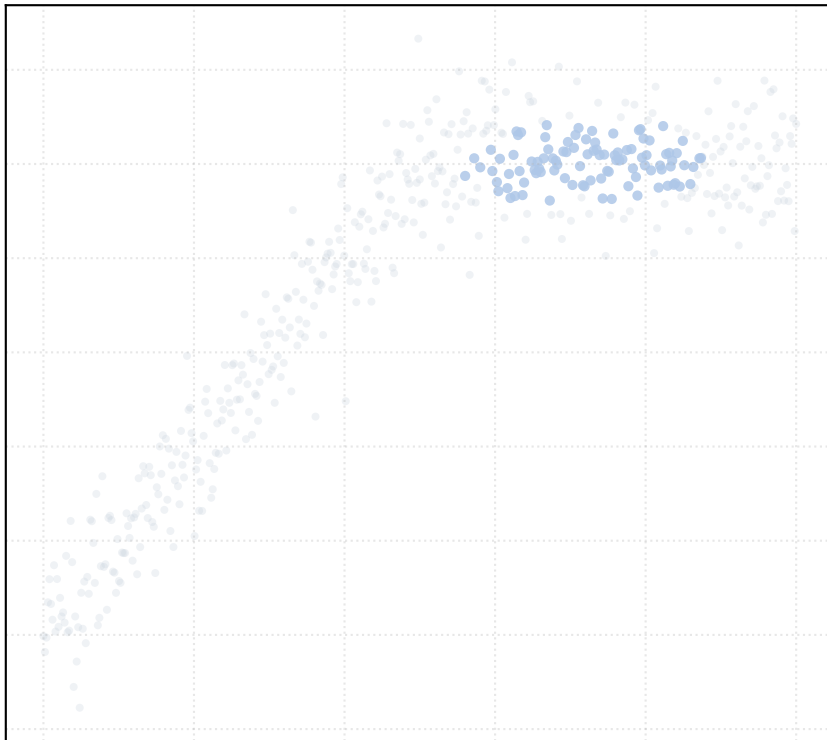


alpha_MAD = 1.0 [1 Clusters]

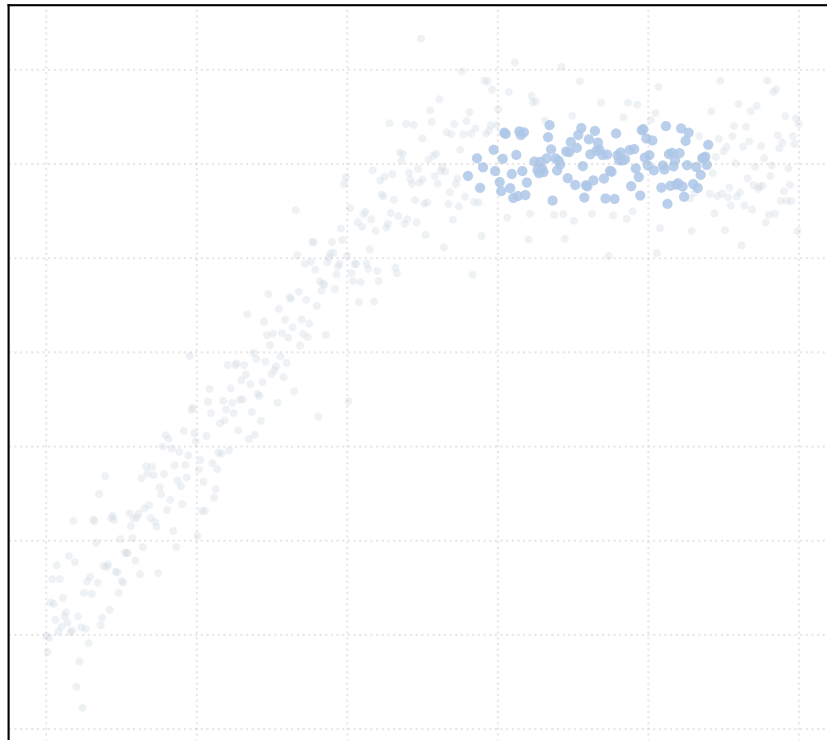


Dataset: kinked_curve_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

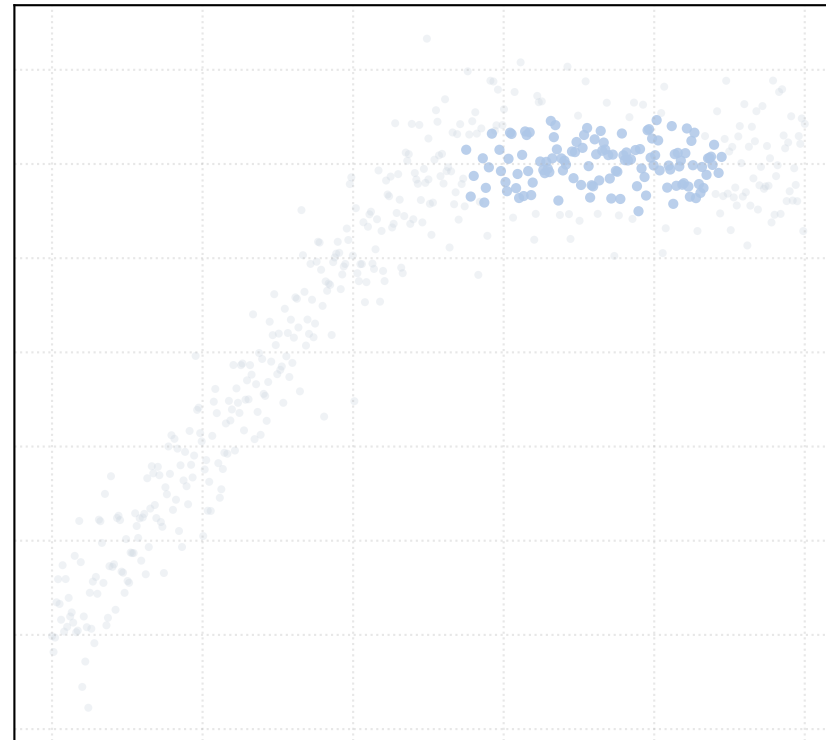
alpha_MAD = 0.25 [1 Clusters]



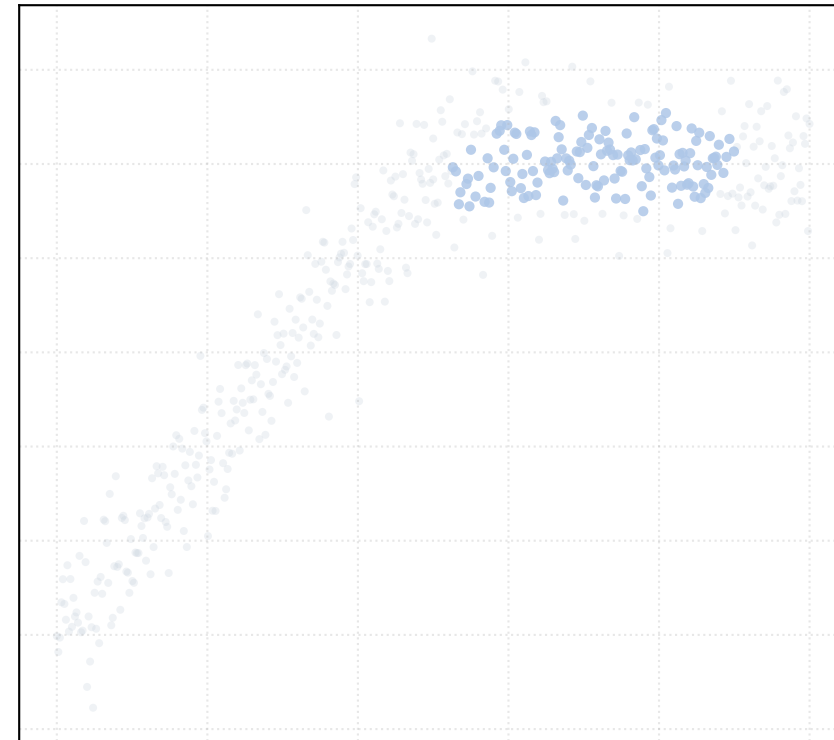
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

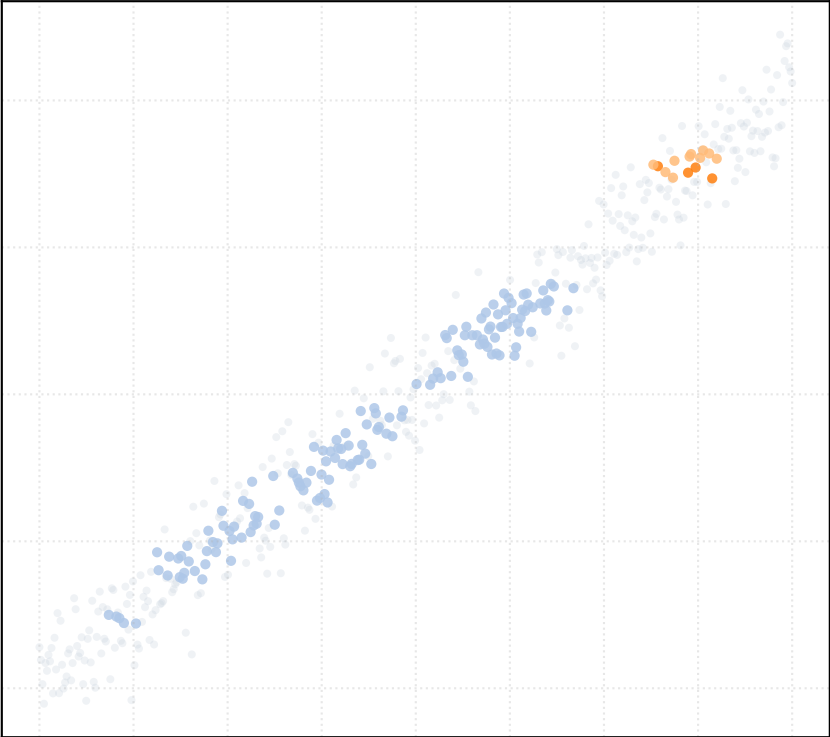


alpha_MAD = 1.0 [1 Clusters]

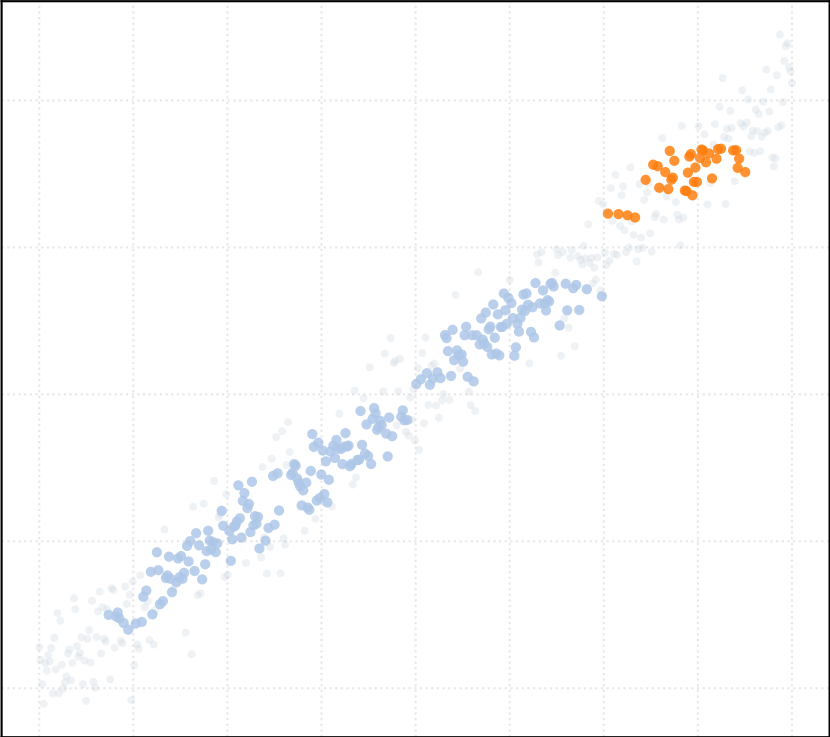


Dataset: linear_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

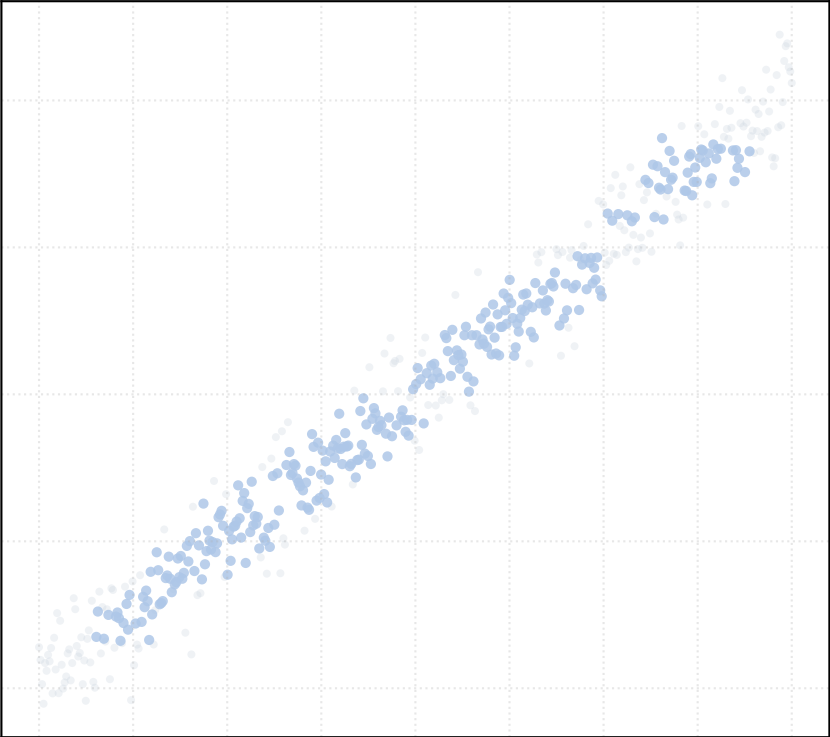
alpha_MAD = 0.25 [3 Clusters]



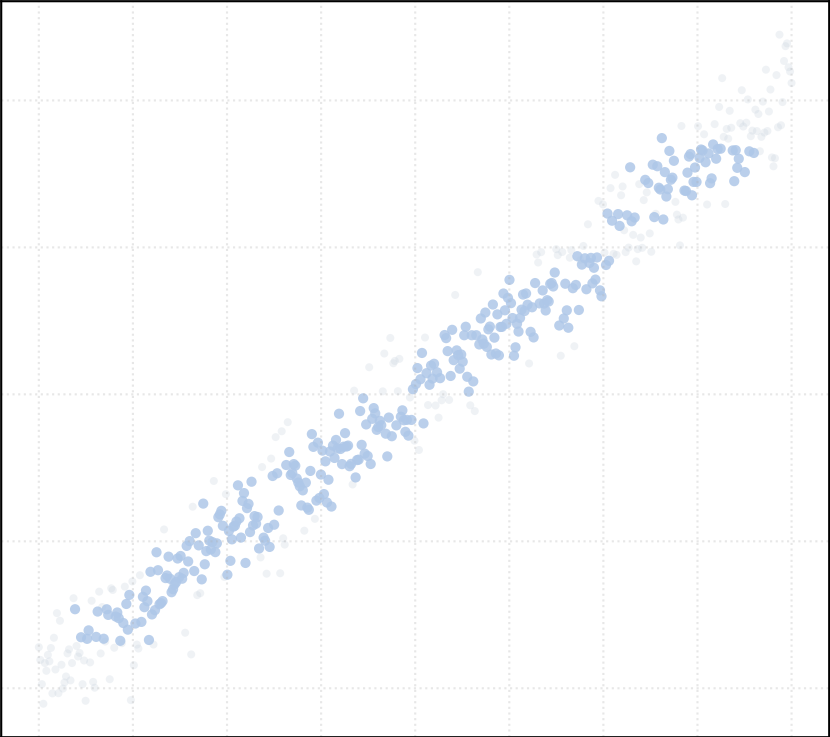
alpha_MAD = 0.5 (Default) [2 Clusters]



alpha_MAD = 0.75 [1 Clusters]

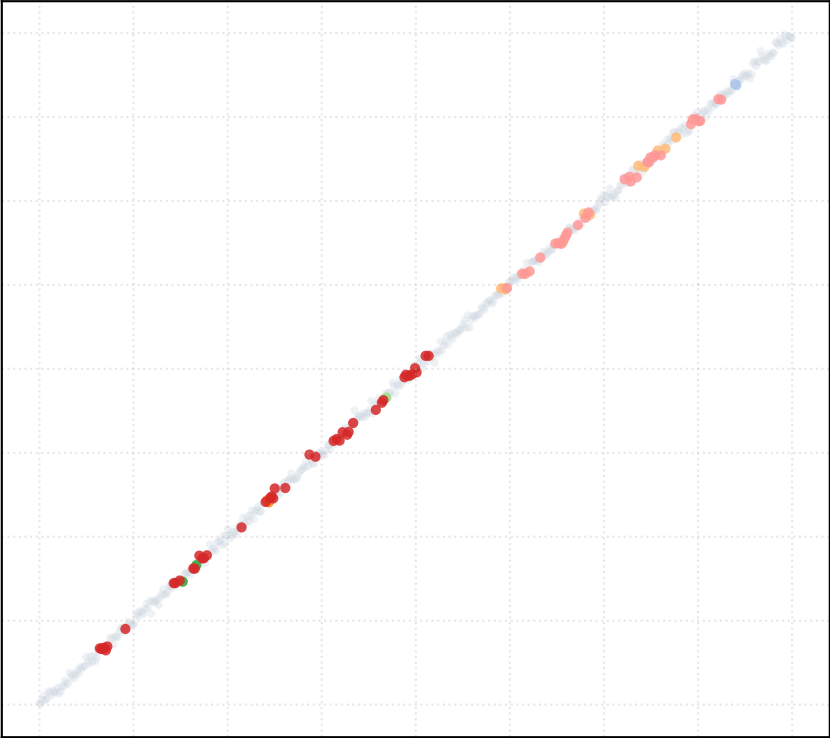


alpha_MAD = 1.0 [1 Clusters]

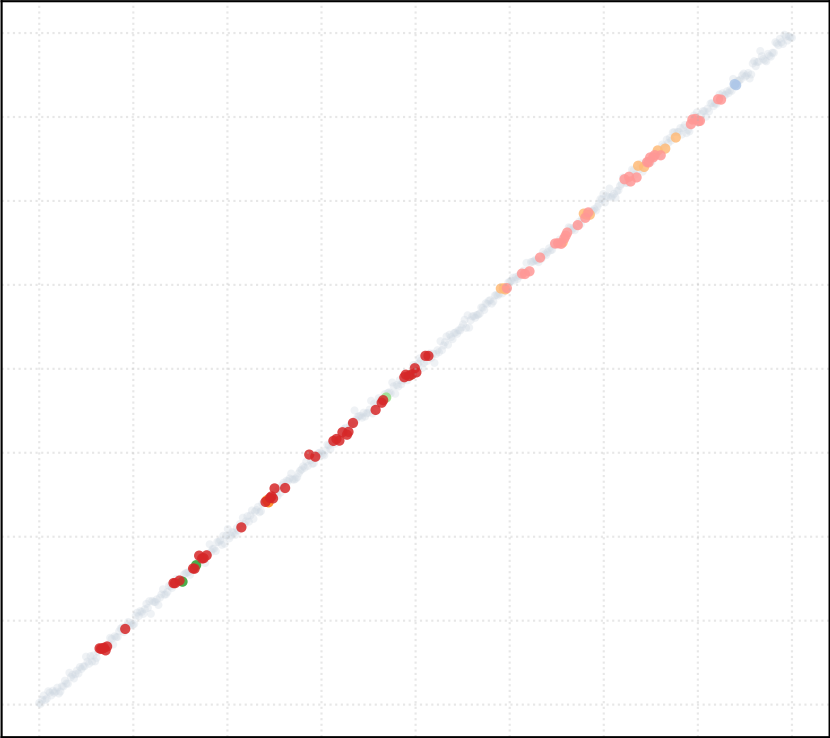


Dataset: linear_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

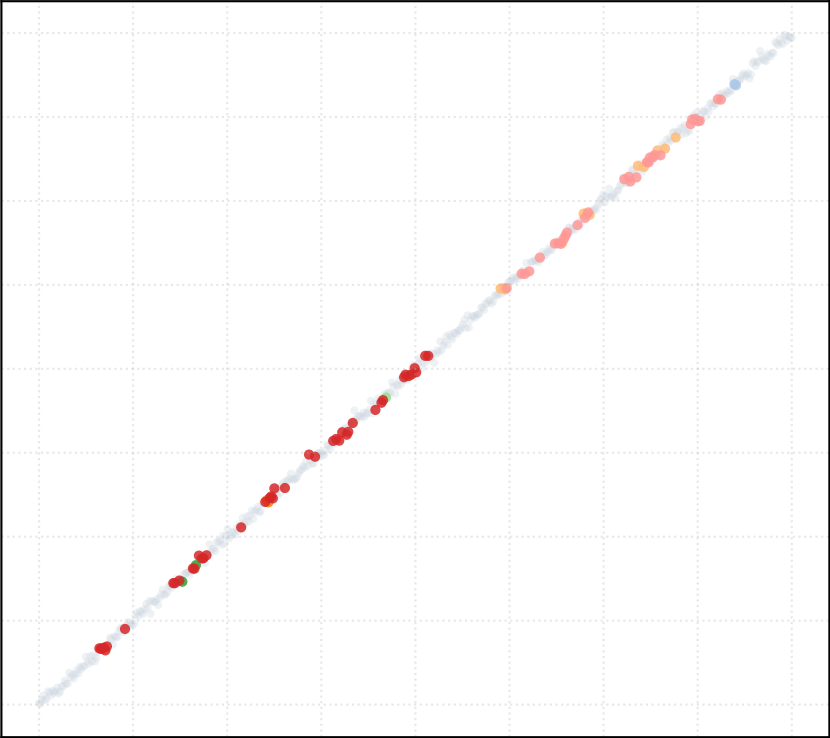
alpha_MAD = 0.25 [7 Clusters]



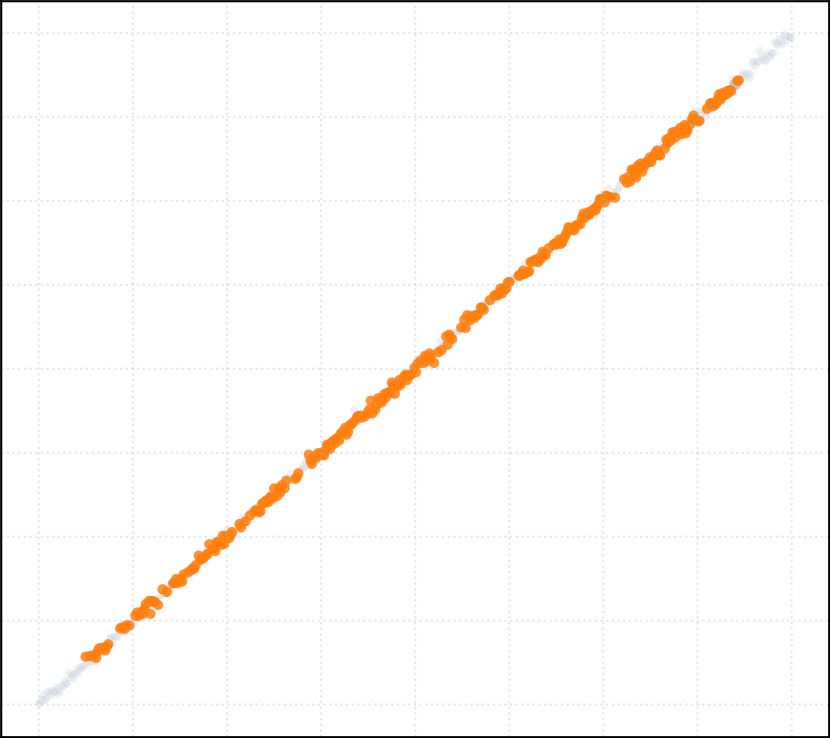
alpha_MAD = 0.5 (Default) [7 Clusters]



alpha_MAD = 0.75 [7 Clusters]

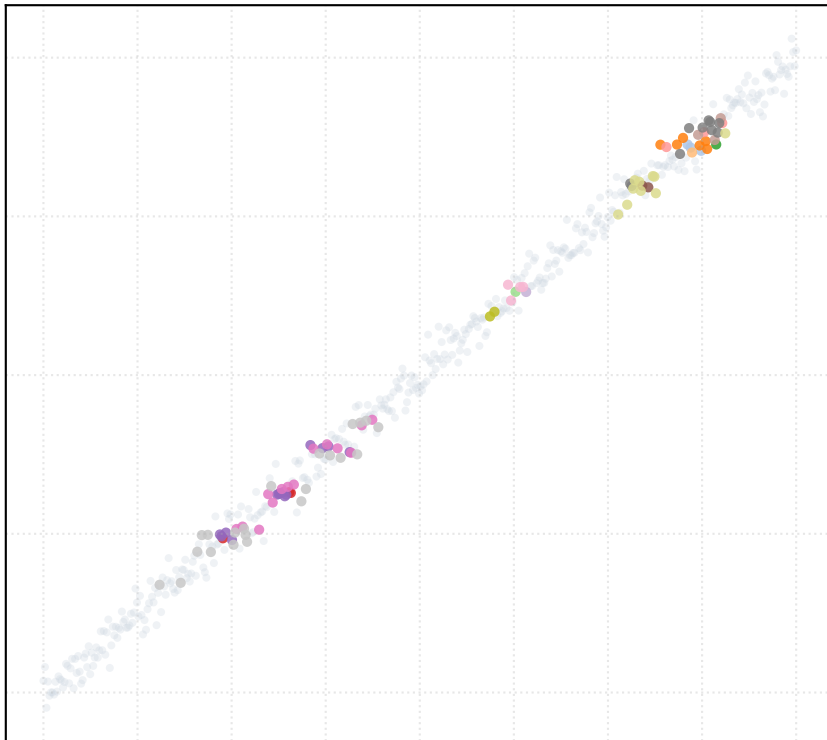


alpha_MAD = 1.0 [2 Clusters]

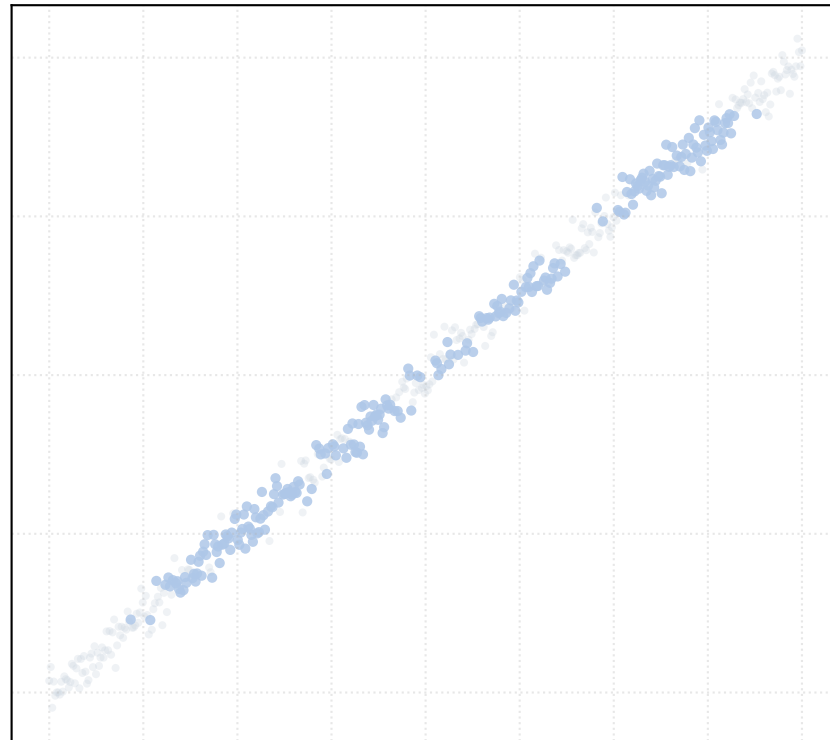


Dataset: linear_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

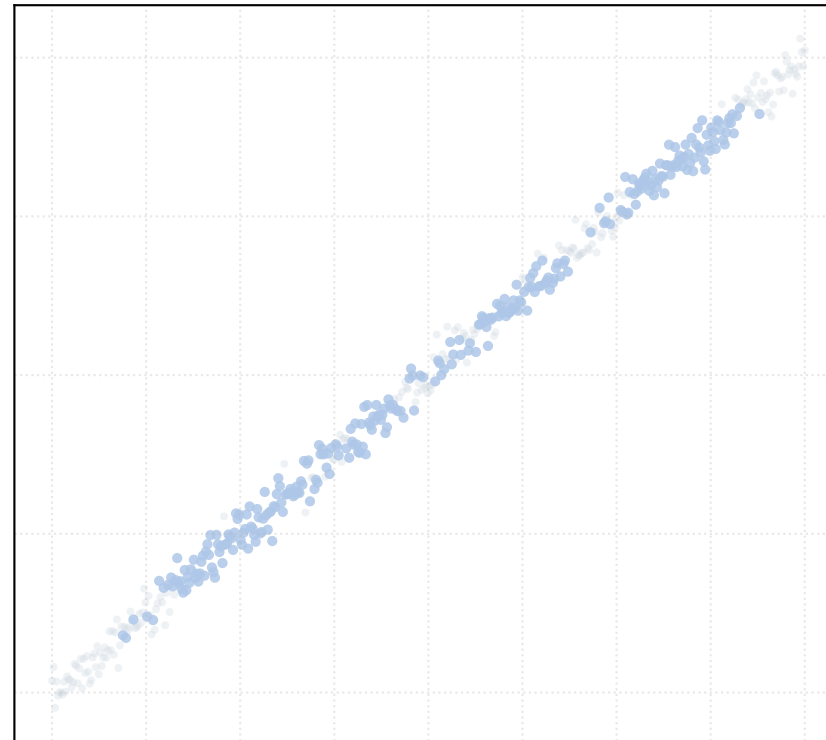
alpha_MAD = 0.25 [17 Clusters]



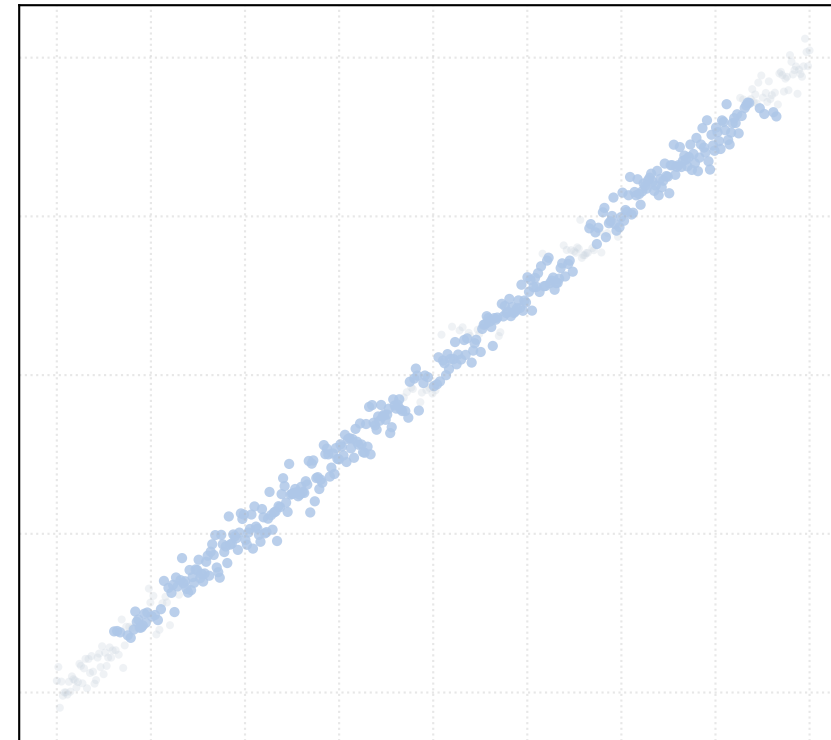
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

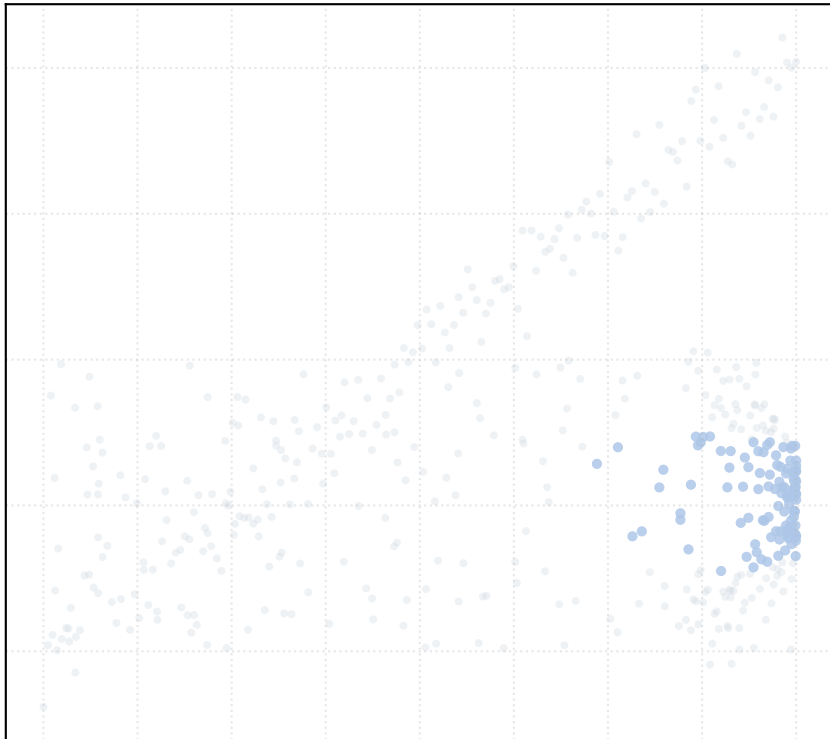


alpha_MAD = 1.0 [1 Clusters]

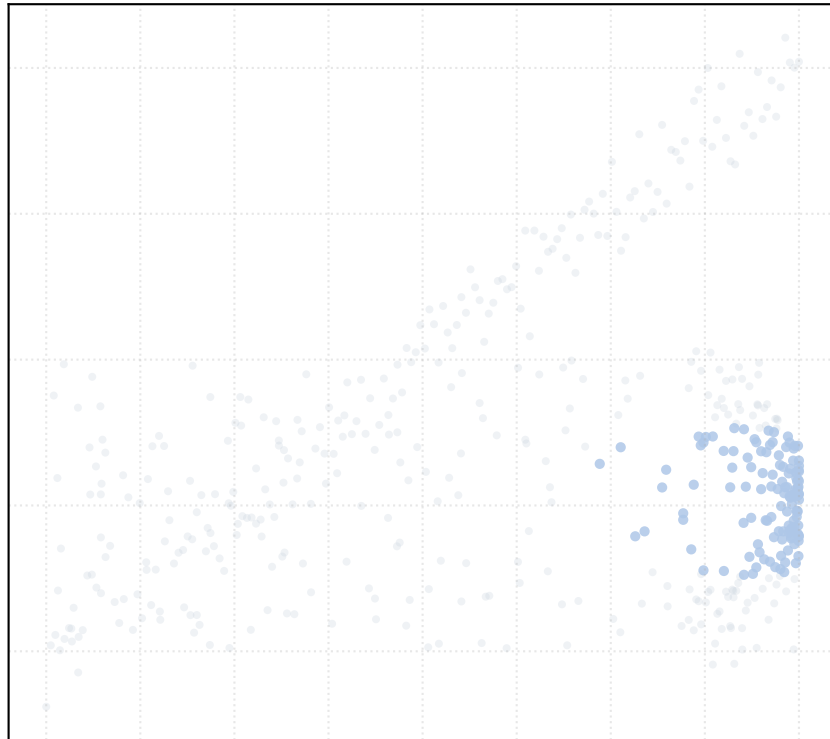


Dataset: mixed_generators_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

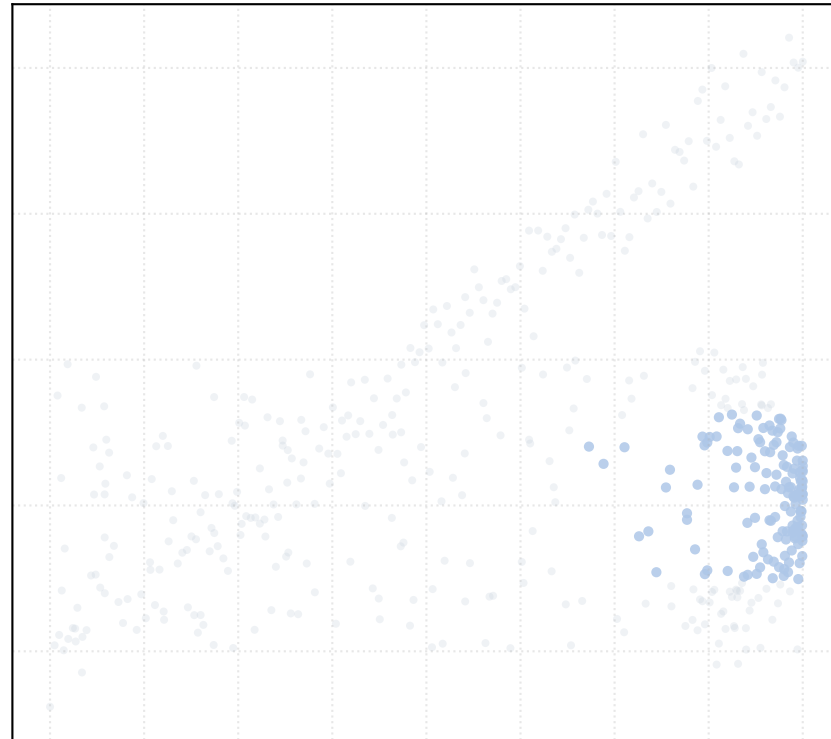
alpha_MAD = 0.25 [1 Clusters]



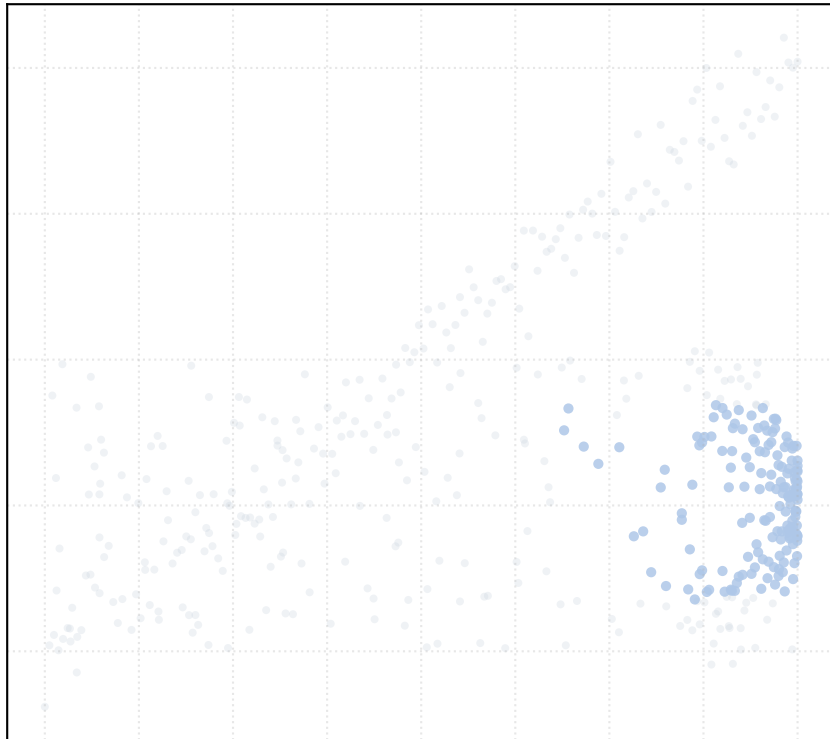
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

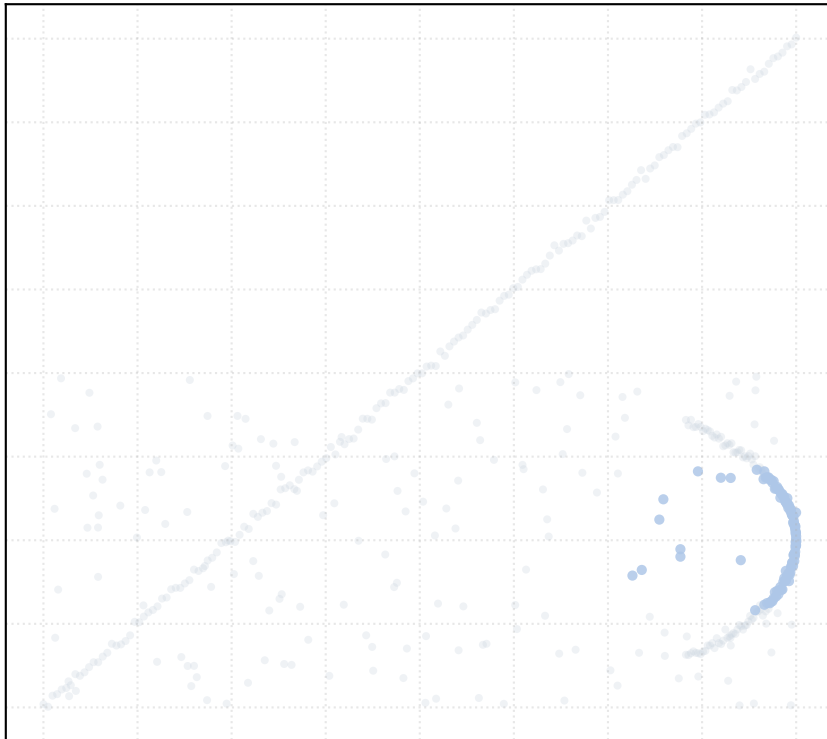


alpha_MAD = 1.0 [1 Clusters]

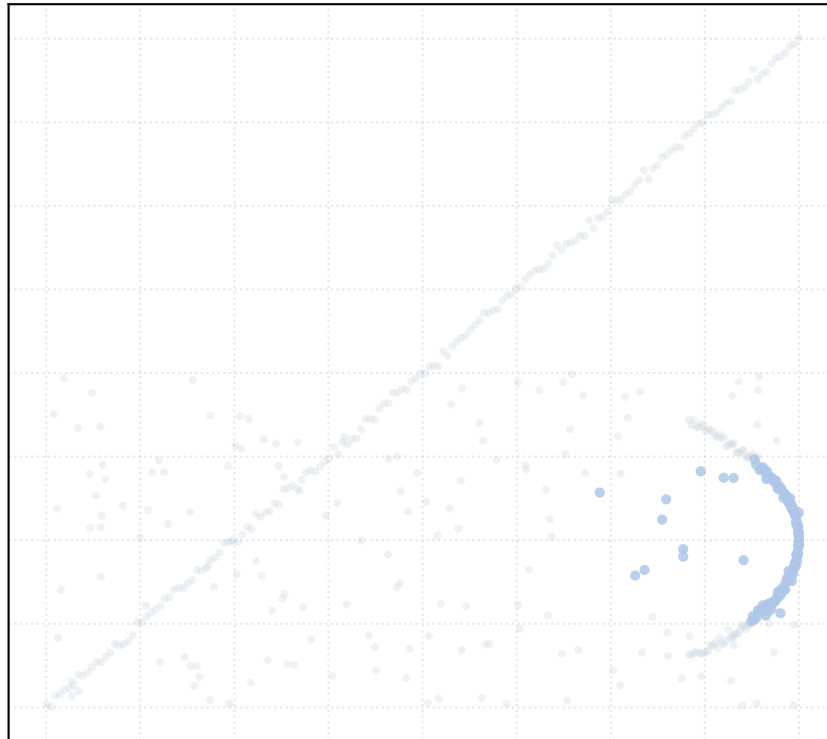


Dataset: mixed_generators_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

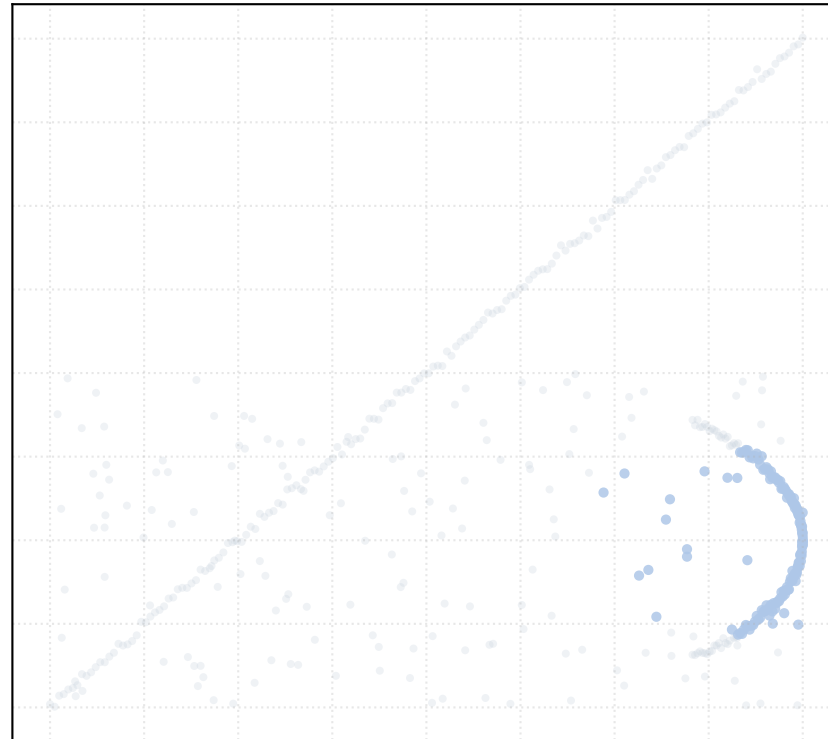
alpha_MAD = 0.25 [1 Clusters]



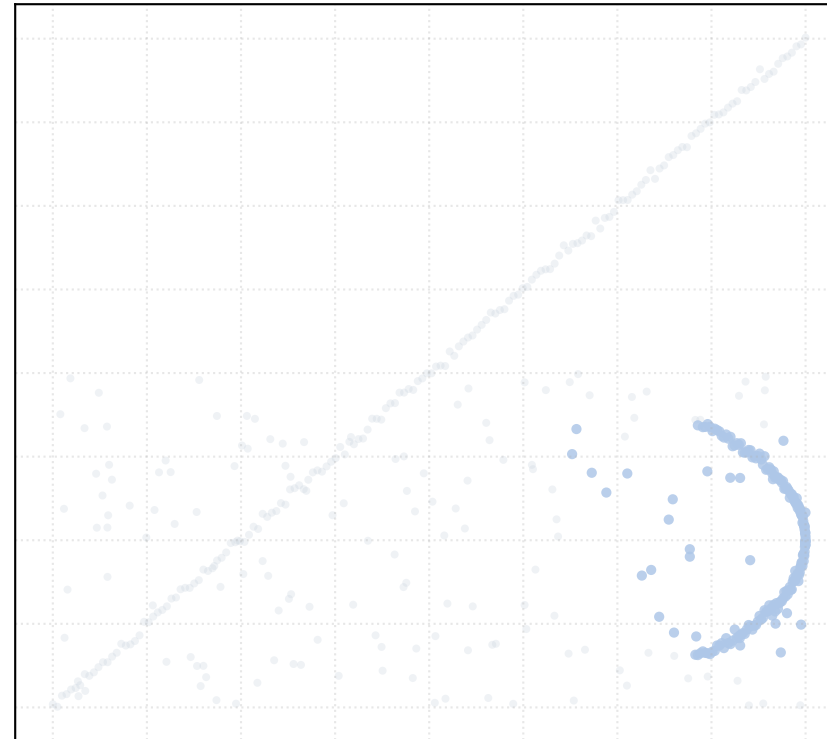
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

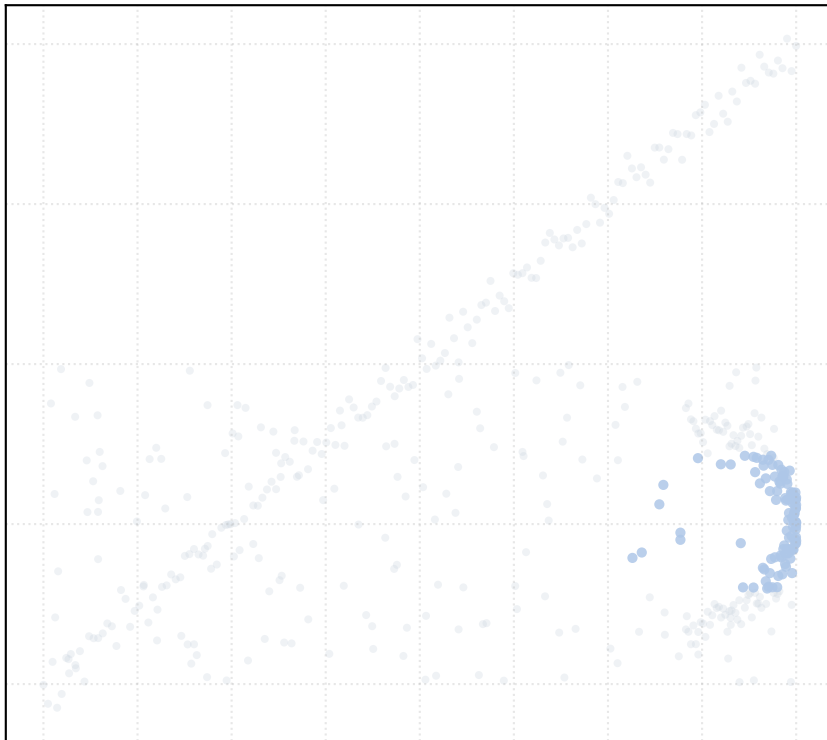


alpha_MAD = 1.0 [1 Clusters]

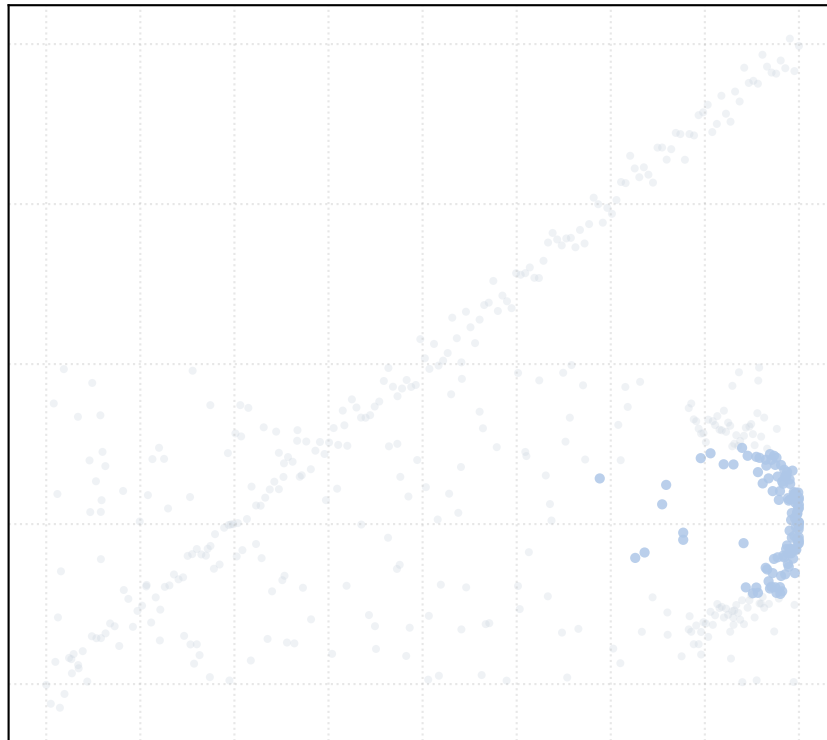


Dataset: mixed_generators_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

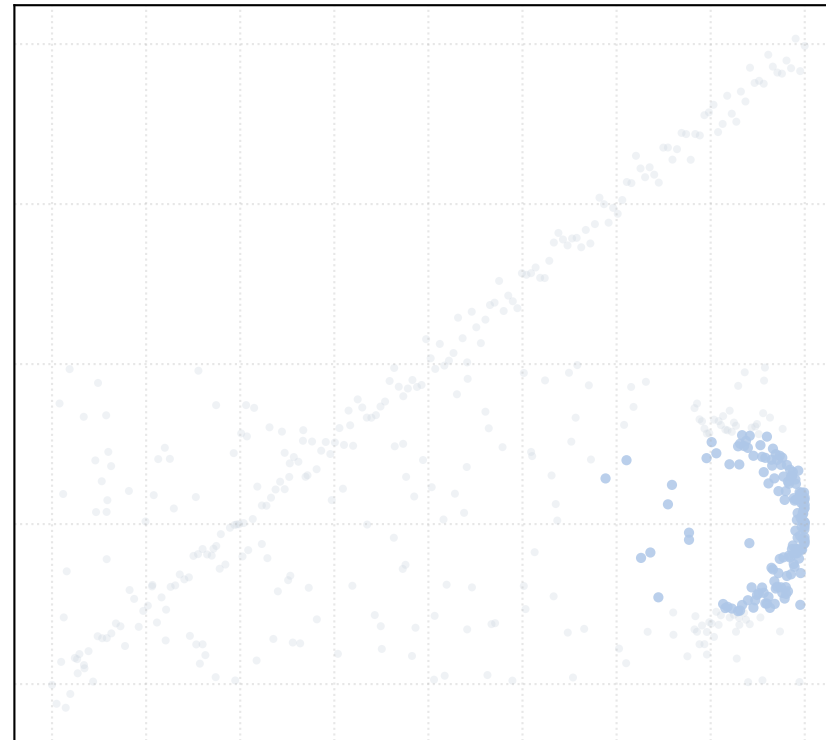
alpha_MAD = 0.25 [1 Clusters]



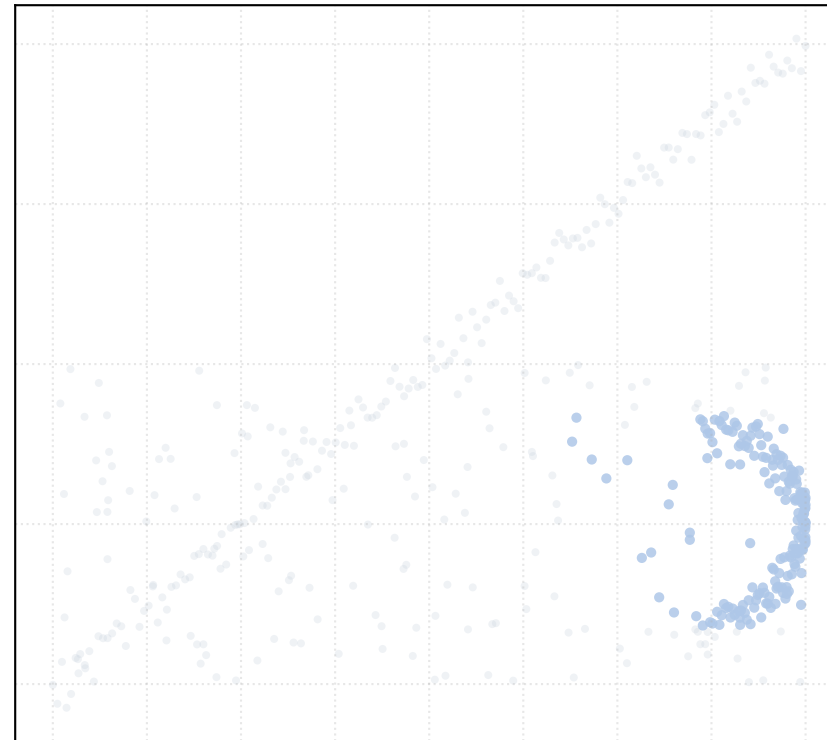
alpha_MAD = 0.5 (Default) [1 Clusters]



alpha_MAD = 0.75 [1 Clusters]

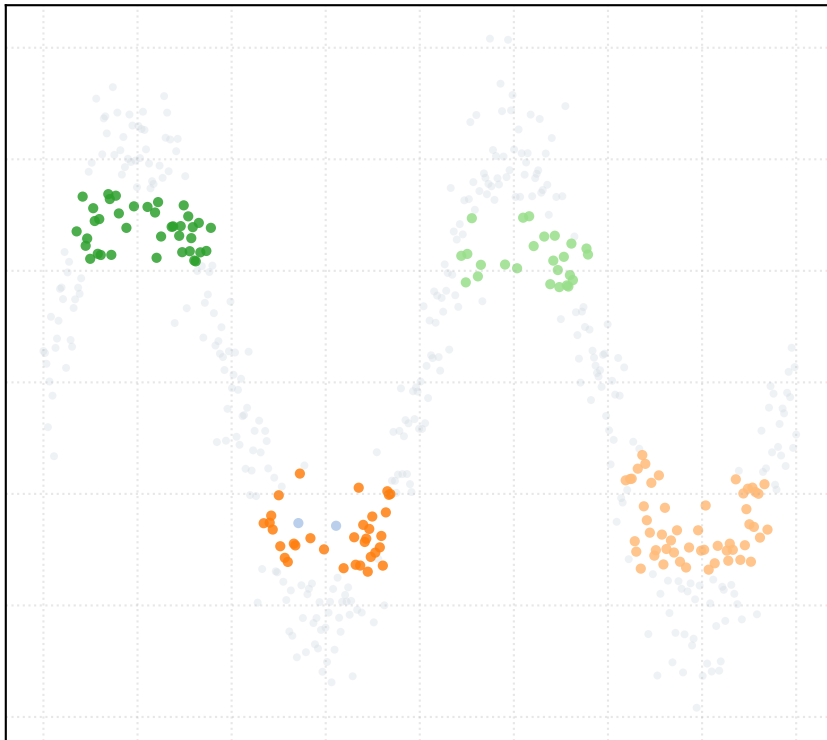


alpha_MAD = 1.0 [1 Clusters]

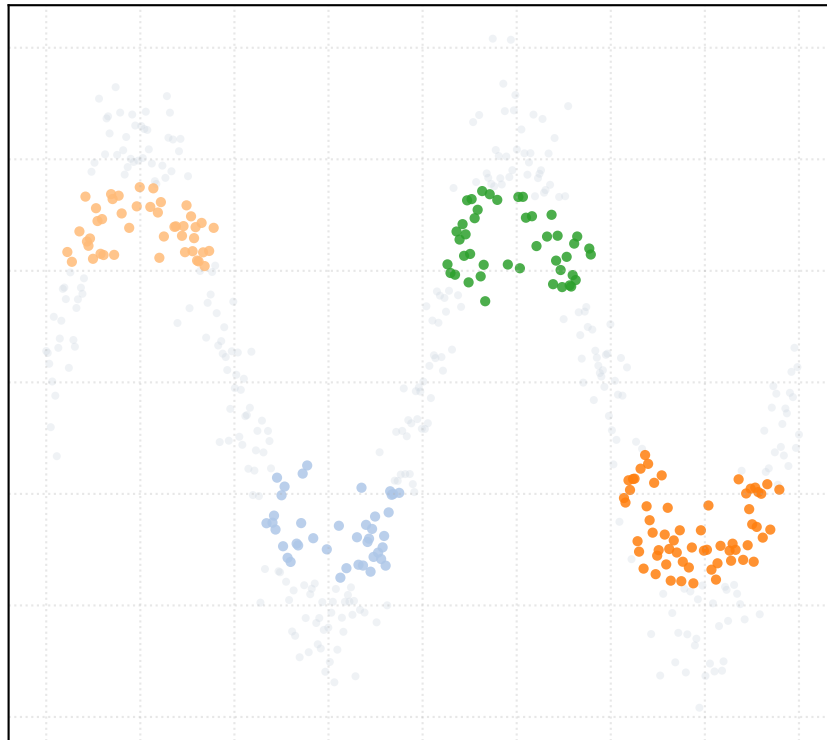


Dataset: s_curve_noise_high | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

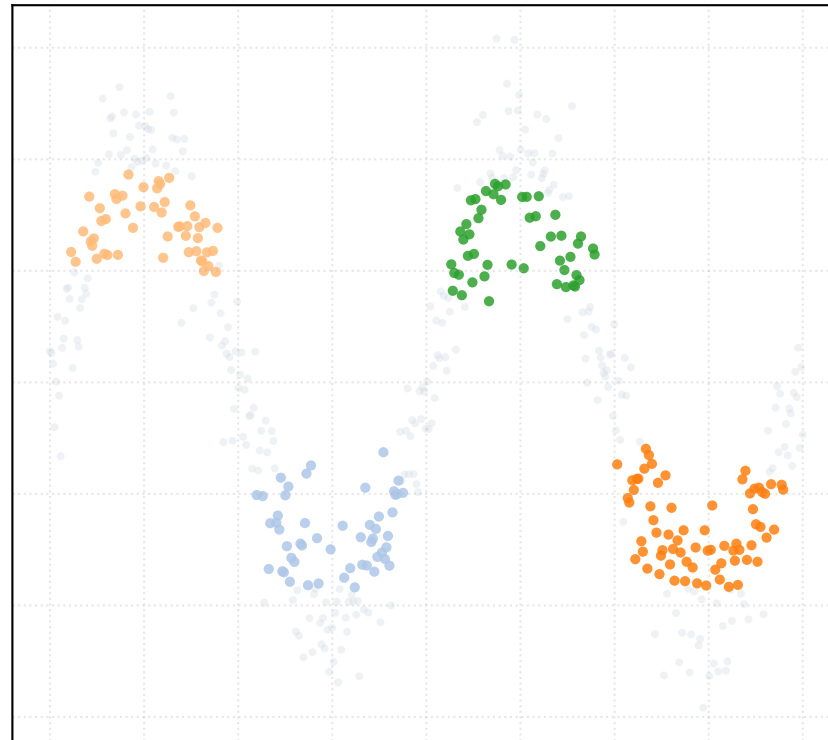
alpha_MAD = 0.25 [5 Clusters]



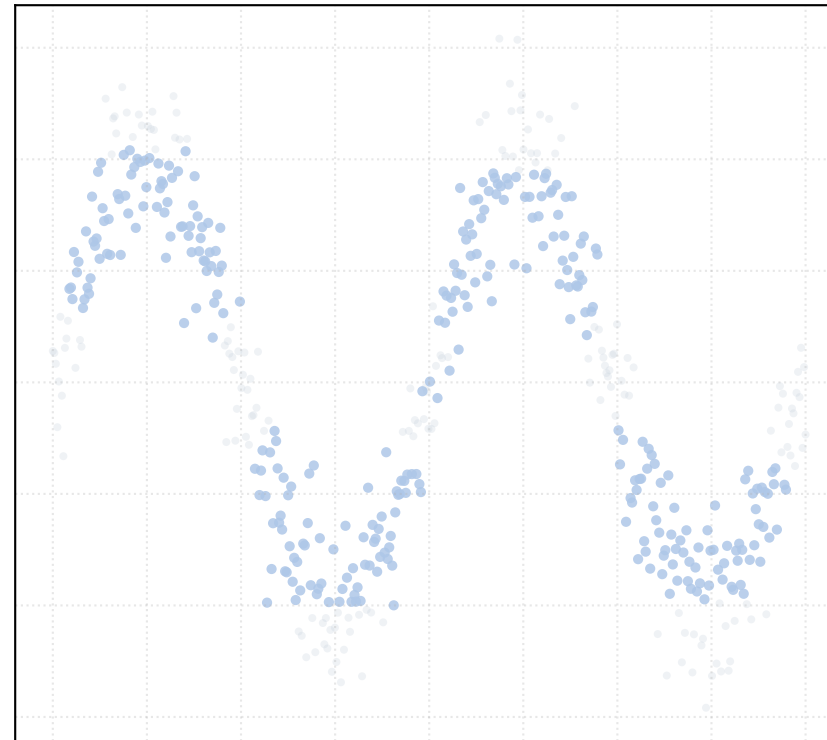
alpha_MAD = 0.5 (Default) [4 Clusters]



alpha_MAD = 0.75 [4 Clusters]

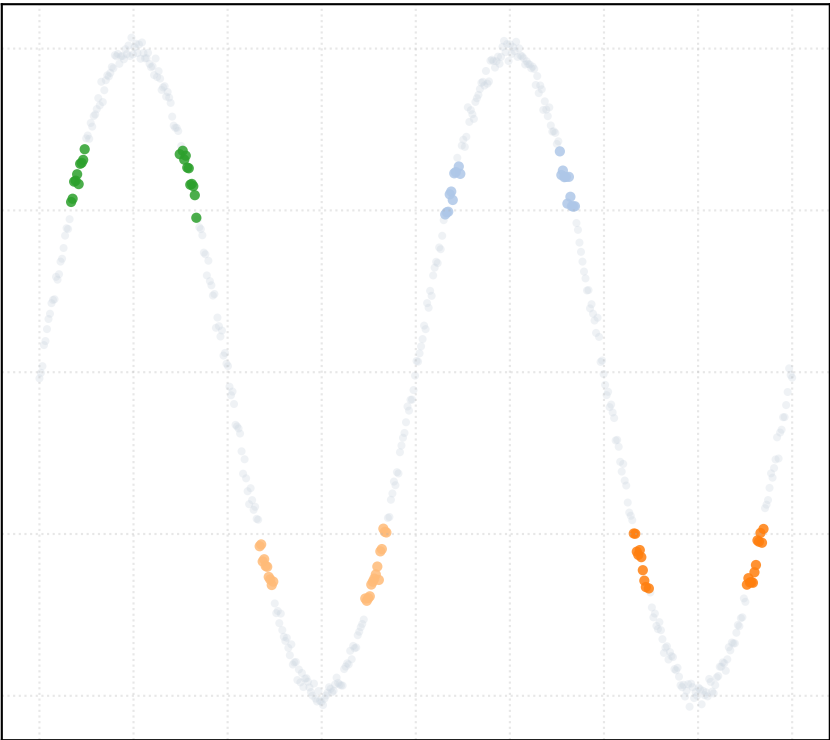


alpha_MAD = 1.0 [1 Clusters]

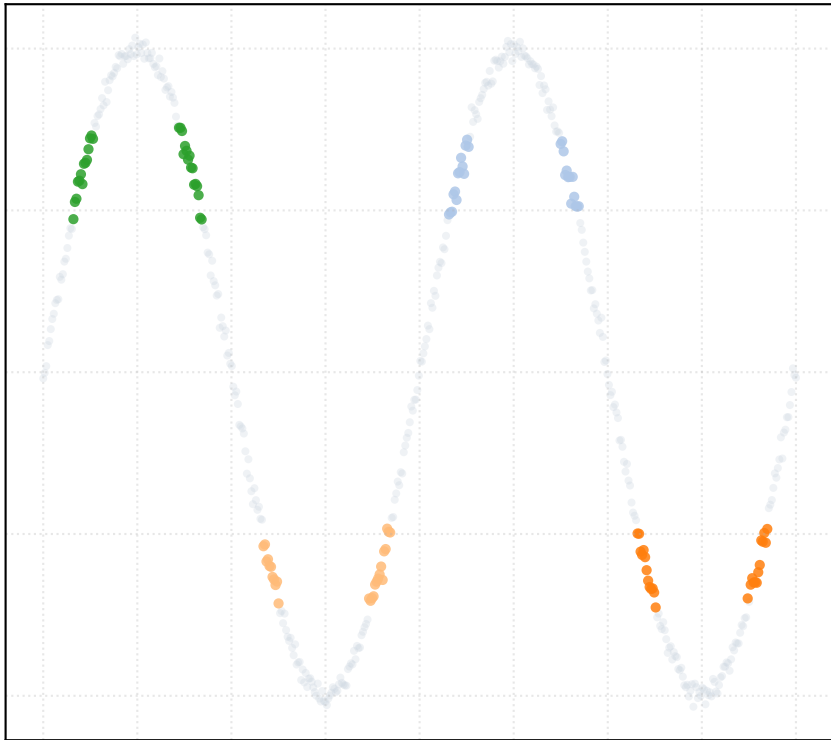


Dataset: s_curve_noise_low | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

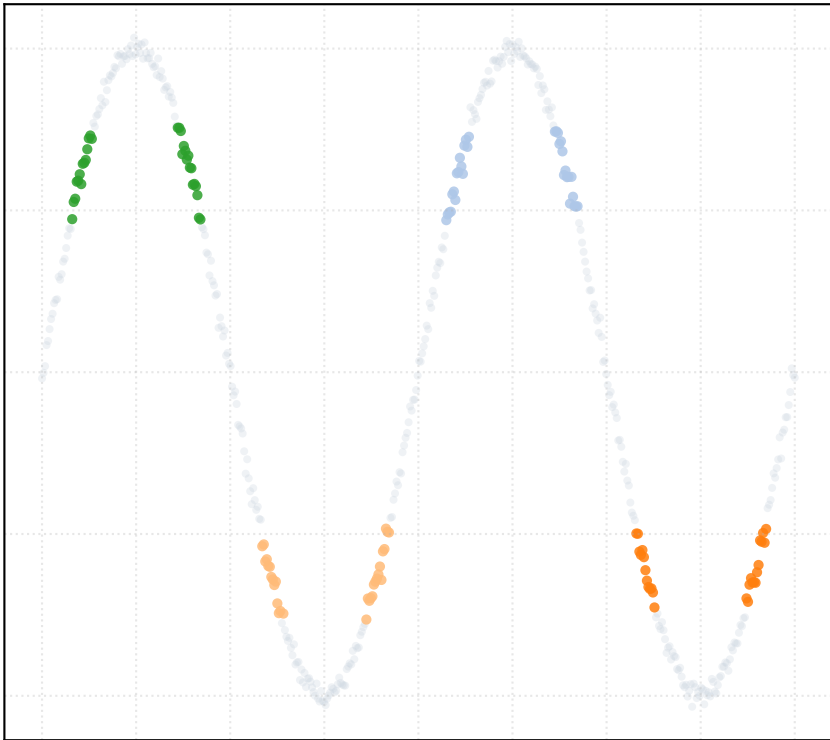
alpha_MAD = 0.25 [4 Clusters]



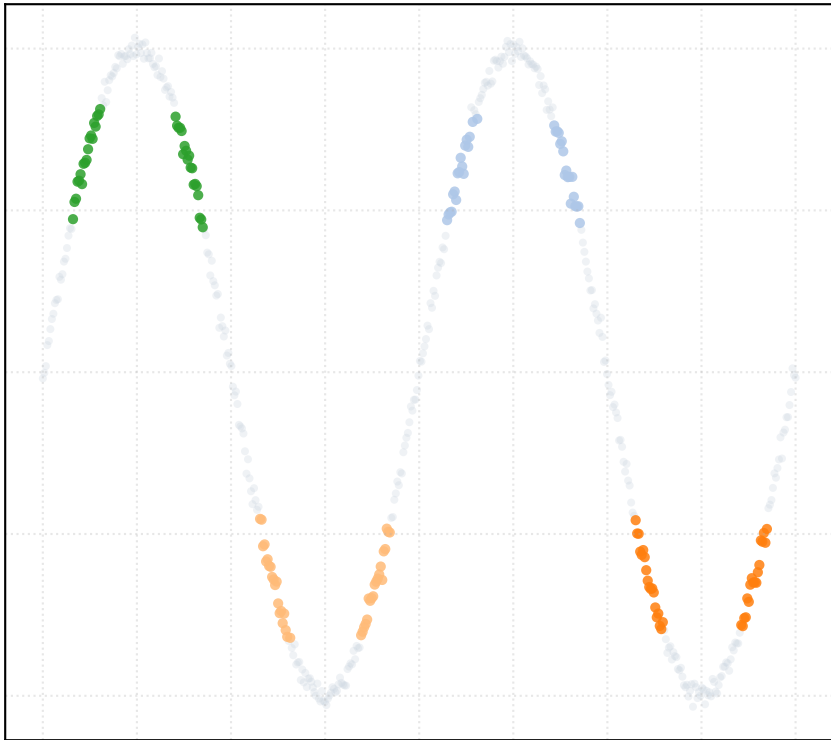
alpha_MAD = 0.5 (Default) [4 Clusters]



alpha_MAD = 0.75 [4 Clusters]

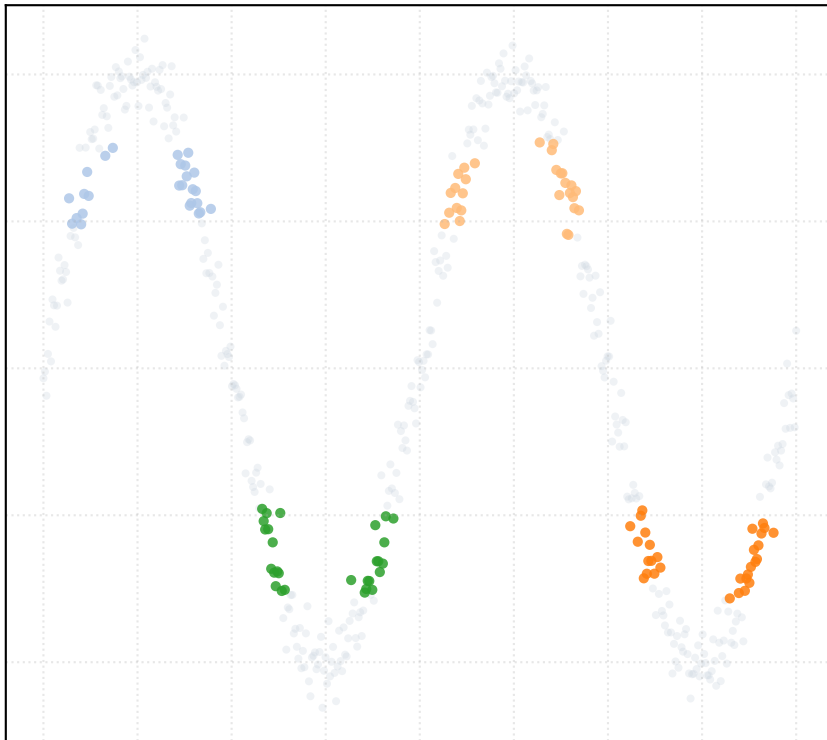


alpha_MAD = 1.0 [4 Clusters]

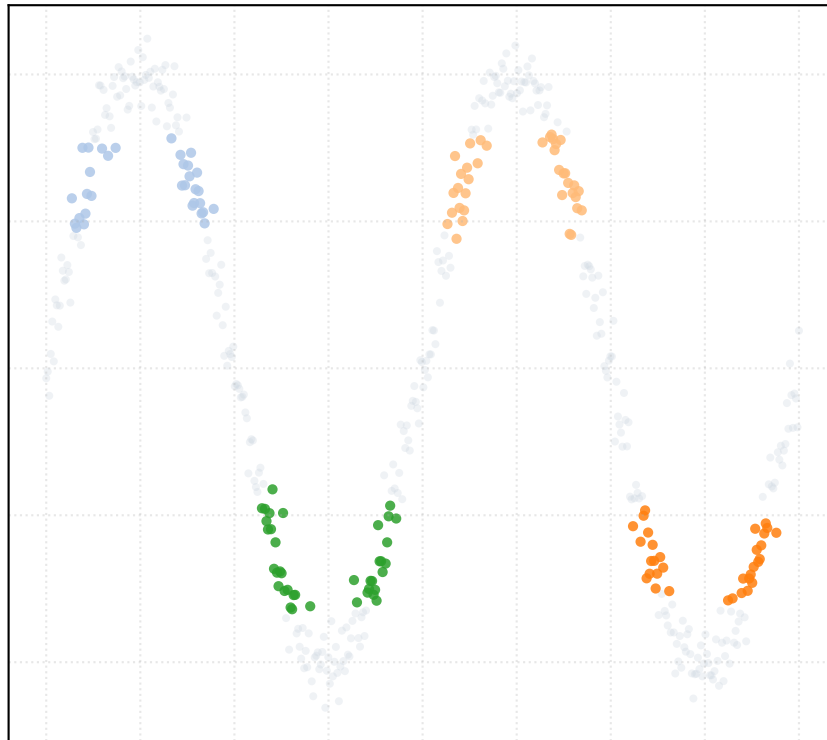


Dataset: s_curve_noise_medium | Surface Density Threshold Sweep (alpha_MAD in grow_ensemble)

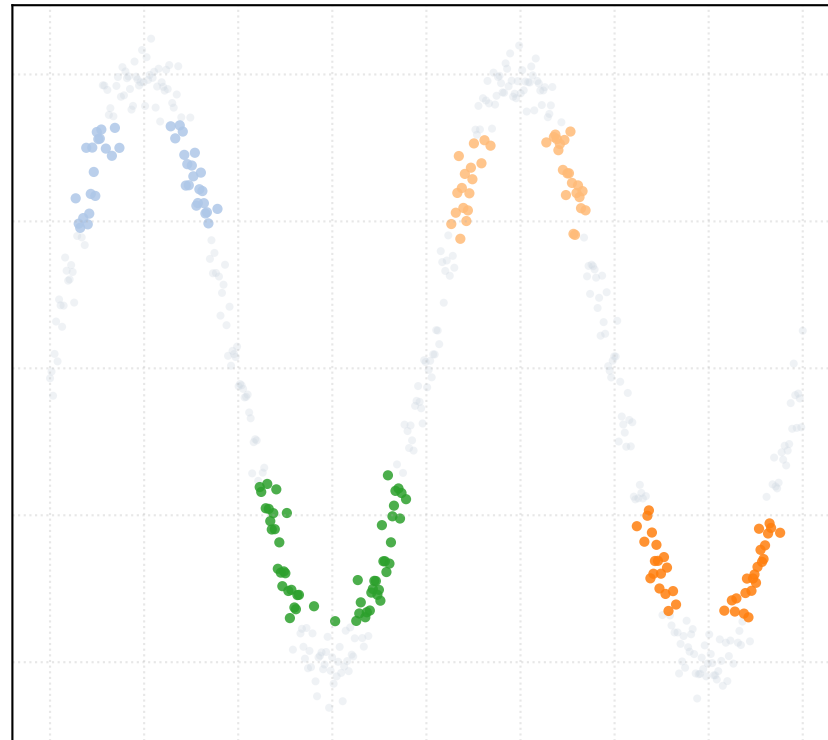
alpha_MAD = 0.25 [4 Clusters]



alpha_MAD = 0.5 (Default) [4 Clusters]



alpha_MAD = 0.75 [4 Clusters]



alpha_MAD = 1.0 [4 Clusters]

